

Classical and Quantum Query Complexity of Scalar Newton Quantities

Abstract

The positive inverse quadratic form $b^*H^{-1}b$ is the squared Newton decrement when H and b are a Hessian and gradient. Its query complexity need not coincide with the cost of preparing or writing a Newton direction. We study relative estimation under sparse and sampling-and-query access, and compare it with coherent block access. For $\kappa^{-1}I \preceq H \preceq I$ and row sparsity d , a classical local estimator uses $\kappa\epsilon^{-2}(d+1)^{O(\sqrt{\kappa}\log(2/\epsilon))}$ queries at constant confidence. Full-sampling lower families separately force the statistical factor and the exponential scale. Explicit clock diagonals and statistical composition give simultaneous lower bounds in stated regimes, without establishing the product of both full factors. Under plain block access of normalization α , a three-dimensional lower bound matches, up to logarithmic factors, the known variable-time upper bound $\tilde{O}(\alpha\kappa/\epsilon)$. We realize hard scalars as LP and SOCP values, optimizer coordinates, and Newton decrements, preserving the promised input interfaces. We also give scalar trajectory and endpoint constructions, together with examples and residual-certified reuse bounds explaining why one-shot hardness cannot simply be multiplied by an iteration count. Finally, a sparse-base solver with sampled low-rank corrections yields scalar estimates and solution samples for structured Lorentz and generalized-power Newton systems under explicit spectral and data-access assumptions. All bounds distinguish query cost, arithmetic, input acquisition, and the requested output.

1 Introduction

A Newton method may need only one number before it decides how to move: the squared decrement $b^*H^{-1}b$. It measures local centrality and Newton progress under the specified objective and barrier; by itself it is not a general certificate of global optimality. A linear-system routine may return a vector, a sample from its squared coordinates, or a normalized quantum state. These outputs carry different information. Normalization can discard the scale needed by a scalar estimate, while writing a whole vector can cost much more than estimating one of its observables. The distinction matters when assessing quantum improvements to interior-point methods, whose work includes acquiring iterate data, solving Newton systems, choosing steps, and reporting a final answer.

This paper studies the scalar itself and then asks when its complexity survives an optimization formulation. Our central problem is relative estimation of

$$q(H, b) = b^*H^{-1}b, \quad \kappa^{-1}I \preceq H \preceq I.$$

Classical algorithms receive local sparse matrix access and sampling-and-query access to b : coordinate values, its norm, and samples proportional to squared coordinate magnitudes. Lower bounds remain valid when the matrix also supplies exact row and column norms and the associated sampling operations. Quantum comparisons separately specify coherent entry access or a normalized matrix block inside a unitary. The distinction between these two quantum interfaces is necessary even for diagonal matrices of dimension three.

1.1 Results and contribution

The results have three connected parts. First, we identify the relative positive-form bounds justified by these interfaces. Second, we construct optimization outputs that retain hardness and examine what information can be reused across a path. Third, we give structured classical solves that delimit the scope of the hard constructions. Table 1 summarizes the central one-scalar rates; later sections state the optimization and structured-system contracts separately.

Relative positive forms and simultaneous lower bounds. A Chebyshev residual polynomial, local sparse evaluation, and coordinate importance sampling give Theorem 3.4. Positivity allows the Kantorovich inequality to bound the estimator’s relative second moment by $O(\kappa)$, yielding the statistical factor $\kappa\epsilon^{-2}$ rather than a bound quadratic in κ . These are established ingredients; the result makes their relative-error and access implications explicit. Theorem 4.1 transfers the matrix-function hardness of Montanaro–Shao to real positive definite forms, verifies every full-SQ operation, and realizes the result at an LP analytic center. Theorem 4.5 supplies an independent two-sparse family with sharp statistical and confidence dependence, including the finite-population cutoff.

Separate lower families do not justify multiplying their costs. Section 5 proves a distributional composition theorem for a single positive form and calculates the inverse diagonals of an explicit witness clock. The diagonal determines the baseline against which relative error measures a hidden signal. This yields, for example, a simultaneous $\epsilon^{-2}s^{\Omega(\sqrt{\kappa})}$ lower bound up to polynomial denominators, and a different κ^2 product in a restricted high-accuracy regime. We also prove contrast limitations for specified cyclic, two-cluster, and Schur-complement constructions. These calculations identify what the constructions establish and which stronger products they do not establish.

For plain coherent block access, Theorem 6.1 combines the known variable-time inverse-power norm algorithm with a matching $\Omega(\alpha\kappa/\epsilon)$ block-query lower bound. Its explicit completion also explains why the lower bound does not transfer to exact entry queries. For general coherent sparse access, the proved lower and upper bounds remain $\Omega(\sqrt{\kappa}/\epsilon)$ and $\tilde{O}(d\kappa/\epsilon)$, with state preparation charged separately.

Optimization outputs and repeated queries. Section 7 constructs LPs and SOCPs with hard optimizer coordinates, ordinary values, or Newton decrements. An exact cyclic inverse supplies public norms and a tunable conditioning–sparsity–accuracy relation. A public inverse-transpose norm gives a larger value signal; an affine-slice LP attains a matching classical upper and lower exponential dependence under its stated value-accuracy contract. The barrier parameter of that slice is one, but eliminating its equalities requires acquiring the hidden information. No free reduced objective is assumed.

Section 8 arranges independent inputs in successive scalar increments and ordinary SOCP predictor decrements. It also compiles promised overlaps into one bounded optimizer coordinate. The resulting direct-sum and endpoint bounds charge the information requested by those outputs. Section 9 gives the complementary boundary: a sparse KKT system can have the same hard normalized solution at every central multiplier, even while reaching a small objective gap requires many bounded local movements. One acquisition can then serve every checkpoint. Residual-certified span reuse gives exact and approximate refresh bounds, with current residual tests and explicit-output costs charged separately.

Structured Newton systems. Theorem 10.1 handles $N = S + CXLX^T C^T$, where $S = CC^T$ is a well-conditioned sparse full-rank base and L is a small, possibly singular or indefinite correction. Its inputs are samples from source vectors for the columns of X . The algorithm never needs to normalize a nearly zero transformed column. A sampled small system, local polynomial evaluation, and raw rejection proposals produce one fixed approximate solution, with a reusable setup and bounded-cost flagged draws. Corollary 10.2 gives relative inverse-form estimation. Lorentz and generalized-power barriers satisfy the required structure under explicit eccentricity, sparse-base conditioning, and scalar-access promises. The need to acquire the power-cone geometric mean is a real restriction, for which we prove a query lower bound. Eccentricity-profile and supplied-treewidth solves give separate comparisons when a full classical direction is required.

To the best of our knowledge, the additional contributions are the full-SQ positive-form lower bounds, the explicit single-form composition and baseline calculations, the parameterized scalar optimization and temporal realizations with their reuse boundaries, and the sparse-full-rank correction theorem under the specified source-vector interface. This is a claim about these precise statements. It is not a priority claim for inverse-form estimation, sparse polynomial dequantization, Forrelation matrix constructions, scalar optimization hardness, low-rank cone algebra, or subspace recycling.

Access and problem	Bound	Regime and location
Classical sparse H , SQ(b); relative scalar	$O(\kappa\epsilon^{-2} \log(1/\zeta)(d+1)^{O(\sqrt{\kappa} \log(2/\epsilon))})$ upper	$\kappa \geq 1$, $0 < \epsilon \leq 1/2$; Theorem 3.4
Classical full SQ; two-sparse family	$\Omega(\min\{M, \kappa\epsilon^{-2}\})$ lower in dimension $2M$; $\Omega(\kappa\epsilon^{-2} \log(1/\zeta))$ for sufficiently large M	$\kappa \geq 4$, $\epsilon \leq 1/128$; Theorem 4.5
Classical full SQ; s -sparse clock family	$\Omega\left(\frac{((s-1)/2)^{\Omega(\sqrt{\kappa} \log(1/\epsilon))}}{\log(s)\sqrt{\kappa} \log(1/\epsilon)}\right)$ lower	$\kappa \geq 4$, $s \geq 5$, small ϵ ; Theorem 4.1
Classical full SQ; one-form composition	$\Omega((\chi/\epsilon)^2 L_g)$ lower, where χ is the signal-to-baseline ratio and L_g the inner randomized query cost	$0 < \chi \leq 5/3$, $\epsilon \leq \chi/200$; Theorem 5.2
Quantum plain block encoding of normalization α	$\Omega(\alpha\kappa/\epsilon)$ lower and $\tilde{O}(\alpha\kappa/\epsilon)$ upper in block calls	$\alpha \geq 1$, $\kappa \geq 4$, $\epsilon \leq 1/16$; Theorem 6.1
Quantum coherent sparse access	$\Omega(\sqrt{\kappa}/\epsilon)$ lower and $\tilde{O}(d\kappa/\epsilon)$ upper; the canonical two-sparse sign-block family attains the lower rate	$d \geq 2$, $\kappa \geq 4$, $\epsilon \leq 1/128$; Corollary 4.6, (33)

Table 1: Selected relative inverse-form query bounds. Lower-bound dimensions are selected by their constructions; the population and promise restrictions in the cited theorems apply. Quantum rows charge input-state preparation separately and use constant failure probability. The clock lower bound first estimates two public forms and hence lower-bounds a routine for an arbitrary supplied form. The composition row requires the theorem’s oracle-preserving inner family. The rows are separate statements: their lower bounds must not be multiplied.

1.2 Relationship to prior work

Inverse quadratic forms are standard numerical quantities. Li–Sra–Jegelka develop Gaussian quadrature for matrix inverse forms [30], and Bizas–Mitrouli–Turek give further analytic and heuristic estimates [10]. Those methods charge matrix–vector products. Our classical model instead explores neighborhoods of sampled coordinates, trading dimension dependence for exponential

dependence on polynomial degree.

Gharibian–Le Gall already give sparse polynomial overlap estimation from sampling access [22, Theorem 4.1]. Montanaro–Shao relate matrix-function query complexity to approximation degree and construct the weighted Forrelation clocks used here [32, Theorem 1.8]. Cifuentes et al. classify matrix-function tasks under several access and measurement models [19]. Edenhofer–Hasegawa–Le Gall apply sparse polynomial evaluation to spectral sums, including reciprocal traces [20, Theorem 3.1 and Section 3.3]. Our target is an arbitrary-vector positive inverse form with relative error, whose baseline and variance must be controlled. The polynomial principle and its approximation-theoretic origin are shared with this prior work. Le Gall’s robustness results already cover approximate length-squared sampling for sparse and low-rank dequantization [29]; our finite-error transfer is a sufficient condition for the particular estimator, not a general new robustness framework.

For quantum algorithms, block encodings and polynomial transformations are established tools [23], and our block-access upper bound specializes Chakraborty–Gilyén–Jeffery’s norm estimator [15, Theorem 33 in the full version]. Alase et al. study scalar observables of linear-system solutions [2]. Their tight observable-query bound concerns the queried observable block and additive expectation error; our sharp block bound charges the system matrix for relative estimation of $b^*H^{-1}b$. Orsucci–Dunjko study positive definite solution-state preparation and stronger access structures that can improve conditioning dependence [36]. A normalized solution state, a shifted-matrix oracle, and the plain block interface used here are distinct contracts. Section 6 details these comparisons.

Grønlund–Larsen establish exponential quantum versus SQ separation for linear-system solution output [25]. Apers–Gribling already prove full-SQ hardness for an ordinary LP optimal value [5, Lemma 8.5]. Our optimization constructions therefore address parameter control, scalar normalization, genuine Newton realizations, and input acquisition rather than the first occurrence of scalar or optimization hardness. An adjacent result of Zhao et al. studies normalized solution observables and Forrelation embeddings under streaming-space and data-sample restrictions [40, Task F.1 and Sections E.5–F]. Here the input is static and classical storage is unrestricted; those space restrictions are not used. General direct-sum, composition, and XOR ingredients are also prior [8, 11, 13, 16]. Our contribution in that part is their realization in the specified optimization outputs and the proof of all interface simulations.

Low-rank SQ matrix arithmetic and solution sampling are established [18]. The structured theorem here has a sparse full-rank base and a small correction, and avoids supplying normalized transformed columns as inputs. Cone Hessian identities and sparse implementations are likewise established [3, 17, 24, 26, 38]. We prove the spectral and sampling conditions needed to use that algebra in a local-query theorem. Krylov recycling and greedy reduced bases [9, 37], as well as sparse and low-treewidth elimination [21, 39], supply the classical context for our reuse and full-output comparisons.

1.3 Scope and organization

All oracle and ideal-arithmetic assumptions are stated in Section 2. Sections 3–5 give classical algorithms, lower families, and composition. Section 6 treats coherent access. Sections 7–9 connect scalar outputs to optimization and distinguish temporal composition from reuse. Section 10 gives the structured cone-system comparisons.

The results are query and arithmetic statements under specified inputs. They do not certify a generic quantum interior-point speedup, a uniform bit-complexity bound, or free maintenance of iterate-dependent sampling oracles. The general joint classical bound and the general coherent sparse bound are not fully matched. These are boundaries of the classifications proved here; none of the optimization or structured results assumes their resolution.

2 Problems and access models

All vector norms are Euclidean and all matrix norms are spectral unless specified otherwise. The adjoint is denoted by $*$. A Hermitian matrix $H \succ 0$ is supplied with public bounds

$$\kappa^{-1}I \preceq H \preceq I, \quad \kappa \geq 1. \quad (1)$$

Here κ is a certified upper bound on its condition number; statements requiring equality say so explicitly. For a nonzero $b \in \mathbb{C}^N$, our main target is

$$q(H, b) = b^* H^{-1} b. \quad (2)$$

An estimate has relative error ϵ if $|\hat{q} - q| \leq \epsilon q$. If $H = \nabla^2 F(x)$ and $b = \nabla F(x)$, this is the squared Newton decrement at x . More generally, b may be a linear-objective perturbation of the barrier gradient. A theorem about an arbitrary pair (H, b) becomes an optimization theorem only after specifying an objective, an interior point, and their input oracles. We give such realizations separately.

For the optimization sections, a barrier ϕ is a C^3 convex function on the relative interior of its feasible affine slice, expressed in independent coordinates and with positive definite restricted Hessian. Its local norm is $\|h\|_x^2 = h^T \nabla^2 \phi(x) h$, with dual norm $\|g\|_{x,*}^2 = g^T \nabla^2 \phi(x)^{-1} g$. A self-concordant barrier satisfies $|D^3 \phi(x)[h, h, h]| \leq 2\|h\|_x^3$ and diverges at the boundary; a parameter ν additionally bounds $|D\phi(x)[h]| \leq \sqrt{\nu}\|h\|_x$. Parameters stated below refer to the specified barrier and coordinates, not to every formulation of the same feasible set. For maximizing $c^T x$, the barrier subproblem is $F_\eta(x) = \phi(x) - \eta c^T x$, $\eta > 0$. A central point minimizes F_η ; an analytic center minimizes ϕ . The Newton direction is $-\nabla^2 \phi(x)^{-1} \nabla F_\eta(x)$ and its squared decrement is $\|\nabla F_\eta(x)\|_{x,*}^2$. At a central point, changing the multiplier by $\Delta\eta$ gives a predictor squared decrement $(\Delta\eta)^2 \|c\|_{x,*}^2$. Equality-constrained versions use the Hessian and gradient restricted to the tangent space, or the explicitly stated KKT equations. These intrinsic quantities need not have the condition number of an augmented KKT matrix.

Definition 2.1 (Sparse access and vector sampling). Row-sparse access to a matrix supplies the nonzero count in each requested row, the location of its j th nonzero, and its value, at unit query cost. The ordering is fixed for each input. A known bound d limits the count. Column-sparse access has the analogous meaning. For Hermitian matrices, column access follows from row access by conjugation.

For $v \neq 0$, SQ(v) supplies coordinate queries, the norm $\|v\|$, and independent samples from $p_v(i) = |v_i|^2 / \|v\|^2$. Zero coordinates have zero sampling probability. These are exact oracles in our main model. Full matrix SQ additionally supplies row and column norms, the Frobenius norm, within-row and within-column squared-magnitude samples, and samples of rows or columns proportional to their squared norms. A zero row or column is identified by its zero norm and has no conditional sampling request. This stronger interface includes sparse locations when a lower bound explicitly grants both interfaces.

Specifying these oracles matters: constructing a sampling data structure from an arbitrary explicit vector generally entails an initial pass over that vector. We do not charge this preprocessing to an algorithm whose input contract already grants SQ(v). Conversely, a lower-bound reduction must simulate every granted oracle; it cannot ignore hidden information in norms or sampling probabilities. Matrix SQ is a classical interface and by itself does not grant a coherent quantum state-preparation unitary.

Our coherent statements specify a unitary U whose designated top-left block is H/α , together with controlled calls to U, U^* and a specified preparation of $b/\|b\|$. The normalization α and the allowed unitary completion are part of that access model. An exact sparse-value oracle, which returns an entry as a number, is a different interface. A black-box block-encoding lower bound need not hold for it.

We count oracle invocations and ideal real or complex arithmetic operations. Independent uniform random draws from $[0, 1]$ and exact comparisons with computed real probabilities are unit-cost primitives in this model. In particular, the exact rejection samplers below use ideal randomness; their arithmetic bounds do not count a finite-bit implementation of these random choices. An index is one word; its bit length is $\lceil \log_2 N \rceil$. Thus “dimension-independent” means no dependence on N in these counts, not an N -independent bit cost or input-loading cost. The notation $\tilde{O}$ suppresses logarithmic factors only in the displayed parameters. Our direct classical upper bounds below use ordinary O notation, with unspecified absolute constants in polynomial degrees and local-walk exponents. Unless stated otherwise, $0 < \epsilon \leq 1/2$ and $0 < \zeta < 1/2$ denote accuracy and failure probability. Unless a different contract is stated, every classical or quantum lower bound requires success probability at least $2/3$ on every promised input and charges the worst-case number of queries. Explicit confidence parameters and distributional expected-cost contracts take precedence over this convention.

Besides one scalar, we consider a solution-sampling output: the algorithm provides coordinate access and samples exactly from $p_{\tilde{x}}$ for one fixed vector satisfying $\|\tilde{x} - A^{-1}b\| \leq \epsilon \|A^{-1}b\|$. This is weaker than writing all coordinates, and stronger than an unspecified approximate output distribution. It does not include an exact norm oracle for $\tilde{x}$. A lower bound using a rare event must also specify whether unflagged algorithmic failure may contaminate that event. These distinctions prevent transferring bounds between incompatible output contracts.

Finally, a public interval $\mu I \preceq H \preceq LI$ reduces to (1) by using H/L with $\kappa = L/\mu$. The resulting inverse quadratic form must be divided by L .

3 Classical local algorithms

The ingredients are classical: Chebyshev residual polynomials, local sparse-walk evaluation, and importance sampling. Sparse-polynomial overlap estimation already appears in Gharibian–Le Gall [22]; matrix-function entry estimation is studied by Montanaro–Shao [32] and Cifuentes et al. [19]. The specialization needed here preserves relative error for a positive inverse quadratic form and uses positivity to reduce the statistical factor from κ^2 to κ . This section proves that specialization, then records the stronger output and nonsymmetric variants needed to compare the optimization constructions.

3.1 A relative inverse polynomial

Lemma 3.1 (Chebyshev residual). *For $K > 1$ and $0 < \eta < 1$, there is a real polynomial P of degree at most $m - 1$, where*

$$m = \left\lceil \frac{\log(2/\eta)}{\log((\sqrt{K} + 1)/(\sqrt{K} - 1))} \right\rceil = O(\sqrt{K} \log(2/\eta)), \quad (3)$$

such that $|1 - \lambda P(\lambda)| \leq \eta$ for $\lambda \in [K^{-1}, 1]$. Consequently, for $K^{-1}I \preceq H \preceq I$,

$$(1 - \eta)H^{-1} \preceq P(H) \preceq (1 + \eta)H^{-1}, \quad \|I - HP(H)\| \leq \eta. \quad (4)$$

For $K = 1$, one may take $P = 1$.

Proof. Put $a = K^{-1}$, $c = (1 + a)/(1 - a)$, and $x(\lambda) = (1 + a - 2\lambda)/(1 - a)$. If T_m is the degree- m Chebyshev polynomial, set

$$r_m(\lambda) = \frac{T_m(x(\lambda))}{T_m(c)}, \quad P(\lambda) = \frac{1 - r_m(\lambda)}{\lambda}.$$

Since $x(0) = c$, the numerator vanishes at zero, so the quotient is a polynomial. On $[a, 1]$, $|T_m(x(\lambda))| \leq 1$. With $\theta = (\sqrt{K} - 1)/(\sqrt{K} + 1)$, the identity $T_m(c) = (\theta^{-m} + \theta^m)/2$ gives $|r_m(\lambda)| \leq 2\theta^m$. The displayed choice of m makes this at most η . Finally $\log((1 + u)/(1 - u)) \geq 2u$ for $u = 1/\sqrt{K}$ proves the degree estimate; the matrix inequalities follow by diagonalizing H . $\square$

Lemma 3.2 (Local evaluation). *If H is d -row-sparse, one coordinate of $P(H)v$, for a specified degree- k polynomial P , can be evaluated with $(d + 1)^{O(k+1)}$ sparse and coordinate queries and arithmetic operations. If row and column sparsity are both bounded by d , a full row or column of $P(H)$ can be materialized in the same bound. More generally, a polynomial $q(A^*A)A^*$ of degree k in A^*A has this property under row and column access to A , with walk length at most $2k + 1$.*

Proof. Expand a requested coordinate of $H^j v$ into matrix walks of length j starting from that coordinate. Each step has at most d choices. Summing for $0 \leq j \leq k$ costs at most $O((k + 1)^2(d + 1)^{k+1})$ operations, including multiplying path weights and aggregating equal endpoints. The same expansion without the terminal coordinate of v materializes the corresponding row; reverse walks materialize a column. One can avoid sorting by retaining the walk list and aggregating equal endpoints by comparisons in quadratic time, still within the stated bound. For $q(A^*A)A^*$ the walks alternate between column and row queries to A . No length- N vector is formed. $\square$

3.2 Positive quadratic forms

Lemma 3.3 (Positive-form second moment and the Kantorovich inequality). *If $P \succ 0$ has spectrum in $[\alpha, \beta]$ and $b \neq 0$, sample $I \sim p_b$ and define*

$$X = \|b\|^2 \frac{(Pb)_I}{b_I}. \quad (5)$$

Then

$$\mathbb{E}X = b^*Pb, \quad \frac{\mathbb{E}|X|^2}{(b^*Pb)^2} \leq \frac{\|b\|^2 \|Pb\|^2}{(b^*Pb)^2} \leq \frac{(\alpha + \beta)^2}{4\alpha\beta}. \quad (6)$$

The last inequality is the classical Kantorovich inequality in an equivalent form; see Lin [31, Eq. (1.1)]. It is sharp.

Proof. The sampling identity follows from $|b_i|^2/b_i = \bar{b}_i$ on the support of b . The second moment is $\|b\|^2 \sum_{i:b_i \neq 0} |(Pb)_i|^2$, which is bounded by the expression shown. Under the spectral weights of $b/\|b\|$, let Y be an eigenvalue of P and $t = \mathbb{E}Y$. Since $(Y - \alpha)(Y - \beta) \leq 0$,

$$\frac{\mathbb{E}Y^2}{(\mathbb{E}Y)^2} \leq \frac{(\alpha + \beta)t - \alpha\beta}{t^2} \leq \frac{(\alpha + \beta)^2}{4\alpha\beta}.$$

The last maximum occurs at $t = 2\alpha\beta/(\alpha + \beta)$. Equality is attained by $P = \text{diag}(\alpha, \beta)$ with squared coordinates of normalized b equal to $\beta/(\alpha + \beta)$ and $\alpha/(\alpha + \beta)$. This proves sharpness for this estimator, without asserting an algorithm-independent lower bound. $\square$

Theorem 3.4 (Relative scalar estimation). *Suppose H satisfies (1), is d -row-sparse, and row-sparse access and $\text{SQ}(b)$ are available. There is a randomized algorithm returning a relative- ϵ estimate of $q(H, b)$ with failure probability at most ζ , using*

$$O\left(\kappa\epsilon^{-2}\log(1/\zeta)(d+1)^{O(\sqrt{\kappa}\log(2/\epsilon))}\right) \quad (7)$$

queries and ideal arithmetic operations. Full matrix SQ is unnecessary.

Proof. For $\kappa = 1$, $H = I$ and the norm oracle returns $q = \|b\|^2$ exactly. Otherwise choose the polynomial in Lemma 3.1 with $\eta = \epsilon/4$ and put $Q = b^*P(H)b$. Then $|Q - q| \leq \eta q$. The eigenvalues of $P(H)$ lie in $[1 - \eta, \kappa(1 + \eta)]$, so their ratio is at most $9\kappa/7$, because $\eta \leq 1/8$. Lemma 3.3 gives $\text{Var}(\text{Re } X) \leq C\kappa Q^2$ for an absolute constant C .

Divide independent samples into an odd number $g = O(\log(1/\zeta))$ of groups, each of size $h = \lceil 64C\kappa/\epsilon^2 \rceil$. Chebyshev's inequality bounds by $1/4$ the probability that a group mean differs from Q by more than $\epsilon Q/4$. The median of the group means fails only if at least half the groups fail; the binomial Chernoff bound makes this probability at most ζ for a suitable absolute constant in g . On success,

$$|\hat{q} - q| \leq (\epsilon/4)Q + (\epsilon/4)q \leq (\epsilon/4)(2 + \epsilon/4)q < \epsilon q.$$

Each sample requires one local polynomial evaluation, covered by Lemma 3.2; multiplying its cost by gh proves the result. $\square$

Unlike standard inverse-form quadrature, which charges full matrix-vector products [30], the algorithm only explores neighborhoods of sampled coordinates. The price is exponential growth in the degree. Its exponential dependence is $\exp(O(\sqrt{\kappa}\log(d+1)\log(2/\epsilon)))$; calling the bound polynomial in κ would therefore be incorrect.

Corollary 3.5 (Bilinear forms). *With the same matrix assumptions, $\text{SQ}(a)$ and coordinate access to b suffice to estimate $a^*H^{-1}b$ to additive error*

$$\epsilon\sqrt{(a^*H^{-1}a)(b^*H^{-1}b)}$$

*with the complexity in (7). If the public promise $|a^*H^{-1}b| \geq \gamma\sqrt{(a^*H^{-1}a)(b^*H^{-1}b)}$ holds for $0 < \gamma \leq 1$, replacing ϵ by $\gamma\epsilon$ gives relative error ϵ .*

Proof. Write $A = a^*H^{-1}a$ and $B = b^*H^{-1}b$. The zero-vector cases return zero. The residual representation gives

$$|a^*(P(H) - H^{-1})b| \leq \eta\sqrt{AB}.$$

Indeed, $P(H) - H^{-1} = H^{-1/2}EH^{-1/2}$ for a Hermitian E with $\|E\| \leq \eta$. Sampling from p_a , the random variable $X = \|a\|^2(P(H)b)_I/a_I$ has mean $a^*P(H)b$ and second moment at most $\|a\|^2\|P(H)b\|^2$. The pointwise spectral inequality $P(H)^2 \preceq \kappa(1 + \eta)^2H^{-1}$ and $\|a\|^2 \leq A$ bound this by $\kappa(1 + \eta)^2AB$. Apply the group-mean argument separately to real and imaginary parts, with tolerances $\epsilon\sqrt{AB}/4$ and failure probabilities $\zeta/2$. The sample count depends on the known upper bound for the ratio of variance to tolerance squared, so it does not require knowing A or B . Taking $\eta = \epsilon/4$ proves the claim. $\square$

3.3 Sampling a sparse inverse solution

Scalar estimation need not construct a solution sampler. For comparison, the following elementary rejection scheme also gives that stronger output.

Lemma 3.6 (Sparse-transform rejection sampler). *Let P be invertible, with at most R nonzeros per row and column and singular values in $[a_P, b_P]$. Given materializable sparse rows and columns of P and $\text{SQ}(v)$, one can sample exactly from p_{Pv} using at most $R(b_P/a_P)^2$ trials in expectation. Each trial materializes at most one row and one column and uses $O(R)$ further coordinate operations.*

Proof. Let $c_j^2 = \sum_i |P_{ij}|^2$ and $S_i = \sum_j |P_{ij}v_j|^2$. Sample $j \sim p_v$, accept with probability c_j^2/b_P^2 , and, if accepted, sample i from column j with probability $|P_{ij}|^2/c_j^2$. Compute $y_i = (Pv)_i$ and S_i from row i , and accept i with probability $|y_i|^2/(RS_i)$. The event $S_i = 0$ has zero probability at this stage and can be defined to reject. Column norm bounds and Cauchy–Schwarz show both acceptance probabilities lie in $[0, 1]$. Summing over j , the probability of outputting i in one raw trial is

$$\sum_j \frac{|v_j|^2}{\|v\|^2} \frac{c_j^2}{b_P^2} \frac{|P_{ij}|^2}{c_j^2} \frac{|y_i|^2}{RS_i} = \frac{|y_i|^2}{Rb_P^2 \|v\|^2}.$$

Conditioning on acceptance gives p_{Pv} . Its total probability is at least $a_P^2/(Rb_P^2)$, proving the expected trial count. $\square$

Theorem 3.7 (Solution sampling). *Let $A \in \mathbb{C}^{N \times N}$ be invertible with $\|A\| \leq 1$, $\sigma_{\min}(A) \geq K^{-1}$, and at most d nonzeros per row and column. Row and column sparse access and $\text{SQ}(b)$ suffice for coordinate access to a fixed $\tilde{x}$ satisfying*

$$\|\tilde{x} - A^{-1}b\| \leq \epsilon \|A^{-1}b\|, \quad (8)$$

and exact sampling from $p_{\tilde{x}}$, in expected cost

$$K^2(d+1)^{O(K \log(2/\epsilon))} \quad (9)$$

per sample. For positive definite A , this improves to $K^2(d+1)^{O(\sqrt{K} \log(2/\epsilon))}$.

Proof. Apply Lemma 3.1 to $B = A^*A$, whose spectrum is contained in $[K^{-2}, 1]$, and let q be its inverse polynomial with residual $r = 1 - \lambda q(\lambda)$ bounded by ϵ . Put $P = q(A^*A)A^*$ and $\tilde{x} = Pb$. If $x = A^{-1}b$, then

$$Pb - x = -r(A^*A)x,$$

which proves (8). The singular value decomposition of A shows that the singular values of P are $|1 - r(\sigma^2)|/\sigma$; thus they lie in $[1 - \epsilon, K(1 + \epsilon)]$ and $\kappa(P) \leq 3K$. Lemma 3.2 materializes each needed row or column in cost $W = (d+1)^{O(K \log(2/\epsilon))}$, and also bounds its nonzero count by W . Lemma 3.6 therefore costs $O(K^2W^2)$ per sample, which has the stated form. For positive definite A , use $P = q(A)$ directly with degree $O(\sqrt{K} \log(2/\epsilon))$. When $K = 1$, the general matrix is unitary and $P = A^*$ is exact; the same rejection proof applies. $\square$

For fixed sparsity and accuracy, this upper bound implies a necessary condition for an N^c classical lower bound under the same access and output contract: $K = \Omega(\log N)$ in general and $K = \Omega(\log^2 N)$ in the positive definite case. These are necessary conditions, not hardness claims for every matrix at those scales. Andoni–Krauthgamer–Pogrow explicitly pose the question of constructing an ℓ_2 sampler for a solution from an ℓ_2 sampler for its right-hand side [4, Section 1.4]; their main results instead give coordinate estimates under different hypotheses and error guarantees. Our proof states the exact sampling contract and rejection cost explicitly.

3.4 Nonsymmetric inverse forms and optimization values

Theorem 3.8 (Norm-sensitive inverse overlap). *Let M satisfy the general hypotheses of Theorem 3.7. Given $\text{SQ}(a)$ and coordinate access to e , the scalar $a^*M^{-1}e$ can be estimated to additive error $\epsilon \|a\| \|e\|$, with failure probability ζ , at cost*

$$O\left(R_e^2 \epsilon^{-2} \log(1/\zeta) (d+1)^{O(K \log(2R_e/\epsilon))}\right), \quad (10)$$

where $1 \leq R_e \leq K$ is any supplied upper bound on $\|M^{-1}e\| / \|e\|$. In the absence of a sharper bound take $R_e = K$. Here the permitted accuracy range is $0 < \epsilon \leq R_e/2$, extending the default absolute accuracy range when $R_e > 1$. The modulus $|a^*M^{-1}e|$ admits the same additive accuracy.

Proof. Take the normal-equation residual polynomial from the preceding proof with $\eta = \epsilon/(4R_e) \leq 1/8$, and put $P = q(M^*M)M^*$. Its error satisfies

$$\|Pe - M^{-1}e\| \leq \eta R_e \|e\|, \quad \|Pe\| \leq (1 + \eta) R_e \|e\|.$$

The estimator $X = \|a\|^2 (Pe)_I / a_I$, with $I \sim p_a$, has mean a^*Pe and second moment at most $(1 + \eta)^2 R_e^2 \|a\|^2 \|e\|^2$. The polynomial bias is at most $\epsilon \|a\| \|e\| / 4$. Separate real and imaginary median-of-means estimates give the claimed statistical error using $O(R_e^2 \epsilon^{-2} \log(1/\zeta))$ samples. Local evaluation costs $(d+1)^{O(K \log(2R_e/\epsilon))}$. The inequality $\|z - \hat{z}\| \leq |z - \hat{z}|$ proves the modulus claim. $\square$

The energy-normalized bilinear guarantee in Corollary 3.5 and the Euclidean-normalized guarantee here answer different questions. Neither is a uniform relative estimate of a potentially cancelling overlap. The parameter R_e is particularly useful when an optimization value equals $|a^*M^{-1}e|$ and the construction supplies a public solution norm.

3.5 What finite precision changes

Robust dequantization under approximate length-squared sampling is studied in Le Gall [29]. The ideal-arithmetic results here do not assert free approximate sampling or bounded input precision. The following transfer statement isolates a sufficient arithmetic condition for the main scalar theorem.

Proposition 3.9 (Finite-error transfer). *Run the polynomial estimator of Theorem 3.4 with $L = O(\kappa \epsilon^{-2} \log(1/\zeta))$ samples. Suppose each requested sample law, conditional on all previous computation, is within total variation δ of p_b . Extend the estimator by the value zero on indices with $b_i = 0$, and have the approximate implementation return zero on those indices. On the support of b , suppose every reported real estimator value $\hat{X}_i$ differs from its exact value $\text{Re } X_i$ by at most $\tau \|b\|^2$. There is a coupling with an exact-oracle execution under which the final medians differ by at most $\tau \|b\|^2$, except on an event of probability at most $L\delta$. In particular, $\tau \leq \epsilon/4$ and $\delta \leq \zeta/L$ retain relative error ϵ and failure at most 2ζ for the constants used above.*

Proof. Successive maximal couplings make every sample agree with the exact-oracle sample except with probability at most $L\delta$, by a conditional union bound. An off-support approximate sample necessarily belongs to the disagreement event. On agreement, every group mean changes by at most $\tau \|b\|^2$; the median is Lipschitz with constant one in the maximum coordinate norm. Since $q \geq \|b\|^2$, this is at most τq . The proof of Theorem 3.4 leaves error at most $17\epsilon q/32$, so the additional $\epsilon q/4$ fits within the claimed tolerance. $\square$

With an exact norm and sampled coordinate, a sufficient condition on a computed polynomial coordinate at an index with $b_i \neq 0$ is

$$|(\widehat{P(H)b})_i - (P(H)b)_i| \leq \tau |b_i|.$$

One may also certify the estimator directly by interval arithmetic, including its division and norm multiplication. For finite rational input data and a rational spectral bound, the residual polynomial has rational coefficients; a finite local-walk sum and its sampled nonzero denominator can therefore be evaluated to any such tolerance, or exactly using rational arithmetic. This observation proves implementability, not a dimension-independent bit bound. Small sampled coordinates, input bit lengths, coefficient growth, and the cost of implementing approximate sampling must all be charged. In particular, bounded polynomial values on the spectral interval do not, by themselves, justify a uniform guard-precision claim for divisions by arbitrarily small b_i . The transfer proposition is the finite-precision claim used here; the oracle bounds remain explicitly ideal-arithmetic bounds.

4 Classical lower bounds with full sampling access

The lower bounds in this section charge queries to hidden Boolean inputs. All other input data are public. This convention is particularly strong: we grant every operation in Definition 2.1, and verify its simulation, rather than infer sampling hardness from entry-query hardness. The dimensions of the clock families are selected by the parameters; no bound exceeding the cost of reading a fixed finite input is asserted.

4.1 The exponential dependence

Write $a = 1/\kappa$, $m = (1 + a)/2$, $h = (1 - a)/2$, and

$$f_\kappa(x) = \frac{a}{m + hx}, \quad \rho = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}. \quad (11)$$

For a continuous real function f on $[-1, 1]$, write $\deg_\tau(f)$ for its least uniform approximation degree at error τ . The classical minimax formula for the reciprocal gives

$$E_n(f_\kappa) = \frac{\kappa - 1}{2\kappa} \rho^n, \quad \deg_\tau(f_\kappa) = \Theta(\sqrt{\kappa} \log(1/\tau)) \quad (\kappa \geq 4, 0 < \tau \leq \tau_0). \quad (12)$$

Here E_n is the best degree- n error, including $n = 0$; see Kraus et al. [28, Corollary 2.2]. Indeed their reciprocal error, after multiplying by a , is $(\sqrt{\kappa} - 1)^2 \rho^{n-1} / (2\kappa)$ for $n \geq 1$, which equals the displayed expression. For $n = 0$ half the range of f_κ gives the same answer. The degree assertion follows because $\log(1/\rho) = \Theta(\kappa^{-1/2})$ and the prefactor lies in $[3/8, 1/2]$. This is an approximation-theoretic input, not a new degree lower bound.

Theorem 4.1 (Relative inverse-form lower bound). *There is an absolute $\epsilon_0 > 0$ such that, for every $\kappa \geq 4$, integer $s \geq 5$, and $0 < \epsilon \leq \epsilon_0$, a family of real SPD matrices satisfying $aI \preceq H \preceq I$, $\kappa(H) = \kappa$, and row sparsity at most s has the following property. Returning relative ϵ estimates of both $q(H, b_+)$ and $q(H, b_-)$ for two public unit vectors of support two, with success probability $2/3$, requires*

$$\Omega \left(\frac{((s-1)/2)^{\Omega(\sqrt{\kappa} \log(1/\epsilon))}}{\log(s) \sqrt{\kappa} \log(1/\epsilon)} \right) \quad (13)$$

classical queries, even with full matrix sampling and query access. Consequently a routine that estimates one form for every supplied b has the same worst-case lower bound, up to an absolute factor. The family has a sparse box-LP realization with public analytic center and a full-SQ constraint oracle using $O(1)$ hidden queries per request.

Proof. We use the matrix-function lower bound of Montanaro and Shao [32, Theorem 1.8, Corollary 5.1 and Section 5.1] (the theorem numbering refers to arXiv:2311.06999). For continuous $f : [-1, 1] \rightarrow [-1, 1]$, their construction gives a d -sparse real symmetric contraction A_y and distinct public indices i, j for which additive error $\tau/4$ in $(f(A_y))_{ij}$ costs

$$\Omega \left(\frac{(d/2)^{(\deg_{2\tau}(f)-1)/6}}{\log(d) \deg_{\tau}(f)} \right). \quad (14)$$

For arbitrary integer sparsity budgets, take the largest admissible power-of-two gate size. Its loss is an absolute constant in the exponential rate for $s \geq 5$; we use this interpretation of the Ω in (13).

For clarity, we describe the input structure needed to strengthen their oracle conclusion. Their matrix is a weighted Feynman path

$$A_y = \sum_{t=1}^T \beta_t (|t\rangle\langle t-1| \otimes U_t + |t-1\rangle\langle t| \otimes U_t^T). \quad (15)$$

The positive weights β_t are public. Each real orthogonal U_t is a public block Walsh–Hadamard gate, a public identity, or a diagonal hidden-sign gate. The latter implements one query to either of two Boolean functions on n bits. The source problem distinguishes

$$\Phi(y) = \langle 0 | W D_1 W D_2 W | 0 \rangle \geq 3/5 \quad \text{from} \quad |\Phi(y)| \leq 1/100, \quad W = H_2^{\otimes n}, \quad (16)$$

where H_2 is the normalized two-dimensional Hadamard matrix. Its randomized query lower bound is $\Omega(2^{n/2}/n)$ [1, Theorem 1]. Factoring each W into local Hadamard layers produces the sparsity dependence in (14). The diagonal orthogonal gauge with blocks $V_0 = I$ and $V_t = U_t \cdots U_1$ conjugates A_y to $J \otimes I$, where J is the public scalar Jacobi matrix with weights β_t . Thus its spectrum is public, and the relevant entry has the form $c\Phi(y)$ with a public $c \geq \tau$ (a public endpoint sign can be absorbed). For the even-function branch the queried clock positions are the second and penultimate vertices; identity gates at both ends give the same formula. This is why the diagonal entries at both queried positions are also independent of y .

Set $H_y = mI + hA_y$. Then $aI \preceq H_y \preceq I$ and $f_{\kappa}(A_y) = aH_y^{-1}$. Adjoin the public scalar blocks $a, 1$ to make the condition number exactly κ , with right-hand sides zero on these blocks. The shift adds one nonzero per row. For $b_{\pm} = (e_i \pm e_j)/\sqrt{2}$, polarization gives

$$(f_{\kappa}(A_y))_{ij} = \frac{a}{2}(q(H_y, b_+) - q(H_y, b_-)).$$

Each form lies in $[1, \kappa]$, so relative ϵ errors propagate to additive error at most ϵ . Set $\tau = 4\epsilon$ in (14), use (12), and amplify the two scalar calls by a constant number of repetitions. Choosing ϵ_0 sufficiently small makes the degree numerator positive uniformly in κ . This proves the query bound.

It remains to justify the stronger oracle. In every clock row, the supports of the two adjacent transitions and the diagonal are disjoint. A unitary transition has squared row and column mass one; a Hadamard row is uniform on a public support and a sign-gate row has one public location. Hence the squared row norm of H_y is $m^2 + h^2(\beta_t^2 + \beta_{t+1}^2)$, omitting nonexistent endpoint edges. Conditional samples first choose one of these three public masses, then a uniform Hadamard location

or the single sign location. Nonzero counts, locations and their ordering are public. Requested values need at most one sign query. Row-norm sampling first chooses a public clock position and then a uniform work index; the Frobenius norm is a public sum. The same argument applies to columns and the scalar pads. The public weight table can be preprocessed freely in this hidden-input query model. Both right-hand sides have public exact SQ. Therefore no granted norm or sampling request reveals any extra hidden information. The LP realization follows from Proposition 4.2 below. $\square$

Proposition 4.2 (Sparse analytic-center realization). *Every clock $H = mI + hA_y$ above has a factor $C_y^T C_y = H$ with $O(s)$ nonzeros per row and column and public full-SQ metadata. The box LP $\max_x \eta b^T x$ subject to $-\mathbf{1} \leq C_y x \leq \mathbf{1}$ has at $x = 0$ the barrier Hessian $2H$ and squared Newton decrement $\eta^2 q(H, b)/2$.*

Proof. Put $d_0 = \sqrt{m}$, $e_t = h\beta_t/d_{t-1}$, and $d_t = \sqrt{m - e_t^2}$. These are positive: their squares are the pivots of the scalar positive definite tridiagonal matrix $mI + hJ$. The lower bidiagonal block matrix B with $B_{tt} = d_t I$ and $B_{t,t-1} = e_t U_t$ obeys $BB^T = H$; its diagonal blocks are $(d_t^2 + e_t^2)I = mI$ and its off-diagonal blocks are $e_t d_{t-1} U_t = h\beta_t U_t$. Take $C_y = B^T$, and factor the public scalar pads separately. Only a diagonal block and one transition meet any row or column. Their disjoint supports and public magnitudes prove both sparsity and the full-SQ simulation, exactly as above; no product $U_t \cdots U_1$ appears in this factor. The paired logarithmic barrier $\phi(x) = -\sum_j \log(1 - (C_y x)_j^2)$ has gradient zero and Hessian $2C_y^T C_y$ at zero. The factor has full column rank, so zero is the unique analytic center. For $\phi(x) - \eta b^T x$, the Newton gradient there is $-\eta b$, giving the asserted decrement. Choosing $\eta = 1/(4\sqrt{\kappa})$ makes its square at most $1/32$ for unit b . The barrier parameter grows with the number of interval constraints; this is a local diagnostic lower bound, not an iteration lower bound. $\square$

4.2 Finite rational instances and public preprocessing

The existential public clock does not require infinite information about the hidden input. The following statement deliberately distinguishes finite description from efficient uniform preparation.

Proposition 4.3 (Rational realization). *For rational κ, ϵ and $s \geq 9$, the clock weights may be chosen dyadic with $O(\log(\kappa/\epsilon))$ fractional bits each, while retaining (13) with the harmless base replacement $(s-1)/2 \mapsto \max\{4, (s-1)/8\}$. The matrix and two unit right-hand sides can be rational. An exactly rational $O(s)$ -sparse constraint factor also exists. The public tables admit a terminating uniform generator; this statement makes no polynomial-time preprocessing or factor-bit-length claim.*

Proof. Let the ideal public coefficient be $c \geq \tau$, where $\tau = \Theta(\epsilon)$ will have a sufficiently large absolute constant. Set $\theta = \tau/(128\kappa)$ and choose L with $2^{-L} \leq \tau/(256\kappa)$. Round each edge by

$$\hat{\beta}_t = 2^{-L} \lfloor 2^L (1 - \theta) \beta_t \rfloor.$$

The Jacobi matrix is nonnegative and bipartite, so its norm is its Perron root. Entrywise monotonicity gives $\|\hat{J}\| \leq 1 - \theta$, and the tridiagonal row-sum bound gives $\|J - \hat{J}\| \leq 2(\theta + 2^{-L})$. For symmetric contractions X, Y , the resolvent identity yields

$$\|f_\kappa(X) - f_\kappa(Y)\| \leq ah \|(mI + hX)^{-1}\| \|(mI + hY)^{-1}\| \|X - Y\| \leq \frac{\kappa}{2} \|X - Y\|. \quad (17)$$

Thus the coefficient changes by at most $3\tau/256$, and $|\hat{c}| \geq 253\tau/256$. An additive $\tau/4$ estimator of $\hat{c}\Phi$ estimates Φ with error at most $64/253$, strictly below half the promise gap 0.59 . In particular,

rounding cannot disconnect the two queried clock positions. The gauge identity makes this perturbation bound uniform over every hidden input.

Choose a power-of-four Hadamard block size, so every normalized gate entry is rational; the largest allowed size is at least both 4 and $(s-1)/8$ for $s \geq 9$. Use the rational unit vectors $(3e_i \pm 4e_j)/5$. Their difference gives the entry through

$$a(H^{-1})_{ij} = \frac{25a}{48}(q_+ - q_-),$$

with propagated error at most $25\epsilon/24$. Taking, for example, $\tau = 5\epsilon$ suffices. The scalar pads are rational. Squared norms and sampling probabilities are rational and public; their square roots are public algebraic numbers, without hidden information.

For completeness an exact rational factor avoids taking irrational Cholesky square roots. Write the rounded off-diagonal coefficient as $w_t = h\hat{\beta}_t$. The block LDL^T factorization uses rational pivots

$$D_0 = m, \quad \ell_t = w_t/D_{t-1}, \quad D_t = m - w_t^2/D_{t-1} > 0, \quad L_{t,t-1} = \ell_t U_t,$$

with identity diagonal blocks. For $D_t = p_t/q_t$ in lowest positive terms, the four-square theorem gives integers $z_{t,r}$ such that $p_t q_t = \sum_{r=1}^4 z_{t,r}^2$. Replace the block row $\sqrt{D_t}(L^T)_{t,*}$ by the four rational block rows $(z_{t,r}/q_t)(L^T)_{t,*}$. Their Gram matrix is exactly LDL^T . Each row has one identity block and one gate block; column sparsity increases by at most four. Zero rows can be deleted publicly. All metadata remain public. The scalar pads are treated in the same way.

The finite table has $O(T \log(\kappa/\epsilon))$ fractional bits, where $T = \Theta(\sqrt{\kappa} \log(1/\epsilon))$. A terminating generator can enumerate dyadic tables of this clock length and test, using exact rational arithmetic, the strict spectral margin and the normalized coefficient condition

$$a \left| [(mI + h\hat{J})^{-1}]_{i,j} \right| > 9\tau/10.$$

Here i, j are the queried clock vertices: the endpoints in the endpoint branch, and the second and penultimate vertices in the even branch. Equivalently, after absorbing the known public endpoint sign, the test is $a[(mI + h\hat{J})^{-1}]_{i,j} > 9\tau/10$. Positivity of rational matrices is decidable by exact principal-minor or elimination tests; the tested quantity is the corresponding normalized entry of $f_\kappa(\hat{J})$. The rounded witness proves the search nonempty. A four-square search subsequently produces the constraint factor. This proves computability, but supplies neither a polynomial bound for this search nor a stable efficient inverse-spectral construction. Public preparation is therefore unpriced only because the theorem counts hidden-input queries. $\square$

The structured clock also illustrates a coherent-output qualification. Its forms have the public representation $q_\pm = D \pm C\Phi$, with $|C| \leq D \leq \kappa$ by the scalar clock's positive-semidefinite Cauchy-Schwarz inequality. Put $q_* := \min_{\Phi \text{ in the source promise}} \pm(D \pm C\Phi)$; positivity gives $q_* \geq 1$ for the realizable inputs. Coherent amplitude estimation of the Forrelation circuit to additive error $O(\epsilon q_*/|C|)$ costs $O(1 + |C|/(\epsilon q_*))$ hidden-sign queries, and suffices for relative estimates of the two forms. If $|C| \leq \epsilon q_*$, outputting the public baseline D already suffices with no hidden query. On clocks calibrated to $c = \Theta(\epsilon)$ we have $|C| = \Theta(\kappa\epsilon)$, so the nontrivial cost is $O(\kappa/q_*)$. It is $O(1/\epsilon)$ when $\kappa\epsilon = O(1)$, or under an additional noncancellation bound $q_* = \Omega(|C|)$. A constant-query quantum algorithm for distinguishing the Forrelation promise alone does not imply the same numerical-estimation cost. No uniform noncancellation estimate for all minimax clocks is assumed here.

4.3 The statistical factor, confidence, and finite population

Lemma 4.4 (Two Hamming spheres). *For integers $1 \leq t \leq M/4$ and $1 \leq \Delta \leq t/4$, distinguishing weights t and $t + \Delta$ with success probability $2/3$ requires $\Omega(\min\{M, tM/\Delta^2\})$ randomized bit queries.*

Proof. Use the uniform measures on the two spheres. Conditional on any transcript with j distinct queries and z observed ones, the next fresh bit has probabilities $(t - z)/(M - j)$ and $(t + \Delta - z)/(M - j)$. Adaptively choosing its address does not change these probabilities. Stop a T -query transcript if $z \geq t/2$, where $T \leq cM$ for a small absolute c . The chance of this stopping event is at most $2T(t + \Delta)/(Mt) \leq (5/2)c$ under either measure, by the expected number of ones in sampling without replacement. Before stopping, both probabilities are at least $t/(2M)$, and both are bounded away from one. Their difference is $\Delta/(M - j)$. The elementary bound $D_{\text{KL}}(\text{Ber}(p) \parallel \text{Ber}(q)) \leq (p - q)^2/(q(1 - q))$ therefore bounds the stopped transcript's divergence by $CT\Delta^2/(tM)$, by the chain rule. Pinsker's inequality bounds its total variation by the square root of half this quantity. Restoring the stopped suffix changes total variation by at most the sum of the two stopping probabilities. For T a sufficiently small constant times $\min\{M, tM/\Delta^2\}$ the resulting total variation is below $1/3$. No such transcript distinguishes the hypotheses with average success $2/3$. Internal randomness can be included in the transcript; equivalently apply the distributional minimax principle. $\square$

Theorem 4.5 (Statistical lower bound). *For $\kappa \geq 4$, $0 < \epsilon \leq 1/128$, and every integer $M \geq 1$, there are two-sparse real SPD matrices of dimension $2M$ and one public unit vector for which relative inverse-form estimation costs*

$$\Omega(\min\{M, \kappa/\epsilon^2\}) \quad (18)$$

classical full-SQ queries. If $0 < \zeta \leq 1/10$ and $M \geq C\kappa\epsilon^{-2}\log(1/\zeta)$, failure at most ζ requires $\Omega(\kappa\epsilon^{-2}\log(1/\zeta))$ queries. Both conclusions hold for a two-sparse box-LP constraint oracle at its public analytic center.

Proof. For $z \in \{0, 1\}$ set

$$B_z = \begin{pmatrix} m & (-1)^z h \\ (-1)^z h & m \end{pmatrix}, \quad H_x = \bigoplus_{j=1}^M B_{x_j}, \quad b = M^{-1/2} \bigoplus_{j=1}^M (1, 1)^T / \sqrt{2}.$$

Each block has eigenvalues $a, 1$. Its supports, entry magnitudes, row and column norms and Frobenius norm are independent of the sign. Thus every matrix or vector SQ operation is public, except a queried entry sign, which uses one input bit. The plus vector has eigenvalue one for $z = 0$ and a for $z = 1$, giving the exact identity

$$q(H_x, b) = 1 + (\kappa - 1)|x|/M. \quad (19)$$

For $M \geq \kappa/(8\epsilon)$ put $u = M/\kappa$, $t = \lfloor u \rfloor$ and $\Delta = \lceil 8\epsilon u \rceil$. Then $u \geq 1/(8\epsilon)$, $t \leq M/4$, and $\Delta \leq 8\epsilon u + 1 \leq (u - 1)/4 \leq t/4$. The last inequality follows by taking the worst endpoint $\epsilon = 1/128$, $u = 16$, or directly by rearranging. The value at weight t is below two, and the increment is at least $8\epsilon(\kappa - 1)/\kappa \geq 6\epsilon$. Therefore the two relative-error intervals are disjoint. Moreover $t = \Theta(u)$ and $\Delta = \Theta(\epsilon u)$, so Lemma 4.4 proves (18). For smaller M , weights zero and one give an increment exceeding 6ϵ and again disjoint intervals; locating a uniformly hidden one costs $\Omega(M)$ queries, as required.

For confidence, use independent Bernoulli bits with parameters $p_0 = 1/\kappa$ and $p_1 = (1 + 16\epsilon)/\kappa$. Under the population assumption in the theorem, Chernoff bounds give $||x| - Mp_j| \leq 2\epsilon M/\kappa$ with

probability at least $1 - \zeta/10$ for either j , by increasing C . On these events the scalar bands are separated by at least $12\epsilon(\kappa - 1)/\kappa \geq 9\epsilon$ and are below three; their relative- ϵ bands are disjoint. Thresholding the estimator thus tests the two product distributions with each error at most 1.1ζ . Each fresh addressed bit has the same Bernoulli law regardless of prior queries, and its KL divergence is at most $C'\epsilon^2/\kappa$. The transcript divergence for T queries is at most $C'T\epsilon^2/\kappa$. Data processing to the binary test bounds it below by $\Omega(\log(1/\zeta))$ uniformly for $\zeta \leq 1/10$. This proves the high-confidence assertion, including randomized algorithms by placing their random seed in the transcript. Public SQ requests contribute no divergence.

Finally take $u_0 = (1 + \sqrt{a})/2$, $v_0 = (1 - \sqrt{a})/2$ and $C_z = \begin{pmatrix} u_0 & (-1)^z v_0 \\ (-1)^z v_0 & u_0 \end{pmatrix}$. Then $C_z^2 = B_z$. Its full-SQ metadata are sign-independent and it is two-sparse. The box LP with factor $\bigoplus_j C_{x_j}$ has Hessian $2H_x$ and decrement $\eta^2 q(H_x, b)/2$ at zero, as before. $\square$

Corollary 4.6 (Coherent counting comparison). *For $\kappa \geq 4$, $0 < \epsilon \leq 1/128$, and $M \geq C\kappa/\epsilon$, the canonical coherent versions of the sign-block oracles require $\Omega(\sqrt{\kappa}/\epsilon)$ quantum queries, and $O(\sqrt{\kappa}/\epsilon)$ suffice on this family.*

Proof. The sparse-value oracle is reversibly simulable with a constant number of bit queries; all other unitaries are fixed public circuits. This specifies their completion and excludes unspecified side information. The weight-decision consequence of Nayak and Wu [35, Theorem 1.1] is $\Omega(\sqrt{M/\Delta} + \sqrt{t(M-t)}/\Delta)$ for the two weights used above. In the stated population range, its second term is $\Omega(\sqrt{\kappa}/\epsilon)$. For the upper bound prepare a uniform block index, query its bit and rotate an ancilla with success probability $1/\kappa$ or one according to that bit. The average success probability is $p = q(H_x, b)/\kappa \geq 1/\kappa$. Amplitude estimation [12, Theorem 12] estimates p relatively with $O(1/(\epsilon\sqrt{p}))$ calls and constant success probability. This proves the bound. Its public common eigenbasis is essential to this simple upper bound; the corollary does not give that complexity for general sparse inverse forms. $\square$

The exponential and statistical theorems certify different aspects of the upper bound. Their separate existence does not multiply their costs. The next section proves products that hold on a single scalar instance, and states exactly which stronger multiplication they do not establish.

5 When statistical and local-query costs multiply

Direct sums preserve spectra and sparsity, but do not automatically multiply query lower bounds. The missing quantity is the relative variation of a *single positive form*, rather than an off-diagonal entry obtained by subtraction. We prove a distributional product that avoids pointwise two-level promises, and then quantify two construction classes that cannot supply the stronger condition-dependent product.

5.1 Averaging and contrast

Suppose G_y satisfies $aI \preceq G_y \preceq I$ with a public unit vector v . For M inputs define

$$H_y = \bigoplus_{j=1}^M G_{y^{(j)}}, \quad b = M^{-1/2} \bigoplus_{j=1}^M v, \quad q(H_y, b) = M^{-1} \sum_j q(G_{y^{(j)}}, v). \quad (20)$$

If all inner instances have identical public norm tables and magnitude patterns, global SQ first selects a uniform block and then uses its inner SQ interface. Norms, supports and sparsity are

unchanged within a block. The same statement applies to sparse constraint factors: their direct sum realizes (20) as one analytic-center squared decrement, multiplied by the public factor $\eta^2/2$.

For exact levels $q_0, q_1 = q_0(1 + D)$ labelled by a partial Boolean function g , an input with label-one fraction p has scalar $q_0(1 + Dp)$. In the high-contrast range $D \geq 1$, choosing $p = \Theta(1/D)$ keeps the scalar of order q_0 , while relative error ϵ resolves a change $\Theta(\epsilon/D)$ in p . Lemma 4.4 then gives an outer cost $\Omega(\min\{M, D/\epsilon^2\})$; ordinary sampling supplies the corresponding upper bound up to confidence factors. Thus a κ/ϵ^2 statistical factor from this averaging mechanism requires contrast $1 + D = \Theta(\kappa)$, rather than merely a large absolute difference between two forms of order κ .

We call a realization *oracle-preserving* if every granted sparse count, location and value request, vector coordinate request, norm request, and sampling request can be simulated with $O(1)$ queries to the hidden input, with all remaining data public. This requirement also covers every granted constraint-factor request. The constant is uniform over the problem parameters and the requested operations.

Proposition 5.1 (Conditional high-contrast compiler). *Let g be a nonconstant partial Boolean problem with bounded-error randomized query complexity $R(g) = L$. Suppose an oracle-preserving sparse SPD family with identical public full-SQ metadata has the exact levels just described. For $D \geq 1$ and sufficiently small ϵ , one relative inverse-form estimate on a direct sum of $M = \Theta(D/\epsilon^2)$ blocks requires $\Omega(DL/\epsilon^2)$ queries.*

Proof. Choose a small fixed $\gamma > 0$, a weight $t = \lfloor \gamma M/D \rfloor$, and $\Delta = \lceil C_1 \epsilon M/D \rceil$, with C_1 an absolute constant large enough to separate relative output intervals. Choose $M = C_2 D/\epsilon^2$ rounded to an integer, and then choose ϵ sufficiently small and C_2 sufficiently large. These choices ensure $1 \leq \Delta \leq t/4$, $t \leq M/4$, $q_t = q_0(1 + Dt/M) = \Theta(q_0)$, and $q_{t+\Delta} - q_t \geq C_1 \epsilon q_0$. The outer partial function F distinguishes weights $t, t + \Delta$. Lemma 4.4 gives $R(F) = \Theta(M)$. Theorem 2 of Chakraborty et al. [16] explicitly allows a partial outer function and proves $R(F \circ g^M) = \Omega(R(F)R(g))$ whenever $R(F) = \Theta(M)$. Equation (20) and the disjoint relative output intervals reduce this composed decision problem to the scalar estimate with constant query overhead. $\square$

If levels have relative half-width η , the weighted average can move by $\eta q_0(1 + Dp)$ even with all labels fixed. Thus the pointwise reduction remains valid for $\eta \leq c\epsilon$ with a sufficiently small absolute c . With one absolute half-width δq_1 for both levels, it requires $\delta \leq c\epsilon/(1 + D)$ instead. A constant-gap Forrelation promise supplies neither of these narrow-level conditions. An unrestricted assertion $R(F \circ g) = \Omega(R(F)R(g))$ is also false for partial functions [8]; the linear outer complexity in Proposition 5.1 is material.

5.2 A distributional product for a single positive form

We recall precisely the distributional consequence of the composition results that will be used. A noisy bit query of bias γ returns the correct bit with probability $(1 + \gamma)/2$ and costs γ^2 . Write $\text{noisyR}(F)$ for the minimum worst-case expected cost of a bounded-error algorithm in this model. Theorems 24 and 35, Definitions 33–34 and the minimax observation following Definition 34 of Ben-David and Blais [8] imply the following: for every nonconstant partial Boolean inner function g , there are distributions μ_0, μ_1 on its two fibers such that, for every outer F , there is a distribution ν on $\text{Dom}(F)$ for which computing F under

$$z \sim \nu, \quad Y_j \sim \mu_{z_j} \text{ independently conditional on } z \quad (21)$$

requires $\Omega(\text{noisyR}(F)R(g))$ expected queries. Here the inner distributions can be chosen to witness the Shaltiel-free complexity of g , which Theorem 24 lower-bounds by $\Omega(R(g))$. This order of

quantifiers is important: the same μ_0, μ_1 work before ν is selected. The proof of their Theorem 35 establishes this statement for that fixed pair; it is stronger than simply maximizing the final composed complexity over unstructured distributions. Worst-case query-bounded estimators are covered by this expected-cost lower bound. We use the theorem numbering of arXiv:2002.10809v2.

Theorem 5.2 (Distributional scalar product). *Let g be a nonconstant partial Boolean problem, and let its oracle-preserving SPD realization have a public unit v and*

$$q(G_y, v) = d + C\Phi(y), \quad C > 0, \quad d > 0, \quad \chi = C/d \leq 5/3, \quad (22)$$

where $|\Phi| \leq 1/100$ on zero inputs and $3/5 \leq \Phi \leq 1$ on one inputs. Assume the identical public full-SQ metadata needed for (20). If $0 < \epsilon \leq \chi/200$, there is a family of $M = 4\lceil \chi/\epsilon \rceil^2$ blocks for which relative ϵ estimation of the single averaged form requires

$$\Omega((\chi/\epsilon)^2 R(g)) \quad (23)$$

classical full-SQ queries. A factor realization transfers the same statement to one box-LP squared decrement.

Proof. Set $n = \lceil \chi/\epsilon \rceil \geq 200$, $M = 4n^2$, and let F distinguish the weights $M/2 \pm 24\sqrt{M} = 2n^2 \pm 48n$. They are integers in $[0, M]$. The Hamming-sphere argument proving Lemma 4.4 applies equally for $t = \Theta(M)$ bounded away from both endpoints: before a small fraction of M queries, the two Bernoulli probabilities are bounded away from zero and one, so the per-query divergence is $O(1/M)$. Thus $R(F) = \Theta(M)$. Observation 23 of Chakraborty et al. [16] gives $\text{noisyR}(F) = \Omega(R(F)^2/M) = \Omega(M)$. The reverse bound follows by reading all bits.

Choose the distributions in (21). Let $\Delta = \mathbb{E}_{\mu_1} \Phi - \mathbb{E}_{\mu_0} \Phi \geq 59/100$. For each fixed outer input of one of the two weights, the conditional mean of $M^{-1} \sum_j \Phi(Y_j)$ depends only on that weight. Their means differ by $48\Delta/\sqrt{M} = 24\Delta/n$. Hoeffding's inequality, since every Φ lies in $[-1, 1]$, gives

$$\mathbb{P} \left[\left| M^{-1} \sum_j \Phi(Y_j) - \mathbb{E}[M^{-1} \sum_j \Phi(Y_j) \mid z] \right| > \frac{24\Delta}{4\sqrt{M}} \mid z \right] \leq 2e^{-24^2 \Delta^2 / 32} < 0.004.$$

Hence the two mean scalar values are separated by $S = 24\Delta C/n$, and each realized scalar is within $S/8$ of its mean except on the displayed event. Because $n \leq (201/200)\chi/\epsilon$, we have $S > 14\epsilon d$. Every scalar is at most $(1 + \chi)d < 3d$. Amplify the estimator's success to 11/12 by a constant number of independent repetitions and a median. Its absolute error is then at most $3\epsilon d$, and $S/8 + 3\epsilon d < S/2$. Thresholding at the midpoint of the two public means computes F with distributional error below 1/3. This threshold may depend on the fixed hard distributions; such nonuniform dependence is allowed in the distributional lower bound. The composition theorem therefore requires $\Omega(MR(g))$ queries. All scalar-input queries have constant-cost hidden-input simulations, proving the result. $\square$

For the clock of Theorem 4.1, gauge equivalence makes the average of its two queried inverse diagonals a public number d . Its positive polarization has exactly $q_+(y) = d + \kappa c \Phi(y)$, where c is the public normalized endpoint coefficient. Thus $C = \kappa c$ and $\chi_c = \kappa c/d$. We have $d \leq \kappa$, so $\chi_c \geq c$. Positive definiteness bounds the absolute cross entry by the average of its two diagonals; applying it to a high input with $\Phi \geq 3/5$ gives $\chi_c \leq 5/3$. The Montanaro–Shao clock at a sufficiently small coefficient scale c has source hardness

$$L_c = \Omega \left(\frac{((s-1)/2)^{\Omega(\sqrt{\kappa} \log(1/c))}}{\log(s) \sqrt{\kappa} \log(1/c)} \right). \quad (24)$$

One can use actual attainable coefficients within an absolute factor of the desired scale, which changes no bounds. The explicit construction below establishes this fact without assuming a continuum of clock sizes. Theorem 5.2 gives $\Omega((\chi_c/\epsilon)^2 L_c)$ for a single form whenever $\chi_c \geq 200\epsilon$. In particular a fixed small coefficient gives $\epsilon^{-2} s^{\Omega(\sqrt{\kappa})}$ up to the displayed denominator, on one instance. For coefficient scales $c = c_0 \epsilon^\alpha$, $0 \leq \alpha < 1$, the conservative $\chi_c \geq c$ gives

$$\epsilon^{-2(1-\alpha)} s^{\Omega(\sqrt{\kappa}[1+\alpha \log(1/\epsilon)])},$$

again with the denominator of (24) and sufficiently small ϵ . This is a simultaneous-cost statement; it does not establish a product of both full factors in the upper bound.

5.3 A witness clock with an exact diagonal

The following calculation determines the signal-to-baseline ratio for an explicit clock. It uses a restricted interval to produce a valid full-domain dual witness, rather than asserting that the restricted approximant is minimax on all of $[-1, 1]$.

Proposition 5.3 (Explicit witness diagonal). *For every $\kappa \geq 4$ and integer $r \geq 1$, there is an odd-order zero-diagonal Jacobi contraction J of order $2r + 5$ with positive edges, whose second and penultimate coordinates satisfy*

$$(f_\kappa(J))_{2,2r+4} = c_r, \quad (f_\kappa(J))_{2,2} = (f_\kappa(J))_{2r+4,2r+4} = d', \quad (25)$$

where

$$c_r = \frac{C\delta^r}{A^2 - 1}, \quad d' = \frac{CA}{A^2 - 1} = \frac{(\kappa + 1)(\kappa^2 + 14\kappa + 1)}{4\kappa(\kappa^2 + 6\kappa + 1)},$$

with $A = 4(m/h)^2 - 3$, $C = 4am/h^2$, and $\delta = A - \sqrt{A^2 - 1}$. In particular $1/4 < d' < 3/5$, $\chi_{c_r} = c_r/d' = \Theta(c_r)$, and $r = \Theta(\sqrt{\kappa} \log(1/c_r))$ at sufficiently small scales.

Proof. A full derivation is given here because the diagonal, unlike the approximation degree alone, controls statistical composition. The even part of f_κ is $am/(m^2 - h^2 x^2)$. On $x^2 \in [1/2, 1]$, set $t = 4x^2 - 3$; it becomes $q(t) = C/(A - t)$. Put $\omega = \sqrt{A^2 - 1}$. The degree- r reciprocal minimax error is $E_r = C\delta^r/\omega^2$. Write T_j and U_j for the first- and second-kind Chebyshev polynomials, respectively: $T_j(\cos \varphi) = \cos(j\varphi)$ and $U_j(\cos \varphi) = \sin((j+1)\varphi)/\sin \varphi$, with continuous extension at the endpoints. The scalar polynomial U_j is distinguished from the matrix-valued clock gate U_t by its argument and context. The normalized error is

$$V_r(t) = \frac{N_r(t)}{A - t}, \quad N_r(t) = (At - 1)T_r(t) + \omega(t^2 - 1)U_{r-1}(t).$$

To verify rather than assume this expression, write $t = \cos \varphi$. The half-angle identity $\tan(\psi/2) = \sqrt{(A+1)/(A-1)} \tan(\varphi/2)$ gives $V_r(t) = \cos(r\varphi + \psi)$. Also $N_r(A) = \omega^2(A + \omega)^r$, so $C - E_r N_r(t)$ vanishes at $t = A$; dividing by $A - t$ yields a degree- r polynomial. The phase increases strictly from zero to $(r+1)\pi$, producing $r+2$ alternating extremal points. The alternation theorem therefore verifies the minimax claim.

Let these points be $t_0 < \dots < t_{r+1}$, and define

$$R_r(t) = (At - 1)U_{r-1}(t) + \omega T_r(t), \quad L_r(t) = (t^2 - 1)R_r(t).$$

The interior extrema are exactly the r roots of R_r (use $\sin(r\varphi + \psi) = 0$). Thus the t_j are the simple roots of L_r , whose leading coefficient is positive, and $V_r(t_j) = \text{sgn } L'_r(t_j) =: \sigma_j$. Set $\mu_j = |L'_r(t_j)|^{-1} / \sum_k |L'_r(t_k)|^{-1}$. Partial fractions at infinity show

$$\sum_j \frac{P(t_j)}{L'_r(t_j)} = 0 \quad (\deg P \leq r). \quad (26)$$

Hence the signed weights $\sigma_j \mu_j$ annihilate every polynomial of degree at most r and pair with q to E_r . At $x_j = \sqrt{(t_j + 3)/4}$, put signed mass $\sigma_j \mu_j / 2$ on each of $\pm x_j$. Odd moments cancel, and even moments through degree $2r$ vanish by (26). This is a full-domain dual witness supported inside $[-1, 1]$.

We now realize its cross entry explicitly, in the form underlying Montanaro and Shao [32, Lemma 2.5]. Put $B = (2 \sum_j \mu_j / x_j^2)^{-1}$ and choose the positive symmetric probability measure with mass $1/2$ at zero and $B \mu_j / (2x_j^2)$ at each $\pm x_j$. Gram–Schmidt on $1, x, \dots, x^{2r+4}$ gives its Jacobi matrix J . The symmetry makes its diagonal zero, positivity of the distinct-node measure makes all edges positive, and its eigenvalues are the support points, so it is a contraction. Since $\int x^2 dw = B$, the second orthonormal polynomial is $x/\sqrt{B}$; its spectral masses at $\pm x_j$ are $\mu_j/2$ and at zero are zero. Interpolate the values σ_j by a degree- $(r+1)$ polynomial $S(t)$. Equation (26) implies that $xS(4x^2 - 3)/\sqrt{B}$ is orthogonal to every lower odd polynomial and, by symmetry, every even polynomial. Its norm is one, since $S(t_j)^2 = 1$. Its leading coefficient is positive: the divided difference of the ordered alternating values is a sum of positive terms. Thus it is exactly the penultimate orthonormal polynomial, of degree $2r+3$. Its spectral masses agree with those of the second coordinate, and their signed cross masses are $\sigma_j \mu_j / 2$. The cross entry is consequently $E_r = c_r$, and both diagonals are $d' = \sum_j \mu_j q(t_j)$.

To evaluate that average, partial fractions of N_r/L_r give

$$S_0 := \sum_j |L'_r(t_j)|^{-1} = \frac{N_r(A)}{L_r(A)}, \quad S_1 := \sum_j \frac{q(t_j)}{|L'_r(t_j)|} = -C \left(\frac{N_r}{L_r} \right)'(A).$$

Indeed $N_r(t_j)/(A - t_j) = \sigma_j$, and differentiating the partial-fraction expansion gives the second identity. If $\mathcal{S} = T_r(A) + \omega U_{r-1}(A) = (A + \omega)^r$, then $N_r(A) = \omega^2 \mathcal{S}$ and $L_r(A) = \omega^3 \mathcal{S}$. Using $T'_r = rU_{r-1}$ and $\omega^2 U'_{r-1}(A) = rT_r(A) - AU_{r-1}(A)$ gives

$$\frac{N'_r(A)}{N_r(A)} = \frac{r}{\omega} + \frac{A}{\omega^2}, \quad \frac{L'_r(A)}{L_r(A)} = \frac{r}{\omega} + \frac{2A}{\omega^2}.$$

It follows that $d' = S_1/S_0 = CA/(A^2 - 1)$, as claimed. The rational expression and $1/4 < d' < 3/5$ follow by substitution and multiplication of positive denominators. Finally $\log(1/\delta) = \text{arcosh}(A) = \Theta(\kappa^{-1/2})$ and $C/(A^2 - 1) = \Theta(1)$ for $\kappa \geq 4$. $\square$

In the lifted clock use public identity gates before and after the hard circuit, so its second and penultimate positions carry the Forrelation amplitude. Fitting an unpadded depth- $\Theta(r)$ circuit between these positions gives the exponent in (24); a bounded number of identity gates adjusts the residue of the length. Successive coefficients c_r have ratio δ , bounded below by an absolute constant for $\kappa \geq 4$, so there is always an attainable coefficient within a constant factor of any sufficiently small requested scale. The inverse baseline is $d = \kappa d'$. Consequently this explicit family has statistical factor $\Theta((c_r/\epsilon)^2)$, without a hidden condition-dependent gain.

5.4 A constant-weight path at high accuracy

Another clock gives a smaller baseline and a different product.

Theorem 5.4 (High-accuracy path product). *There are absolute constants $c_* > 0$ and κ_0 such that for $\kappa \geq \kappa_0$, $0 < \epsilon \leq c_* \kappa^{-3/2}$, and $q = 2^r$ with integer $r \geq 1$, a full-SQ family with matrix sparsity at most $2q + 1$, condition exactly κ , and one public right-hand side requires*

$$\Omega\left(\frac{\kappa^2 q^{\ell/2}}{r\ell}\right), \quad \ell = \Theta\left(\sqrt{\kappa} \left[1 + \log \frac{1}{\kappa^{3/2}\epsilon}\right]\right) \quad (27)$$

queries to estimate its inverse form to relative error ϵ . It has an $O(q)$ -sparse box-LP realization with the same query simulation.

Proof. Take an n -vertex scalar path J_n with all off-diagonals $1/2$. Lift its transitions to the Forrelation circuit as in (15). Let $H = mI + hJ_y$, and append the scalar pads $a, 1$. Set $z_0 = m/h = (\kappa + 1)/(\kappa - 1)$, $R = z_0 + \sqrt{z_0^2 - 1} = (\sqrt{\kappa} + 1)/(\sqrt{\kappa} - 1)$, and $\theta = \log R$. The determinant recurrence, with $D_0 = 1, D_1 = m$, gives $D_n = (h/2)^n U_n(z_0)$. Cofactors give the normalized endpoint coefficient and diagonal

$$c_n = a|(H^{-1})_{1n}| = \frac{2a}{hU_n(z_0)}, \quad d'_n = a(H^{-1})_{11} = \frac{2aU_{n-1}(z_0)}{hU_n(z_0)}, \quad \chi_n = \frac{c_n}{d'_n} = \frac{1}{U_{n-1}(z_0)}. \quad (28)$$

The opposite endpoint has the same diagonal. Choose the sign of the public two-sparse vector to cancel the alternating cofactor sign; its form is $\kappa d'_n + \kappa c_n \Phi$. Since $U_j(z_0) = \sinh((j+1)\theta)/\sinh \theta$, we obtain

$$c_n = \frac{16\sqrt{\kappa}}{(\kappa - 1)^2} \frac{1}{R^{n+1} - R^{-(n+1)}}, \quad \chi_n = \frac{4\sqrt{\kappa}}{\kappa - 1} \frac{1}{R^n - R^{-n}}, \quad \frac{\chi_n}{c_n} = \frac{\kappa - 1}{4} \frac{\sinh((n+1)\theta)}{\sinh(n\theta)}.$$

Uniformly for $n\theta \geq 1$ and $\kappa \geq 16$ this implies $c_n = \Theta(\kappa^{-3/2}e^{-n\theta})$, $d'_n = \Theta(\kappa^{-1})$ and $\chi_n = \Theta(\kappa c_n)$.

To make increasing length genuine source hardness, take a work register of $r\ell$ qubits, decompose each of the three full Hadamards into ℓ disjoint r -qubit layers, and use the two hidden sign queries. There are exactly $3\ell + 2$ transitions and $n = 3(\ell + 1)$ vertices. Choose the largest such n with $c_n \geq \epsilon$. For sufficiently small c_* this length has $n\theta \geq 1$. Consecutive admissible lengths differ by three and the displayed formula bounds their coefficient ratio by an absolute constant. Hence $c_n = \Theta(\epsilon)$, and solving the exponential formula gives the stated ℓ . Further, $\chi_n \geq (\kappa - 1)\epsilon/4$. Choose, for example, $\kappa_0 = 1024$ so $\chi_n \geq 200\epsilon$. The source Forrelation problem has randomized complexity $\Omega(2^{r\ell/2}/(r\ell))$. Apply Theorem 5.2 to $M = 4\lceil\chi_n/\epsilon\rceil^2 = \Theta(\kappa^2)$ blocks. This proves (27).

The unlifted path has norm less than one, so the affine shift obeys the promised spectral bounds, and the public pads make the condition exact. Every transition has public squared row and column mass $1/4$. Interior squared row norms of H are $m^2 + h^2/2$ and boundary norms are $m^2 + h^2/4$. All conditional and global samples are public mixtures of uniform Hadamard supports, a single sign location, and the diagonal; values cost one hidden query. Sparsity is at most $2q + 1$. Proposition 4.2 supplies the sparse factor with the same simulation. Direct sums preserve these properties. $\square$

The larger constant κ_0 is only convenient bookkeeping for the concentration theorem. This refinement is restricted to high accuracy: the constant path's boundary prefactor $\Theta(\kappa^{-3/2})$ shortens the admissible hard-circuit length, relative to the optimized witness at the same coefficient scale, by order $\sqrt{\kappa} \log \kappa$. Its explicit κ^2 outer factor therefore does not imply an improved leading exponential envelope.

5.5 Proved limitations of particular contrast constructions

Proposition 5.5 (Local cyclic-clock dilution). *Let U be unitary on an L -position clock, and let e satisfy $U^L e = \sigma e$ for $\sigma \in \{1, -1\}$, with $e, Ue, \dots, U^{L-1}e$ orthonormal. For $0 < \gamma < 1$ set $A = I - \gamma U$, $G = AA^*/(1 + \gamma)^2$, and $K = (1 + \gamma)/(1 - \gamma)$. Then $\kappa(G) \leq K^2$ and*

$$\frac{q_+}{q_-} = \left(\frac{1 + \gamma^L}{1 - \gamma^L} \right)^2 \leq (1 + K/L)^2, \quad q_\sigma = e^* G^{-1} e. \quad (29)$$

Proof. The singular values of $I - \gamma U$ lie between $1 - \gamma$ and $1 + \gamma$. The finite geometric identity gives $A^{-1}e = (1 - \sigma\gamma^L)^{-1} \sum_{t=0}^{L-1} \gamma^t U^t e$. Since A is normal, $q_\sigma = (1 + \gamma)^2 \|A^{-1}e\|^2$. Orthogonality makes the numerator $\sum_t \gamma^{2t}$ common to both signs, proving the equality. Write the ratio as $\coth^2((L/2) \log((K+1)/(K-1)))$. The inequalities $\log((K+1)/(K-1)) \geq 2/K$ and $\coth x \leq 1 + 1/x$ prove the upper bound. $\square$

For a real circuit this is an ordinary real SPD construction. If each transition is q -sparse in rows and columns, G is $(2q + 1)$ -sparse. Public scalar pads can make its condition exactly K^2 . Equation (29) bounds contrast by $O(1 + K^2/L^2)$ even for an *exact* holonomy sign. Thus contrast of order the final condition requires bounded L in this construction. This is a limitation of the stated first-order cyclic resolvent, not of all sparse linearizations.

Proposition 5.6 (Two-cluster and Schur-complement limitations). *The following two assertions hold.*

1. If $J_y = J_y^T$, $J_y^2 = I$, and orthogonal public unit vectors x, z satisfy $x^T J_y x = z^T J_y z = 0$, put $u_y = x^T J_y z$ and $G_y = mI + hJ_y$. For $v = (x + z)/\sqrt{2}$,

$$q(G_y, v) = 1 + \frac{\kappa - 1}{2}(1 - u_y). \quad (30)$$

A high-amplitude interval $u_y \in [3/5, 1]$ therefore fails to force an $O(1)$ scalar uniformly as κ grows. Such a uniform bound requires $1 - u_y = O(1/\kappa)$.

2. Let $G \succ 0$ have largest eigenvalue one and condition K_0 . For an SPD extension $\mathcal{H} = \begin{pmatrix} \alpha & w^T \\ w & G \end{pmatrix}$ with Schur complement $S = \alpha - w^T G^{-1} w > 0$,

$$e_0^T \mathcal{H}^{-1} e_0 = S^{-1}, \quad \kappa(\mathcal{H}) \geq \frac{1 + \|G^{-1}w\|^2}{S}. \quad (31)$$

If $w = tv$ with $\|v\| = 1$, $v^T G^{-1} v \geq cK_0$, $w^T G^{-1} w \geq c_0 > 0$, and $S = \Theta(1/R)$, then $\kappa(\mathcal{H}) = \Omega(K_0 R)$.

Proof. Since $m^2 - h^2 = a$, $(mI + hJ)^{-1} = (mI - hJ)/a$. Substitution gives (30). At $u = 3/5$ its value is $\kappa/5 + 4/5$, while for $|u| \leq 1/100$ it is of order κ ; the uniform contrast between those intervals is only constant. For the second assertion, block inversion gives the inverse entry. The vector $(1, -G^{-1}w)$ has Rayleigh quotient $S/(1 + \|G^{-1}w\|^2)$, and interlacing gives $\lambda_{\max}(\mathcal{H}) \geq 1$, proving (31). Finally Cauchy–Schwarz for the spectral measure of v gives $v^T G^{-2} v \geq (v^T G^{-1} v)^2$. Consequently

$$\|G^{-1}w\|^2 = t^2 v^T G^{-2} v \geq (w^T G^{-1} w)(v^T G^{-1} v) \geq c_0 c K_0.$$

Combine this with (31) and $S = \Theta(1/R)$. $\square$

Under the hypotheses of the Schur construction, a final condition budget $\bar{K}$ enforces $R = O(\bar{K}/K_0)$. If its formal clock factor is $\exp(A\sqrt{K_0})$ for fixed $A > 0$, optimizing the condition-dependent part of the proposed product gives

$$\max_{1 \leq K_0 \leq \bar{K}} \frac{\bar{K}}{K_0} e^{A\sqrt{K_0}} = \max\{\bar{K}e^A, e^{A\sqrt{\bar{K}}}\}.$$

Indeed the logarithm, in $t = \sqrt{K_0}$, is $\log \bar{K} - 2 \log t + At$, which is convex and hence maximized at an endpoint. This algebra explains the condition-budget loss for this specific compiler. It is not a general upper bound for sparse SPD query complexity, and it does not rule out the common ϵ^{-2} factor already proved distributionally.

There is also a simple circuit-resource qualification. If a chosen coherent repetition or polynomial amplification procedure uses r_{amp} invocations of a depth- D source circuit, placing that procedure in a clock of length T requires $r_{\text{amp}}D \leq O(T)$. Standard repetition to reduce a constant decision error to $O(\epsilon/\kappa)$ uses $r_{\text{amp}} = O(\log(\kappa/\epsilon))$ invocations. One cannot count the original depth- T Forrelation exponent and then amplify its promise at no clock-length cost. This bookkeeping does not assert that every structure-specific amplification must have that cost; nor does a narrow acceptance probability by itself supply the condition-preserving positive-form realization in Proposition 5.1.

Our conclusions are therefore separate, explicit lower families: (13), (18), (23), and (27). For worst-case complexity over the corresponding dimensions their maximum is a valid lower envelope. We do not claim the full product $\kappa\epsilon^{-2}s^{\Omega(\sqrt{\kappa}\log(1/\epsilon))}$. It would require additional information, for example a hard single-form family with condition-scale contrast and the narrow levels of Proposition 5.1. The proved products and the exact construction limitations stand independently of that stronger question.

6 Coherent access: a sharp bound and its oracle boundary

The distinction between a matrix entry and a normalized block of a unitary is decisive for quantum lower bounds. We make that distinction explicit before transferring scalar hardness to optimization.

An $(\alpha, a, \delta_{\text{in}})$ block encoding is a unitary U_H such that

$$\|H - \alpha(|0^a\rangle \otimes I)U_H(|0^a\rangle \otimes I)\| \leq \delta_{\text{in}}.$$

The interface includes controlled uses of U_H and $U_H^\dagger$. A second controlled unitary prepares $|b\rangle = b/\|b\|$, and $\|b\|$ is known. In the *plain block-encoding model*, no additional entry oracle is supplied. The normalization $\alpha \geq 1$ is part of the input contract; it is not the condition number.

Theorem 6.1 (Relative inverse forms under block access). *Let $\kappa^{-1}I \preceq H \preceq I$, $\kappa \geq 4$, and $0 < \epsilon \leq 1/16$. At constant failure probability, relative ϵ estimation of $Q = b^*H^{-1}b$ has worst-case block-query complexity*

$$\Omega(\alpha\kappa/\epsilon) \quad \text{and} \quad \tilde{O}(\alpha\kappa/\epsilon).$$

The upper bound is a specialization of the variable-time negative-power norm estimator of Chakraborty, Gilyén, and Jeffery [15, Theorem 33 in the full version]. Writing T_H, T_b for implementation costs of the two controlled unitaries, its cost at failure probability ζ is

$$O\left(\frac{\log^3 \kappa}{\epsilon} \left[\alpha\kappa(T_H + a) \log^2 \frac{\kappa}{\epsilon} + \sqrt{\kappa} T_b \right] \log \frac{\log \kappa}{\zeta} \right). \quad (32)$$

Their sufficient encoding accuracy specializes to $\delta_{\text{in}} = o(\epsilon/[\kappa \log^3(\kappa/\epsilon)])$. The lower bound already holds for exact encodings in dimension three, with public b .

Proof. Set $R_b = \|H^{-1/2}|b\rangle\|$. The cited theorem with negative-power parameter $c = 1/2$ estimates R_b multiplicatively. If $|\hat{R}_b/R_b - 1| \leq \epsilon/3$, then $|\hat{R}_b^2/R_b^2 - 1| \leq \epsilon$ for $\epsilon \leq 1$. Thus $\|b\|^2 \hat{R}_b^2$ estimates Q , with the displayed cost. This invocation estimates a norm directly; preparing the normalized inverse state alone would not provide Q .

For the lower bound, let

$$H_z = \text{diag}(\kappa^{-1}, 2(1 + 4\epsilon z)/\kappa, 1), \quad b = e_2, \quad z \in \{0, 1\}.$$

Both matrices have condition number exactly κ , and $Q_0 = \kappa/2$, $Q_1 = \kappa/[2(1 + 4\epsilon)]$. Their relative-error intervals are disjoint, since $(1 - \epsilon)(1 + 4\epsilon) > 1 + \epsilon$. For a Hermitian contraction A , define the exact unitary completion

$$W(A) = \begin{pmatrix} A & \sqrt{I - A^2} \\ \sqrt{I - A^2} & -A \end{pmatrix}.$$

Take $U_z = W(H_z/\alpha)$. The sole changing scalar signal is at most $5/8$, and $x \mapsto \sqrt{1 - x^2}$ has bounded derivative there. Consequently $\|U_0 - U_1\| = O(\epsilon/(\alpha\kappa))$. Replacing the oracle calls one at a time bounds the Euclidean distance between final purified states of any T -query algorithm by $T\|U_0 - U_1\|$; inverse and controlled calls satisfy the same bound. Measurements and adaptive operations can be deferred or purified. Constant-bias distinction of the disjoint output intervals requires a constant trace distance and hence $T = \Omega(\alpha\kappa/\epsilon)$. $\square$

This lower bound does not hold for the same family with exact sparse-value access: one query to the second diagonal entry reveals z . A standard coherent d -sparse encoding has normalization at most $d\|H\|_{\max} \leq d$ [23]. Together with Corollary 4.6, the justified generic comparison is

$$\Omega(\sqrt{\kappa}/\epsilon) \leq Q_{\text{sparse}} \leq \tilde{O}(d\kappa/\epsilon), \quad (33)$$

for $d \geq 2$, $\kappa \geq 4$, $0 < \epsilon \leq 1/128$, and the sufficiently large dimensions required by the counting family, with input-state preparation charged separately. The lower and upper bounds in (33) do not match in general. The lower bound already holds for the two-sparse sign-block family. The commuting sign-block family has public eigenvalues; hidden input only changes their weights in b . Its tight $\Theta(\sqrt{\kappa}/\epsilon)$ law therefore does not contradict the unknown-eigenvalue obstruction in Theorem 6.1.

6.1 Why a single polynomial transform loses a square-root factor

A useful but weaker construction block-encodes $B = H^{-1/2}/(2\sqrt{\kappa})$, whose success probability on $|b\rangle$ is

$$p_b = \frac{\langle b|H^{-1}|b\rangle}{4\kappa} \in [1/(4\kappa), 1/4].$$

Amplitude estimation [12] gives additive error $O(\sqrt{p_b}/L + L^{-2})$ after L uses of this circuit. Relative error ϵ therefore needs $O(\sqrt{\kappa}/\epsilon)$ uses in the worst case. A top-block error $\xi = O(\epsilon/\sqrt{\kappa})$ is sufficient, because its induced probability error is at most $2\xi\sqrt{p_b} + \xi^2$. Globally bounded negative-power approximation [23] supplies this transform in $\tilde{O}(\alpha\kappa)$ block queries. This procedure costs $\tilde{O}(\alpha\kappa^{3/2}/\epsilon)$ overall. Variable stopping times in (32) avoid multiplying the two worst-case costs.

Proposition 6.2 (Bounded-transform obstruction). *Suppose $\kappa \geq 8$, $\alpha \geq 1$, and a degree- n complex polynomial P satisfies $|P(x)| \leq 1$ for every $x \in [-1, 1]$. If P approximates $1/(2\sqrt{\alpha\kappa x})$ to absolute error $1/32$ at both $x_1 = 1/(\alpha\kappa)$ and $x_2 = 4/(\alpha\kappa)$, then $n = \Omega(\alpha\kappa)$. The same conclusion holds for a bounded Laurent transform of the signal phase with bandwidth n .*

Proof. The target values are $1/2$ and $1/4$, so the real part of P changes by at least $3/16$ on an interval of length $3/(\alpha\kappa)$. At some $\xi \in [x_1, x_2] \subset [0, 1/2]$, its derivative has magnitude at least $\alpha\kappa/16$. Bernstein’s polynomial inequality gives $|\operatorname{Re} P'(\xi)| \leq n/\sqrt{1-\xi^2} \leq 2n/\sqrt{3}$. For a Laurent transform $F(e^{i\theta})$ bounded by one on the unit circle, the two phases $\arccos x_1, \arccos x_2$ are separated by $\Theta(1/(\alpha\kappa))$. The trigonometric Bernstein inequality $\|dF/d\theta\|_\infty \leq n \|F\|_\infty$ gives the same bound. $\square$

The elementary Bernstein inequalities are used here only for bounded polynomials and trigonometric polynomials. This is an interior signal obstruction: the physical interval is a small part of the full unitary signal domain. Removing QSVT’s parity restriction, including by generalized signal processing [34], preserves global boundedness and does not remove the obstruction. This proposition is not a lower bound for variable-time, rational, or postselected algorithms.

The access dependence is also central to Orsucci–Dunjko [36, Proposition 6 and Sections 4.2–4.3]. They prove a worst-case $\Omega(\kappa)$ query lower bound for positive-definite solution-state preparation under their matrix-oracle contracts. With a normalized encoding of $I - \eta A$, their inverse approximation instead has degree $\tilde{O}(\sqrt{\kappa})$ for fixed $\eta > 0$: the steep part of the approximation is near a signal endpoint. Such a normalized shifted encoding is an additional access assumption; obtaining it from an arbitrary encoding of A is not free. Their solution-state output and the shifted-access construction differ from our relative scalar output and the unshifted bounded transform above. The square-root polynomial degree alone also does not include their cost of preparing the solution state.

6.2 Relation to earlier scalar quantum algorithms

Scalar readout of a linear-system solution is established prior work. Alase et al. [2] study additive estimation of x^*Cx for $Ax = b$. Their expectation-value subproblem for a supplied normalized state has tight complexity $\Theta(\alpha_C/\epsilon)$ in queries to an encoding of the *observable* C , and supplies a lower bound for the linear-system scalar task. They also explicitly distinguish block and entry access for a fixed computational-basis state: their block-query lower bound persists, whereas one exact-entry query suffices [2, discussion after Theorem IV.13 and Appendix A]. Our lower bound queries the inverse’s matrix H , uses relative error, and exposes an additional κ through a changing eigenvalue. Thus neither a first scalar-output quantum algorithm nor the negative-power upper primitive is claimed here. The contribution of Theorem 6.1 is the matching condition-number dependence for the relative inverse form and an explicit realization of this access distinction for that task. Likewise, a constant-query quantum decision of an underlying Forrelation promise is not automatically a constant-query numerical estimator of every associated inverse form; the q_* qualification following Proposition 4.3 still applies.

7 Scalar outputs of sparse conic and linear programs

The preceding matrix bounds need not disappear when the scalar is required to be a conventional optimization output. This section constructs sparse programs whose designated optimizer coordinate, optimal value, or Newton decrement contains the same hidden amplitude. The construction also identifies exactly what changes when the output is a sample instead of a number. Its clock mechanism builds on the Forrelation matrix-function reductions of Montanaro and Shao [32] and the inverse sampling separation of Grønlund and Larsen [25]; those mechanisms are not claimed as new.

7.1 A cyclic inverse with public norms

Let $q = 2^r$, $r \geq 1$, and let $n = r\ell$ be the number of work qubits. A two-function Forrelation circuit V has three global Walsh–Hadamard transforms and two hidden diagonal sign gates. Its real amplitude $\Phi = \langle 0^n | V | 0^n \rangle$ obeys the promise

$$|\Phi| \leq 1/100 \quad \text{or} \quad \Phi \geq 3/5.$$

The randomized query lower bound is $\Omega(2^{n/2}/n)$ [1]. Split each global transform into ℓ transforms on disjoint r -qubit blocks. Padding by at most one identity makes the forward length T even, where $T = 3\ell + 2$ or $3\ell + 3$. Each layer is q -sparse.

Use a clock of length $L = 3T$. The transitions run V forward, hold its final state for T steps, then run its gates in reverse transposed order. Let U be this real orthogonal clock transition. With W_t the forward product of transitions through clock t , $W_0 = I$, the orthogonal gauge $W = \sum_{t=0}^{L-1} |t\rangle\langle t| \otimes W_t$ satisfies

$$W^T U W = S \otimes I, \quad U^L = I, \quad (34)$$

where $S|t\rangle = |t+1 \bmod L\rangle$. In particular $W_t = V$ for $T \leq t < 2T$. For $K \geq 3$, define

$$\gamma = \frac{K-1}{K+1}, \quad \alpha_K = \log \frac{K+1}{K-1}, \quad M = \frac{I - \gamma U}{1 + \gamma}, \quad e = |0, 0^n\rangle. \quad (35)$$

Here M need not be symmetric; the associated Newton Hessian will be SPD. Since L is even, the cycle spectrum contains 1 and -1 . Thus

$$\|M\| = 1, \quad \sigma_{\min}(M) = K^{-1}, \quad \kappa_2(M) = K, \quad \text{nnz}(M_{i,:}), \text{nnz}(M_{:,j}) \leq q + 1. \quad (36)$$

Lemma 7.1 (Inverse history and full sampling access). *Write $z_0 = \gamma^{3T}$, $S_0 = \sum_{t=0}^{3T-1} \gamma^{2t}$, $S_I = \sum_{t=T}^{2T-1} \gamma^{2t}$, and $z = \gamma^{2T}$. Then*

$$x_* := M^{-1}e = \frac{1+\gamma}{1-z_0} \sum_{t=0}^{3T-1} \gamma^t U^t e, \quad R^2 := \|x_*\|^2 = K \frac{1+z_0}{1-z_0}, \quad (37)$$

$$p := S_I/S_0 = \frac{z}{1+z+z^2}, \quad K_{\text{eff}} := K/R = \sqrt{K \frac{1-z_0}{1+z_0}}. \quad (38)$$

Full SQ access to M , to

$$H_0 = 2M^T M = \frac{2}{(1+\gamma)^2} [(1+\gamma^2)I - \gamma(U + U^T)], \quad (39)$$

and to $M^T e$ requires at most a constant number of hidden sign queries per oracle call. The Hessian has condition number K^2 and sparsity at most $2q + 1$.

Proof. Multiplying the finite geometric series by $I - \gamma U$ proves the inverse formula. Different clock positions are orthogonal and each $U^t e$ is a unit vector. Hence $R^2 = (1+\gamma)^2 S_0 / (1-z_0)^2$; summing the geometric progression and using $(1+\gamma)/(1-\gamma) = K$ gives (37). Partitioning S_0 into three equal clock segments gives (38). Normality of U gives (39) and the spectrum assertion.

For the access claim, the diagonal, forward-clock, and reverse-clock supports are disjoint. A transition is either a public Hadamard block, a hidden diagonal sign, or an identity. Its row and column squared norm is one and its magnitudes and locations are public. Therefore every row and column of M has squared norm $(1+\gamma^2)/(1+\gamma)^2$; every row and column of H_0 has the same public

squared norm $4[(1 + \gamma^2)^2 + 2\gamma^2]/(1 + \gamma)^4$. Their global row distributions are uniform. Conditional samples choose between the disjoint clock blocks with public probabilities and then sample a Hadamard row uniformly, or take a deterministic sign location. The corresponding column rules are identical. Locations and norms need no hidden query; a value uses at most one. The vector $M^T e$ is a public-norm mixture of the clock-zero basis vector and a row of the last transition. The same simulation supplies its SQ interface. $\square$

Fix a sufficiently small constant $\delta_0 > 0$. For $0 < \delta \leq \delta_0$, choose an allowed T nearest to $\log(1/\delta)/\alpha_K$. Allowed lengths have bounded gaps and $\gamma \geq 1/2$, so, with universal constants,

$$p = \Theta(\delta^2), \quad R = \Theta(\sqrt{K}), \quad \ell = \frac{\log(1/\delta)}{3\alpha_K} + O(1), \quad N = 3Tq^\ell. \quad (40)$$

Taking δ_0 small ensures $\ell \geq 1$. All lower bounds below are for these parameter-selected dimensions; they are not asserted for every prescribed N .

7.2 An optimizer coordinate and a feasible-output guarantee

Let $Q_{N+1} = \{(t, u) : t \geq \|u\|\}$ be the Lorentz cone. Consider

$$\max_{x, u} 2e^T u \quad \text{subject to} \quad u = Mx, \quad (1, u) \in Q_{N+1}. \quad (41)$$

Its unique optimizer is (x_*, e) and its value is the public constant two. Thus this unperturbed problem has no value-estimation hardness. Its pulled-back barrier is $\phi(x) = -\log(1 - \|Mx\|^2)$. This restriction has barrier parameter one, even though the canonical logarithmically homogeneous barrier on the entire Lorentz cone has parameter two. Self-concordance is inherited by the affine restriction; in $u = Mx$ coordinates its squared gradient norm in the inverse Hessian is $2\|u\|^2/(1 + \|u\|^2) \leq 1$, proving the sharper parameter. At the public analytic center zero, for $F_\eta(x) = \phi(x) - 2\eta e^T Mx$,

$$\nabla F_\eta(0) = -2\eta M^T e, \quad \nabla^2 F_\eta(0) = H_0, \quad d_\eta = \eta M^{-1} e, \quad \Lambda_\eta^2 = 2\eta^2. \quad (42)$$

For $\eta < 1$, this full Newton step is strictly feasible; for $\eta \leq 1/(4\sqrt{2})$ its decrement is at most $1/4$. The central path is $x(\eta) = \rho(\eta)x_*$, where $\rho/(1 - \rho^2) = \eta$ and $0 < \rho < 1$. These identities concern the reduced Hessian. They make no assertion about an augmented KKT matrix containing redundant free variables.

For $t \in I := \{T, \dots, 2T - 1\}$ set $i_t = (t, 0^n)$ and

$$a = \sum_{t \in I} \frac{\gamma^t}{R\sqrt{S_I}} e_{i_t}, \quad h = \sqrt{p}. \quad (43)$$

Then $\|a\| = 1/R$ and $a^T x_* = h\Phi$. This linear observable can be made a literal optimizer coordinate. Append free variables satisfying

$$y_T = a_{i_T} x_{i_T}, \quad y_{t+1} = y_t + a_{i_{t+1}} x_{i_{t+1}}, \quad w = y_{2T-1}. \quad (44)$$

The chain has $O(T)$ rows and each row has at most three entries. Every selected x column gains one incidence. These equalities and their norms are public, so full SQ access to the augmented formulation is a public mixture with the interface of Lemma 7.1. The maximum row and column sparsity remains $s = \Theta(q)$. Free accumulator variables can be eliminated when differentiating the barrier; they do not add a cone.

Theorem 7.2 (Sparse scalar coordinate separation). *For the parameters in (40), estimating the optimizer coordinate $w_* = h\Phi$ to additive error $c\delta$, for a sufficiently small universal c , requires*

$$Q_{\text{classical}} = \Omega\left(\frac{q^{\ell/2}}{r\ell}\right) \quad (45)$$

full-SQ input queries at bounded error. The same bound applies to returning this coordinate of any exactly feasible point of objective gap $\tau \leq c_0\delta^2/K$, for a sufficiently small constant c_0 . For the source-equivalent coherent input oracle, estimating the exact optimizer coordinate at the stated accuracy takes $O(1)$ hidden queries and $\text{polylog } N$ public gates.

Proof. The coordinate identity follows by taking the overlap of (37) with (43). Its low and high intervals are separated by $(3/5 - 1/100)h = \Theta(\delta)$. Every formulation query has constant source-query overhead, so the Forrelation lower bound proves (45). If $(\hat{x}, \hat{u})$ is exactly feasible and has gap τ , then

$$\|\hat{u} - e\|^2 \leq 2 - 2e^T \hat{u} \leq \tau, \quad |a^T \hat{x} - h\Phi| \leq \frac{K}{R} \sqrt{\tau} \leq \sqrt{K\tau}.$$

Exact accumulator feasibility equates the returned coordinate with $a^T \hat{x}$. The specified τ therefore preserves the promise gap. An approximate equality or cone residual would require an additional error budget; it is not implicit in this statement.

A coherent query to a designated transition entry recovers any desired hidden sign, and conversely each coherent matrix-oracle call uses a constant number of sign queries. Run the source circuit and estimate its real amplitude to a sufficiently small constant error, then multiply by public h . No explicit N -coordinate output is included. $\square$

The last distinction is substantive: a numerical estimate of an optimizer coordinate is a scalar output, whereas producing a complete feasible vector is a different task. The classical lower bound for coordinates of approximately optimal feasible points uses the latter contract only as a sufficient source of an accurate scalar. The matched quantum comparison in this paper is for the scalar estimate, not dense output.

Since $\alpha_K = \Theta(1/K)$, the logarithm of (45) is $\Omega(K \log s \log(1/\delta))$, after retaining the displayed rounding and polynomial denominator. This is an exponent in sparsity, not in ambient dimension. The family has a stronger dimension exponent when a fixed number of hidden sign tables is allowed.

Corollary 7.3 (Fixed-order Forrelation). *Fix $k \geq 2$. Replace V by the k -Forrelation circuit and use the primary promise*

$$|\Phi| \leq \alpha_k := 2^{-5k-1} \quad \text{or} \quad \Phi \geq \beta_k := 2^{-5k}.$$

Then $T = (k+1)\ell + O(k)$ and

$$\ell = \frac{\log(1/\delta)}{(k+1)\alpha_K} + O_k(1).$$

At coordinate accuracy $c(\beta_k - \alpha_k)\delta$, the lower bound is

$$\Omega_k\left(\frac{q^\ell}{r\ell}\right)^{1-1/k} = \tilde{\Omega}_k(N^{1-1/k}). \quad (46)$$

The feasible-point gap sufficient for the same decision is $c_k\delta^2/K$, with $c_k = 2^{-O(k)}$. The coherent scalar algorithm uses $2^{O(k)}$ hidden queries.

Proof. Theorem 1.3 and Corollary 1.4 of Bansal and Sinha [6] give, for fixed k , the stated randomized lower bound and a $2^{O(k)}$ -query bounded-error algorithm. The source promise has a positive high case; allowing either high sign in a magnitude-output problem only enlarges the promise domain, so the lower bound still applies to that problem. The $k + 1$ global Hadamards give the claimed clock length. All remaining identities and oracle simulations are unchanged. Estimating the amplitude to error $O(\beta_k - \alpha_k)$ proves the numerical upper bound, and the feasible-point error $\sqrt{K\tau}$ proves the last accuracy claim. $\square$

7.3 Public norm normalization gives a hard optimal value

The unperturbed value is public, but a public objective tilt makes the value itself informative. Define

$$g = M^{-T}a, \quad G = \|g\|.$$

At first G might appear to contain hidden data. The hold plateau makes it exactly public, which avoids treating the quadratic tilt term as an unknown error.

Lemma 7.4 (Public tilt norm and plateau optimality). *Let $m = (I - \gamma S)/(1 + \gamma)$ and $\bar{a} = \sum_{t \in I} \gamma^t |t\rangle / (R\sqrt{S_I})$. Then*

$$G = \|m^{-T}\bar{a}\| = \Theta(\sqrt{K}), \quad h \leq G \leq K/R \leq \sqrt{K}. \quad (47)$$

For any clock vector $\bar{c}$ supported on I , with $b_c = m^{-T}\bar{c} \neq 0$,

$$\frac{|(b_c)_0|}{\|b_c\|} \leq \left(\sum_{t=0}^{T-1} \gamma^{-2t} \right)^{-1/2} = \Theta(\delta/\sqrt{K}). \quad (48)$$

Thus the readout (43) attains, to constants, the largest normalized inverse-transpose overlap among these plateau-supported clock readouts.

Proof. On the plateau $W_t = V$. Applying (34) gives

$$W^T a = \bar{a} \otimes V^T |0^n\rangle, \quad W^T g = (m^{-T}\bar{a}) \otimes V^T |0^n\rangle.$$

This proves that G is a clock-only quantity. Its value follows from the finite geometric inverse of m^T , requiring no hidden queries. If $b = m^{-T}\bar{a}$, then $b_0 = \bar{a}^T m^{-1} |0\rangle = \sqrt{S_I/S_0} = h$. Since $\bar{a}_t = 0$ for $0 \leq t < T$, the equation $b_t - \gamma b_{t+1} = 0$ gives $b_t = h\gamma^{-t}$ there. Hence

$$G^2 \geq h^2 \sum_{t=0}^{T-1} \gamma^{-2t} = \frac{1-z}{1+z+z^2} \frac{\gamma^2}{1-\gamma^2} = \frac{1-z}{1+z+z^2} \frac{(K-1)^2}{4K}. \quad (49)$$

For the chosen $z = \Theta(\delta^2)$ this is $\Omega(K)$. The upper bound is $\|m^{-1}\| \|\bar{a}\| = K/R$. The same homogeneous recurrence for b_c proves (48); evaluating its geometric sum gives the last order estimate. For any public readout on $(t, 0^n)$ over this plateau, its hidden overlap is $(b_c)_0 \Phi$ and its direction norm is $\|b_c\|$. Our choice has ratio $h/G = \Theta(\delta/\sqrt{K})$. $\square$

Theorem 7.5 (Optimal value and direct Newton decrement). *Replace the objective in (41), retaining (44), by $e^T u + \lambda w$, where $\lambda = 1/(2G)$. Then*

$$\text{OPT}^2 = \frac{5}{4} + \frac{h}{G} \Phi, \quad \frac{1}{2} \leq \text{OPT} \leq \frac{3}{2}. \quad (50)$$

Optimal-value estimation to additive error $c(\beta - \alpha)\delta/\sqrt{K}$ has lower bound (45), or (46), with the corresponding source thresholds α, β . The source-equivalent coherent algorithm attains that accuracy with $O(1)$ or $2^{O(k)}$ hidden queries, respectively.

At the public analytic center, with objective multiplier $\eta > 0$, the squared Newton decrement is

$$\Lambda_\eta^2 = \frac{\eta^2}{2} \left(\frac{5}{4} + \frac{h}{G} \Phi \right). \quad (51)$$

The same lower bounds hold for its additive estimation at scale $c(\beta - \alpha)\eta^2\delta/\sqrt{K}$, even when full SQ access to the reduced Hessian and its right-hand side is supplied directly.

Proof. Eliminating the equalities gives $w = g^T u$. Maximizing $(e + \lambda g)^T u$ on the unit ball gives $\text{OPT} = \|e + \lambda g\|$. Since $e^T g = h\Phi$ and $\lambda^2 G^2 = 1/4$, this proves (50). The bound $h/G \leq 1$ proves its value range. The low and high cases have respectively absolute squared deviations from $5/4$ at most $\alpha h/G$ and at least $\beta h/G$. On $[1/2, 3/2]$ squaring and taking square roots have constant Lipschitz bounds, proving the accuracy transfer. The objective has only the public e and w coefficients, so its access adds no hidden query. The coherent algorithm outputs $\sqrt{5/4 + (h/G)\Phi}$ after estimating Φ to a small multiple of $\beta - \alpha$; clipping $\hat{\Phi}$ to $[-1, 1]$ is harmless.

The reduced gradient is $-\eta c$, where $c = M^T e + \lambda a = M^T(e + \lambda g)$. Multiplying by $(2M^T M)^{-1}$ gives (51). The supports of $M^T e$ and a occupy disjoint public clock regions, and both component norms are public. Thus $\text{SQ}(c)$ is a known mixture, with at most one hidden query for a value. Lemma 7.1 supplies full SQ access to the Hessian. This proves the direct-Newton-system assertion. $\square$

The norm-normalized tilt has bounded eliminated coefficients: $G \geq \|\bar{a}\| = 1/R$, because every singular value of m^{-T} is at least one, so $\|\lambda a\| \leq 1/2$. The optimality statement in Lemma 7.4 is restricted to this plateau and a constant-scale normalized objective direction. It does not rule out other scalar reductions. For comparison, using only $G^2 \leq K$ would permit the smaller public tilt $\lambda = \theta h/K$. Then

$$\text{OPT}_\lambda^2 = 1 + 2\theta h^2 \Phi/K + \theta^2 h^2 G^2/K^2.$$

Taking $\theta \leq c(\beta - \alpha)$ separates the cases at accuracy $\Theta(\theta(\beta - \alpha)\delta^2/K)$, even without computing G . This fallback explains why accounting for the exactly public norm improves the signal from order δ^2/K to $\delta/\sqrt{K}$.

Parameterizing the two-function result by requested value accuracy ε , choose $\delta = \Theta(\varepsilon\sqrt{K})$ within the small- δ regime. The exponential lower dependence becomes

$$\exp \left(\Omega \left(K \log s \log \frac{1}{\varepsilon\sqrt{K}} \right) \right),$$

with (45) retaining the exact denominator. For the reduced Hessian condition $\kappa_H = K^2$, this substitutes $K = \sqrt{\kappa_H}$; it is a high-accuracy structural realization, not a replacement for the wider-range lower bound of Theorem 4.1.

7.4 Two LP realizations with different access implications

The direct decrement also occurs in a box LP. Set $c = M^T e + \lambda a$ as above and consider

$$\max_x c^T x \quad \text{subject to} \quad -\mathbf{1} \leq Mx \leq \mathbf{1}.$$

The paired logarithmic barrier $\phi_{\text{box}}(x) = -\sum_i \log(1 - (Mx)_i^2)$ has analytic center zero and Hessian H_0 . Its decrement is exactly (51). Both the stacked constraint matrix $[M; -M]$ and c have the previously proved full-SQ simulation. Thus the direct scalar diagnostic lower bound already applies to scalar inequalities. This statement does not identify the box LP's optimal value with the SOCP value.

For the unperturbed objective $2e^T Mx$, the box barrier subproblem in $u = Mx$ is

$$F_\eta(u) = -\sum_i \log(1 - u_i^2) - 2\eta e^T u.$$

Separability gives $u(\eta) = \rho(\eta)e$, with $\rho/(1 - \rho^2) = \eta$. Therefore the nonzero path has the same inverse-state ray as (42), and its accumulator coordinate is $\rho h\Phi$. Its Hessian is at least $2I$ in u coordinates. Consequently a barrier-subproblem gap at most τ implies $\|\hat{u} - \rho e\| \leq \sqrt{\tau}$ and accumulator error at most $\sqrt{K\tau}$. For any fixed public $\eta > 0$, the scalar coordinate lower bound survives a centering-subproblem gap $c\rho(\eta)^2(\beta - \alpha)^2\delta^2/K$, with an absolute sufficiently small $c > 0$. Accumulator equalities must again be exact, or the reported scalar must be computed as $a^T \hat{x}$. The box barrier has parameter N , since each paired interval barrier has parameter one; no small iteration bound follows from this realization.

A second LP trades the full-dimensional box for a one-dimensional affine slice. Let $\tilde{a} = Ra$, so $\|\tilde{a}\| = 1$, and set

$$\max_{x,t} \tilde{a}^T x \quad \text{subject to} \quad Mx = te, \quad -1 \leq t \leq 1. \quad (52)$$

Replace a by $\tilde{a} = Ra$ in the accumulator equalities (44) and use w as the objective. This scales only the readout coefficients multiplying x , and makes the objective literally one-sparse without changing the feasible projection.

Theorem 7.6 (Affine-slice value and matched upper exponent). *The LP (52) has a unit objective, full-SQ input access with constant source-query overhead, and exact value*

$$\text{OPT} = Rh|\Phi| = \Theta(\sqrt{K}\delta)|\Phi|. \quad (53)$$

Its core equality operator $A_{\text{eq}} = [M \ -e]$ has $K \leq \kappa_2(A_{\text{eq}}) \leq \sqrt{2}K$. Additive value accuracy $c(\beta - \alpha)\sqrt{K}\delta$ has the classical lower bounds (45) and (46), and the corresponding constant or $2^{O(k)}$ coherent hidden-query upper bounds. In particular, for fixed k and constant additive value accuracy, one can take $\delta = \Theta_k(K^{-1/2})$, yielding lower exponent $\Omega_k(K \log s \log K)$.

For the two-function family, at additive accuracy ε with $\varepsilon/\sqrt{K}$ sufficiently small, the classical lower and upper exponential dependences agree up to constant factors:

$$\log Q = \Theta \left(K \log s \log \frac{\sqrt{K}}{\varepsilon} \right) \quad (54)$$

when the displayed exponent dominates the polynomial and rounding factors. The exact lower bound remains (45), with $\ell = \lceil \log(\sqrt{K}/\varepsilon) + O(1) \rceil / (3\alpha_K) + O(1)$. The constant accuracy calibration inside the numerator changes the length by $O(K)$; only the final $O(1)$ accounts for allowed-length rounding.

Proof. Invertibility forces $x = tM^{-1}e$, so maximizing over $|t| \leq 1$ gives (53). The added column $-e$ changes only one public row norm of M ; its location, norm, and all conditional mixture weights are public. The objective $\tilde{a}$ is also public. Moreover,

$$K^{-2}I \preceq A_{\text{eq}}A_{\text{eq}}^T = MM^T + ee^T \preceq 2I.$$

The K^{-2} eigenspace of MM^T has multiplicity $q^\ell \geq 2$. A vector in it orthogonal to e preserves the smallest eigenvalue. The largest singular value is at least $\|M\| = 1$, proving the condition bounds. No such bound is asserted for the accumulator-augmented KKT matrix. The value identity and the source gap prove the lower and coherent upper bounds exactly as before.

For the classical upper bound, apply Theorem 3.8 to $(M, \tilde{a}, e)$, using the exact public $R_e = R = \Theta(\sqrt{K})$. The theorem permits $0 < \varepsilon \leq R/2$, which contains the chosen small- δ value-accuracy regime even when $\varepsilon > 1/2$. The absolute-value function is 1-Lipschitz, so the theorem estimates (53) at cost

$$O\left(R^2 \varepsilon^{-2} \log(1/\zeta) (q+2)^{O(K \log(2R/\varepsilon))}\right).$$

This yields the upper exponent in (54) without constructing a nullspace basis, an optimizer, or an SQ interface to $M^{-1}e$. Substituting $\delta = \Theta(\varepsilon/\sqrt{K})$ in the lower family yields the matching exponent. $\square$

The affine slice has a parameter-one barrier in its intrinsic coordinate: $\varphi(t) = -\log(1-t^2)$. Indeed,

$$\varphi' = \frac{2t}{1-t^2}, \quad \varphi'' = \frac{2(1+t^2)}{(1-t^2)^2}, \quad \varphi''' = \frac{4t(3+t^2)}{(1-t^2)^3}.$$

The gradient bound is $(\varphi')^2/\varphi'' = 2t^2/(1+t^2) \leq 1$. Squaring the self-concordance inequality $|\varphi'''| \leq 2(\varphi'')^{3/2}$ reduces to $(1-t^2)^2(t^2+2) \geq 0$. At its analytic center, the tangent is $\text{span}\{(M^{-1}e, 1)\}$, its scalar barrier Hessian is two, and its projected objective is $Rh\Phi$. Therefore

$$\Lambda_\eta^2 = \frac{\eta^2}{2} R^2 h^2 \Phi^2 = \frac{\eta^2}{2} \text{OPT}^2. \quad (55)$$

Estimating this squared decrement at scale $c(\beta^2 - \alpha^2)\eta^2 K \delta^2$ has the same source lower bound. Here the scalar projected right-hand side is *not* supplied: it already contains the hard quantity. Supplying a nullspace basis and its objective projection reduces the program to an explicit interval problem. Thus the constant-barrier result isolates equality preprocessing and scalar readout, not repeated high-dimensional barrier work.

7.5 Three-dimensional cones preserve the scalar construction

For completeness, the single large Lorentz cone can be replaced by an elementary binary norm tree. Let D be the smallest power of two at least N , pad u by zeros to D leaves, and for each internal node v impose $(t_v, z_{v,0}, z_{v,1}) \in Q_3$, where its children are leaf coordinates or lower node variables. Fix the root t to one. Recursive use of $t_v^2 \geq z_{v,0}^2 + z_{v,1}^2$ proves that the feasible projection is precisely $\|u\| \leq 1$; choosing each t_v as the subtree norm proves the converse. Public two-sparse consensus equalities can copy shared variables into standard product-cone form. This uses $D-1$ cones, constant incidence per new variable, and only public new SQ data. All optimizer-coordinate and optimal-value results above are therefore unchanged.

The first Newton system can also be computed exactly. In the identified coordinates, the product barrier is

$$\Psi(t, u) = - \sum_{v \text{ internal}} \log(t_v^2 - z_{v,0}^2 - z_{v,1}^2).$$

First leave all D leaves free, with only the root fixed to one. Put $J = \log_2 D$. The analytic center of this full tree at $u = 0$ has common squared variable at depth d given by

$$(t_d^\circ)^2 = \frac{D - 2^d}{2^d(D-1)}, \quad 0 \leq d < J. \quad (56)$$

To verify this, set $y_d = 2^d t_d^2$, $y_0 = 1$, and $y_J = 0$. The barrier on depth-symmetric points, up to a constant, is $-\sum_{d=0}^{J-1} 2^d \log(y_d - y_{d+1})$. Its minimizing gaps are $2^d/(D-1)$, giving (56). This point is stationary in all tree variables: automorphisms are transitive at each depth, so vanishing of each depth derivative makes every coordinate derivative vanish. Strict convexity of the sum of Lorentz barriers on the affine slice then proves it is the unique center; each nonzero variable direction affects at least one strictly convex cone term. All leaf derivatives vanish at $u = 0$. This full-tree center lies on the affine slice fixing the padded leaves to zero, so restriction preserves its minimizing property and uniqueness. This argument does not require the padded problem itself to retain the full tree symmetry.

Each leaf enters quadratically only in its bottom cone. Hence at this center

$$\nabla_{ut}^2 \Psi = 0, \quad \nabla_{uu}^2 \Psi = c_D I, \quad c_D = 2(D-1).$$

After $u = Mx$, the x Hessian block is $c_D M^T M$ with zero cross block. For the objective $2e^T u$, its first x direction is $(2\eta/c_D)M^{-1}e$. For the normalized tilt the squared decrement is

$$\Lambda_\eta^2 = \frac{\eta^2}{c_D} \left(\frac{5}{4} + \frac{h}{G} \Phi \right). \quad (57)$$

The tree-variable block is public, has constant degree, and can be preprocessed without hidden queries. Thus the direct full-SQ Newton simulation survives, but its additive decrement gap is $\Theta((\beta - \alpha)\eta^2\delta/(c_D\sqrt{K}))$. The standard product barrier has parameter $2(D-1)$. Neither a small condition number of the full lifted Hessian nor a polylogarithmic iteration bound follows from the reduced clock block. The norm tree is a standard repeated Lorentz norm representation [3], not a new cone lift.

7.6 Sampling outputs and the rare-event cost

The distinction between scalar and sampling output can be quantified without appealing to an unspecified failure convention. Let Π project onto $(t, 0^n)$ over the hold segment. The normalized inverse history $v = x_*/R$ satisfies

$$\|\Pi v\|^2 = p\Phi^2. \quad (58)$$

If $0 \leq \varepsilon_v < 1$ and $\|\hat{x} - x_*\| \leq \varepsilon_v R$, then

$$\frac{(\sqrt{p}|\Phi| - \varepsilon_v)_+^2}{(1 + \varepsilon_v)^2} \leq \frac{\|\Pi \hat{x}\|^2}{\|\hat{x}\|^2} \leq \frac{(\sqrt{p}|\Phi| + \varepsilon_v)^2}{(1 - \varepsilon_v)^2}.$$

For a fixed source gap $\beta > \alpha$, choosing $\varepsilon_v \leq c(\beta - \alpha)\sqrt{p}$ with a sufficiently small constant makes the low and high Bernoulli event probabilities separated by $c'p$, and both are $O(p)$. Averaging $O(1/p)$ independent samples therefore decides the source promise with constant error, by the variance bound for a Bernoulli mean. It follows that a per-sample algorithm needs

$$\Omega \left(p \frac{q^{\ell/2}}{r\ell} \right) \quad (59)$$

queries in the two-function case. On the scalar family $p = \Theta(\delta^2)$, this carries the explicit δ^2 factor. The argument also permits mixtures of good approximate-vector laws, as long as every component satisfies the same perturbation bound.

Equation (59) assumes the overall sample law has that guarantee. Alternatively, failure may be flagged with bounded expected retry cost, or unflagged contamination must have mass at most $c'p$

for a sufficiently small constant. Constant unflagged failure on each fresh sample call can overwhelm a vanishing plateau and is insufficient. A different contract permits one randomized setup that produces a fixed good sampler with constant probability: conditional on this single good-setup event, repeated draws all have a good sample law. Taking a constant number of independent setups and a majority of their source decisions gives the usual bounded-error reduction, provided setup success is at least $1/2 + c_{\text{setup}}$ for an absolute constant $c_{\text{setup}} > 0$, and each conditional decision has error at most $c_{\text{setup}}/2$. Each setup then gives a correct decision with a fixed positive margin above one half, even if bad setups always err. This does not require vanishing setup failure. The corresponding cost statement is $Q_{\text{setup}} + O(1/p)Q_{\text{draw}} = \Omega(Q_{\text{source}})$; it does not lower-bound a draw while leaving its preprocessing uncharged. The deterministic polynomial approximation in Theorem 3.7 has no setup-approximation failure; its rejection randomness samples a fixed approximate vector. If the sampler is promised relative vector error ε_v , choosing instead $p = \Theta(\varepsilon_v)$ gives the earlier, weaker but useful form

$$\Omega\left(\frac{\varepsilon_v q^{\ell/2}}{r\ell}\right), \quad \ell = \frac{\log(1/\varepsilon_v)}{6\alpha_K} + O(1),$$

under the same contamination condition, for vector error sufficiently small relative to the fixed source gap $\beta - \alpha$. The sharper perturbation inequality above also allows $p = \Theta(\varepsilon_v^2)$ with a sufficiently large fixed prefactor, at the cost of its ε_v^2 repetition factor. Ordinary bounded-error scalar estimation in Theorem 7.2 needs neither repetition penalty nor a vanishing-failure contract.

At fixed sufficiently small vector error and fixed q , choosing $p = \Theta(1)$ gives $T = \Theta(K)$ and $\log N = \Theta(K)$. The fixed- k construction then gives $\tilde{\Omega}_k(N^{1-1/k})$ classical sample hardness at $\kappa(M) = \Theta(\log N)$ and $\kappa(H_0) = \Theta(\log^2 N)$. These attain the necessary conditioning scales following Theorem 3.7. For full first-Newton access use $(H_0, M^T e)$, whose inverse solution is $M^{-1}e/2$ and whose full-SQ simulation was proved above. An exactly feasible point of SOCP objective gap τ has relative vector error at most $K_{\text{eff}}\sqrt{\tau} \leq \sqrt{K}\tau$; thus $\tau \leq \varepsilon_v^2/K$ transfers its sample output to this guarantee.

There are explicit upper bounds for this constructed family. Classically, reading all hidden sign tables costs $O_k(q^\ell)$ queries; computing their circuit prefixes and sampling the public geometric clock gives the exact inverse-state distribution in $q^\ell \text{poly}_k(r, \ell)$ arithmetic. A designated matrix value query recovers each sign, so this upper bound uses the same input interface. Consequently the classical logarithmic complexity is $\Theta(K \log s \log(1/\varepsilon_v))$ whenever its leading term dominates the displayed repetition and denominator factors. Quantumly, prepare the public geometric clock and apply a gate whenever the clock has passed that gate's position. Forward and reverse segments use each hidden sign gate twice in total. This prepares the exact normalized history in the ideal circuit model. For any requested Euclidean state error $0 < \eta < 1$, synthesize the public circuit to operator error at most η : with G_{public} public gates, allocating error at most η/G_{public} to each gate suffices by telescoping. The implementation retains $O(k)$ hidden queries and public gate cost polynomial in $n, T, \log(1/\eta)$. Taking $\eta \leq \varepsilon_v$ identifies the synthesized normalized state with a vector $\hat{x}/R$ satisfying the relative-vector sampling tolerance above. It is an algorithm for this source-equivalent coherent realization; a classical SQ data structure alone does not provide its controlled sign queries.

There is also a generic feasible-*sample* upper envelope. The normal-equation polynomial used in Theorem 3.7 can produce $z = Pe$ with $\|Mz - e\| \leq \xi$, at degree $O(K \log(1/\xi))$. This follows directly from the singular-value residual bound for MP , not just the relative error in z . For $\xi < 1/2$, the normalized point

$$x = z / \|Mz\|, \quad u = Mz / \|Mz\|$$

is exactly feasible for (41) and has objective gap $O(\xi^2)$: its squared angular distance from e is bounded by a constant times $\|Mz - e\|^2$. Scaling z does not change its squared-coordinate sample

law. Setting $\xi = \Theta(\varepsilon_v/\sqrt{K})$ therefore gives a sample from a point of gap $O(\varepsilon_v^2/K)$ with exponential cost $\exp(O(K \log(s+1) \log(\sqrt{K}/\varepsilon_v)))$, up to polynomial factors. This assertion supplies the x -block sample law; it does not supply the normalization or explicit coordinates of the feasible point for free.

7.7 Prior work and the meaning of the optimization separation

Forrelation decision and its near-linear classical lower bounds are due to Aaronson–Ambainis and Bansal–Sinha [1, 6]. The history-state sampling obstruction is already present in Grønlund–Larsen [25]. Ellipsoids, norm trees, and their SOCP formulations are standard. Apers and Gribling [5, Lemma 8.5] already prove a full-SQ classical lower bound for constant-error LP optimal-value estimation. Therefore neither an SQ optimization lower bound nor scalar optimization hardness alone is new here. To the best of our knowledge, the additional content is the combined conditioning–sparsity–accuracy accounting for these specific wrappers, the public inverse-transpose norm that permits the larger value signal, its plateau-restricted optimality, and the exact direct-Newton and parameter-one affine-slice realizations under their stated interfaces.

The coherent upper bounds use explicit structured algorithms. They do not establish a general quantum interior-point method for arbitrary SOCPs or LPs. Quantum SOCP methods that reconstruct classical Newton directions have a different output ledger [27]. Building an explicit matrix table or outputting an entire optimizer has dimension-dependent cost; those tasks are absent from the scalar query contracts studied here.

8 Scalar composition along a path and at its endpoint

The preceding constructions encode one source instance in one scalar. We now distinguish three ways of using independent instances: requesting several path diagnostics, compiling their answers into one endpoint coordinate, and merely revisiting a fixed hidden input. Only the first two supply a direct-sum conclusion. Throughout this section the input is the *formulation*, with the full SQ interface of Section 7; an exact oracle for a later iterate is not supplied without charge.

8.1 A direct sum for classical formulation queries

Let g be the promised two-function Forrelation predicate, with $|\Phi| \leq \alpha = 1/100$ in case zero and $\Phi \geq \beta = 3/5$ in case one. Write $L_g = R_{1/3}(g)$ for its worst-case bounded-error randomized query complexity. The cyclic instances have

$$L_g = \Omega\left(\frac{q^{\ell/2}}{r\ell}\right), \quad a_i^T M_i^{-1} e = h\Phi_i, \quad \|M_i\| = 1, \quad \sigma_{\min}(M_i) = K^{-1},$$

where $h = \Theta(\delta)$ and the one-block dimension is $N_0 = \Theta(Tq^\ell)$. Here T remains the clock length of Section 7; the positive integer B denotes the number of independent blocks.

Lemma 8.1 (Classical direct sum). *Suppose every query to a formulation built from B independent promised inputs is simulated by $O(1)$ classical queries, each touching one named or sampled input block. All choices of sampled block have public probabilities. An output from which all B promise bits can be recovered, with joint success at least $2/3$, requires $\Omega(BL_g)$ formulation queries.*

Proof. Constant-error randomized query complexities are comparable by worst-case amplification. Minimax therefore gives a distribution μ for which any randomized algorithm using fewer than cL_g queries has average success at most $9/16$. Embed a target drawn from μ in a uniform hidden block

I , and fill the other blocks independently from μ . The simulator may sample and retain complete filler inputs for free: only queries to the target count. The assembled input has distribution μ^B and is independent of I . Each of the $O(Q)$ simulated queries made by a Q -query formulation algorithm thus touches the target with average probability $1/B$. This also holds for global SQ samples with nonuniform public block weights. The expected target cost is $O(Q/B)$. Truncate after twelve times that expectation. Markov's inequality loses at most $1/12$ success, leaving average target success at least $7/12$. Since $7/12 > 9/16$, the chosen distribution forces $Q/B = \Omega(L_g)$. No amplification of a distributional average advantage is used. $\square$

This elementary argument is a special case of randomized direct-sum phenomena [11]. It does not apply to one coherent query in superposition over all blocks.

8.2 Successive scalar increments of a box central path

Consider the single LP

$$\max 2 \sum_{i=1}^B \theta_i e^T M_i x_i, \quad -\mathbf{1} \leq M_i x_i \leq \mathbf{1}, \quad \theta_i = \Gamma^{-i}. \quad (60)$$

A public three-sparse partial-sum chain defines the free coordinate $w = \sum_i a_i^T x_i$, as in (44). Define

$$\rho(z) = \frac{2z}{1 + \sqrt{1 + 4z^2}}, \quad \rho(0) = 0.$$

Since $\rho/(1 - \rho^2) = z$, the unique logarithmic central point is $x_i(\eta) = \rho(\eta\theta_i)M_i^{-1}e$. At $\eta_j = \Gamma^{j+1/2}$, $0 \leq j \leq B$, its B increments, indexed by $j = 1, \dots, B$, obey

$$D_j := w(\eta_j) - w(\eta_{j-1}) = h \sum_{i=1}^B d_{j-i} \Phi_i, \quad d_m = \rho(\Gamma^{m+1/2}) - \rho(\Gamma^{m-1/2}). \quad (61)$$

The kernel is nonnegative and has infinite sum one. The identities $\rho(z) \leq z$ and $1 - \rho(z) = \rho(z)/(z(1 + \rho(z))) \leq 1/(2z)$ imply

$$d_0 \geq 1 - \frac{3}{2\sqrt{\Gamma}}, \quad |D_j/h - d_0 \Phi_j| \leq 1 - d_0 =: \tau_\Gamma. \quad (62)$$

These bounds follow by telescoping; restricting the sum to the actual blocks only reduces its off-diagonal mass.

Theorem 8.2 (Scalar-increment trajectory). *For $\Gamma = 2^{20}$, returning all B increments in (61) to additive error $0.21h$, jointly with probability $2/3$, requires $\Omega(BL_g)$ classical full-SQ queries. The canonical coherent source interface permits $O(B)$ queries at this accuracy, or $O(B \log(2B))$ queries at the stronger accuracy $h/100$. The LP has dimension $\Theta(BN_0)$, row and column sparsity $O(q)$, and selected multiplier range $\log(\eta_B/\eta_0) = B \log \Gamma$.*

Proof. The low and high intervals for D_j/h are separated by at least $d_0(\beta - \alpha) - 2\tau_\Gamma > 0.586$. Error 0.21 therefore still identifies the bit. Each matrix query touches one block; the added readout chain and objective coefficients are public. Row and column norms and sampling probabilities follow from public squared magnitudes. The chain has $O(BT)$ rows and constant incidence, so it preserves the claimed dimension and sparsity. Lemma 8.1 proves the lower bound.

For the numerical upper, estimate each Φ_j to error 0.005, with failure probability $1/(3B)$, and return $hd_0\hat{\Phi}_j$. Equation (62) bounds the total error by $0.005h + 0.0015h < h/100$. Constant-accuracy source estimation followed by median amplification costs $O(\log(2B))$ queries per block. For the looser accuracy, use the coherent Boolean recovery theorem of Buhrman et al. [14, Corollary 3]. It recovers all B bits jointly using $O(B)$ controlled calls to their bounded-error constant-query subroutines and their inverses. Return zero for a low bit and $0.8hd_0$ for a high bit. Their errors are at most $(0.01d_0 + \tau_\Gamma)h < 0.012h$ and $(0.2d_0 + \tau_\Gamma)h < 0.202h$, respectively. $\square$

The Boolean recovery theorem is essential for the $O(B)$ claim: individual constant success does not imply joint success. Nor does Boolean recovery supply arbitrary real-valued estimates. In particular, absolute checkpoint values include sums of actual amplitudes. Estimating each amplitude to $O(1/B)$ and failure $O(1/B)$ gives an elementary $O(B^2 \log(2B))$ quantum upper for absolute checkpoint accuracy $O(h)$; no matching linear upper for that stronger output is asserted.

The same lower bound applies to exactly feasible approximate centers with barrier-subproblem gap $\tau_{\text{ctr}} \leq ch^2/(BK)$, for a sufficiently small constant c . Indeed, the interval barrier Hessian is at least $2I$ in $u_i = M_i x_i$ coordinates, so strong convexity gives $\|\hat{u} - u\| \leq \sqrt{\tau_{\text{ctr}}}$. Since $\|M_i^{-T} a_i\| \leq K/R \leq \sqrt{K}$,

$$|\hat{w} - w| \leq \sqrt{BK\tau_{\text{ctr}}}.$$

Differencing two such values preserves the decoding gap. The accumulator equality must hold exactly, or its residual must be included in this error budget. At the analytic center the reduced Hessian has condition K^2 ; there is no uniform condition bound over the whole scaled trajectory. The paired interval barrier has parameter one as established in Section 7, giving parameter BN_0 for this product.

For the fixed- k promise in Corollary 7.3, choose $\sqrt{\Gamma} \geq 12/(\beta_k - \alpha_k)$ and use error $c_k h$, $c_k > 0$ sufficiently small. The same proof gives

$$\tilde{\Omega}_k(BN_0^{1-1/k}) = \tilde{\Omega}_k(B^{1/k} \mathcal{N}^{1-1/k}), \quad \mathcal{N} = \Theta(BN_0),$$

and a numerical coherent upper $2^{O(k)} B \log(2B)$. Even with the positive high promise, its interval $[\beta_k, 1]$ can be much wider than the decoding gap; one Boolean representative need not meet that numerical accuracy.

8.3 Ordinary SOCP predictor decrements

Let $g_i = M_i^{-T} a_i$, and use the public common norm $G = \|g_i\|$ from Lemma 7.4. Put

$$v_i = e + \frac{g_i}{2G}, \quad C = \frac{5}{4}, \quad \xi = \frac{h}{CG}, \quad \|v_i\|^2 = C(1 + \xi \Phi_i), \quad 0 < \xi \leq \frac{1}{4}.$$

The last restriction is obtained by taking the public upper bound on δ small enough. On the product of B unit balls consider

$$\max \sum_i \theta_i \left(e^T u_i + \frac{a_i^T x_i}{2G} \right), \quad u_i = M_i x_i, \quad \|u_i\| \leq 1, \quad \theta_i = \frac{\Gamma^{-i}}{\sqrt{C}}. \quad (63)$$

The barrier is $-\sum_i \log(1 - \|u_i\|^2)$, with the conventional Lorentz parameter $2B$. At its exact central point for $\eta_j = \Gamma^j$, with $j = 1, \dots, B$, propose the relative update $\eta_j \mapsto (1 + \sigma)\eta_j$, where $\sigma = \sigma_0/\sqrt{B}$ and σ_0 is a small constant.

Theorem 8.3 (Predictor-decrement trajectory). *For $\Gamma \geq 256$, the ordinary squared predictor decrements for $j = 1, \dots, B$ satisfy*

$$P_j := \frac{\Lambda_j^2}{\sigma^2} = \sum_i f\left(\Gamma^{j-i} \sqrt{1 + \xi \Phi_i}\right), \quad f(z) = 1 - \frac{1}{\sqrt{1 + 4z^2}}. \quad (64)$$

Returning all B predictor decrements Λ_j^2 to error $\sigma^2 \xi / 100$, with joint success $2/3$, requires $\Omega(BL_g)$ classical full-SQ formulation queries. Their scale is $\sigma^2 \xi = \Theta(\delta / (B\sqrt{K}))$, and every predictor obeys $\Lambda_j^2 \leq \sigma_0^2$. The structured coherent upper is $O(B \log(2B))$. For $\Gamma = 2^{20}$, at the looser error $0.036\sigma^2 \xi$, it is $O(B)$, while the same classical lower bound holds.

Proof. Write $s_i = \|v_i\|$. The central point is $u_i = \rho(\eta\theta_i s_i)v_i/s_i$. At radius r_i , the radial barrier-Hessian eigenvalue is $2(1 + r_i^2)/(1 - r_i^2)^2$. The multiplier update leaves residual $2\sigma\eta\theta_i v_i$. Its squared inverse-Hessian norm is $2\sigma^2 r_i^2 / (1 + r_i^2) = \sigma^2 f(\eta\theta_i s_i)$. This proves (64) and the bound on its size.

Subtract the public baseline $P_j^0 = \sum_i f(\Gamma^{j-i})$ and set $\psi_z(x) = f(z\sqrt{1 + \xi x}) - f(z)$. Direct differentiation gives

$$\psi'_z(x) = \frac{2\xi z^2}{(1 + 4z^2(1 + \xi x))^{3/2}}, \quad \frac{2\xi}{6^{3/2}} \leq \psi'_1(x) \leq \frac{\xi}{4} \quad (|x| \leq 1). \quad (65)$$

The other blocks contribute absolute total at most

$$T_\Gamma \leq \xi \left(\frac{2}{3^{3/2}(\Gamma - 1)} + \frac{2}{\Gamma^2 - 1} \right) \leq \frac{0.4\xi}{\Gamma - 1}, \quad \Gamma \geq 256. \quad (66)$$

For scales $z = \Gamma^m \geq \Gamma$, use $\psi'_z \leq 2\xi/(3^{3/2}z)$; for $z \leq \Gamma^{-1}$, use $\psi'_z \leq 2\xi z^2$, and sum both geometric series. The low/high gap for $P_j - P_j^0$ is at least

$$\xi \left(\frac{2\beta}{6^{3/2}} - \frac{\alpha}{4} - \frac{0.8}{\Gamma - 1} \right) > 0.075\xi.$$

Lemma 8.1 proves the classical lower bound. The two objective pieces in each block occupy disjoint u and x registers, with public squared norm $4\theta_i^2(1 + \|a_i\|^2/(4G^2))$. Thus objective SQ, as well as equality-matrix SQ, has constant source-query simulation cost.

Estimating each active amplitude to error 0.02 and failure $1/(3B)$, then returning $P_j^0 + \psi_1(\hat{\Phi}_j)$, gives error at most $0.005\xi + T_\Gamma < 0.01\xi$. For $\Gamma = 2^{20}$, recover all promise bits jointly and output the midpoint of $P_j^0 + \psi_1([-\alpha, \alpha])$ or $P_j^0 + \psi_1([\beta, 1])$. On the low interval, $\psi'_1 \leq 2\xi/(5 - 0.01)^{3/2} < 0.180\xi$, so the midpoint error is less than 0.0018ξ . On the positive high interval, $\psi'_1 \leq 2\xi/5^{3/2}$, so its midpoint error is at most $0.4\xi/5^{3/2} < 0.035778\xi$. Adding (66) keeps both below 0.036ξ . The gap is still greater than $0.079\xi > 2(0.036\xi)$. $\square$

For fixed k , the same proof uses $\Delta_k = 2\beta_k/6^{3/2} - \alpha_k/4 > 0$, chooses $\Gamma_k \geq \max\{256, 1 + 1.6/\Delta_k\}$, and takes error $c_k\sigma^2\xi$ with $c_k < \Delta_k/8$. It gives the same $\tilde{\Omega}_k(BN_0^{1-1/k})$ classical lower and $2^{O(k)}B \log(2B)$ numerical coherent upper. The formulation dimension is $\Theta(BN_0)$ and sparsity $O(q)$. The reporting points span $(B - 1) \log \Gamma$ in logarithmic multiplier. Intermediate short steps may be inserted; the diagnostic at each reporting point concerns the small update σ , not a jump of size Γ . No full-trajectory Hessian condition bound is asserted.

Corollary 8.4 (Cost of a dynamic block-norm interface). *For an interface constructed from the formulation, the total cost of setup and answering all B active cone-block norm requests $\|u_j(\eta_j)\|$ to additive error $\xi/50$, with joint success at least $2/3$, is $\Omega(BL_g)$ classical queries. For a precompiled interface whose subsequent requests each cost $O(1)$, this bounds setup cost plus $O(B)$.*

Proof. The norm equals $\rho(\sqrt{1+\xi\Phi_j})$. In this argument $\rho < 2/3$ and $\rho'(z) = (1-\rho^2)^2/(1+\rho^2) \geq 25/117$. Consequently its derivative in Φ_j is at least $25\xi/(117\sqrt{5})$. The promise gap in the norm is greater than $0.056\xi > 2\xi/50$, so Lemma 8.1 applies. $\square$

Free exact iterate SQ with conditional block norms would expose this information. Theorem 8.3 therefore cannot be reinterpreted as a lower bound with that stronger input already supplied. The corollary charges the construction of precisely that interface. A normalized quantum state with its norm withheld is a different output.

8.4 Compiling promised overlaps into one bounded optimizer coordinate

A sum of promised amplitudes need not determine a sum of promise bits: intervals corresponding to different bit patterns can overlap. The following constant-size LP removes that obstacle.

Lemma 8.5 (LP threshold compiler). *Suppose $\varphi \in [-1, 1]$ satisfies $|\varphi| \leq \alpha$ or $\varphi \geq \beta$, with $0 \leq \alpha < \beta \leq 1$. Put $c = (\alpha + \beta)/2$, $\Delta = (\beta - \alpha)/2$, $J = 1/\Delta$, $M_0 = 2J$, and $C_0 = 4J$. The LP*

$$\max C_0 z - M_0 r + t, \quad 0 \leq z \leq 1, \quad 0 \leq r \leq 2, \quad 0 \leq t \leq z, \quad t \leq Jr, \quad r \geq z(\varphi - c) \quad (67)$$

has a unique optimizer with $(z, t) = (1, g)$, where g is the promise bit. Every feasible point has objective gap at least $|t - g|$.

Proof. For a low input the last constraint is redundant. For fixed z , the best t is $\min\{z, Jr\}$ and the penalty coefficient $M_0 = 2J$ uniquely sets $r = t = 0$. Then $z = 1$ is unique. Moreover the gap is $C_0(1 - z) + M_0 r - t \geq t$. For a high input set $d = \varphi - c \geq 1/J$. Feasibility forces $r \geq dz$, so the unique best choices are $r = dz, t = z$. The remaining coefficient $C_0 + 1 - M_0 d$ is positive since $d < 1$. Thus $z = t = 1$ uniquely. For every feasible point, $f \leq (C_0 - M_0 d)z + t$ and $C_0 - M_0 d > 0$, giving $f^* - f \geq 1 - t$. $\square$

For each cyclic block, introduce $M_i y_i = z_i e$ and a public sparse readout chain defining $q_i = a_i^T y_i / h$. These linear equalities enforce $q_i = z_i \Phi_i$. Replace the last constraint of (67) by $r_i \geq q_i - cz_i$. There is no product of two optimization variables: Φ_i is implicit input data, and its product with z_i is implemented by linear equations. Give block i the objective weight $\theta_i > 0$. Set $p_1 = t_1$ and, for $2 \leq i \leq B$, impose

$$\begin{aligned} p_i &\leq p_{i-1} + t_i, & p_i &\geq p_{i-1} - t_i, \\ p_i &\geq t_i - p_{i-1}, & p_i &\leq 2 - p_{i-1} - t_i. \end{aligned} \quad (68)$$

These are the convex hull of Boolean XOR triples. Their projection onto (p_{i-1}, t_i) is the whole unit square, and Boolean inputs uniquely force $p_i = p_{i-1} \oplus t_i$.

Theorem 8.6 (A bounded endpoint scalar). *The compiled LP has a unique optimizer and $P^* := p_B^* = g_1 \oplus \dots \oplus g_B$. Its dimension is $\Theta(BN_0)$, row and column sparsity $O(q)$, and its displayed logarithmic barrier has certified parameter $12B - 4$. Estimating P^* to additive error less than $1/3$ requires $\Omega(BL_g)$ classical full-SQ queries and has coherent source query complexity $\Theta(B)$ for two-function Forrelation. For fixed k , the classical lower is $\tilde{\Omega}_k(BN_0^{1-1/k})$ and the coherent complexity is $\Theta_k(B)$. Every exactly feasible point with objective gap τ satisfies*

$$|p_B - P^*| \leq \frac{\tau}{\min_i \theta_i}. \quad (69)$$

Proof. The XOR constraints leave the projection onto all t_i unchanged. Lemma 8.5 fixes each block's optimizer uniquely; then the XOR variables, inverse histories and accumulators are also unique. There are eight scalar inequalities per threshold block and four per later XOR gate. Strict relative feasibility follows by taking $z_i = 3/4$, $r_i = 3/2$, $t_i = p_i = 1/2$, with the equality variables determined accordingly. The resulting scalar product log barrier has parameter at most the number $12B - 4$ of its logarithms, independent of N_0 .

For $B = 1$, the randomized lower is the one-block lower bound immediately. For $B \geq 2$, it follows from the strong XOR theorem of Brody et al. [13, Theorem 1.1]: $\bar{R}_\epsilon(\text{XOR}_B \circ g^B) = \Theta(B\bar{R}_{\epsilon/B}(g))$ for constant outer error, where the bar denotes worst-input expected query cost. Expected constant-error and worst-case constant-error costs are comparable: truncate after a constant multiple of the worst-input expectation and amplify the resulting worst-input success. Hence the theorem implies $\Omega(BR_{1/3}(g))$ also for worst-case queries. For the coherent upper, jointly recover all promise bits using Buhrman et al. [14, Corollary 3] and take their XOR. This costs $O(B)$ or $2^{O(k)}B$ source queries. For the lower, restrict every block to two fixed inputs having opposite promise bits. One source query on this restriction is simulated by $O(1)$ queries to the selecting outer bit. Thus an algorithm computes parity of B independent bits, which needs $\Omega(B)$ quantum queries [7].

Finally, Lemma 8.5 gives $\tau \geq \sum_i \theta_i |t_i - g_i|$. The XOR facets imply $|p_i - p_{i-1}| \leq t_i$ and $|p_i - (1 - p_{i-1})| \leq 1 - t_i$. Induction, using the appropriate inequality for $g_i = 0$ or 1 , yields $|p_i - (g_1 \oplus \dots \oplus g_i)| \leq \sum_{j \leq i} |t_j - g_j|$, proving (69). $\square$

All new coefficients and supports are public. An old row or column gains only constantly many public entries, so its new SQ law is a public mixture of the old law and those entries. Readout chains keep incidence constant. Thus full formulation SQ has constant source-query overhead. Conversely, hidden signs remain at known matrix entries, so the source upper has matched formulation access. The factor $1/h$ is public; no condition-number bound for the entire compiled equality system is claimed.

With $\theta_i = 1$, constant objective gap $\tau < 1/3$ suffices for the same coordinate lower bound. With $\theta_i = \Gamma^{-i}$, the sufficient gap is $\tau < \Gamma^{-B}/3$. Before adding XOR facets the separated threshold blocks have central problems at multipliers $\eta\theta_i$; a public summation readout $\sum_i t_i$ does not affect this property. The XOR facets couple the barrier, so this rescaling identity is not asserted for the bounded-output formulation. Endpoint hardness does not establish an unavoidable charge at each of B path scales.

The compiler tolerates finite public readout precision. If $b_i = a_i/h$ is replaced by $\tilde{b}_i$ with $\|\tilde{b}_i - b_i\| \leq \Delta/(10R)$, then $\|y_i\| \leq R$ makes the readout error at most $\Delta/10$. Use the narrower effective half-gap $9\Delta/10$ when choosing J, M_0, C_0 . This preserves exact optimizer bits: although a rounded amplitude can slightly exceed one, its high-branch value $d = \tilde{\Phi}_i - c$ is still below one because $c \geq \Delta$, so the proof of Lemma 8.5 applies with the narrower gap. Keeping the old threshold constants need not preserve saturation at the high boundary. This statement rounds only public readouts; rounding M_i requires a separate resolvent budget.

8.5 A parity endpoint and the meaning of temporal hardness

A simpler sparse family illustrates the distinction without an inner quantum separation. Let $\sigma_{i,k} \in \{-1, 1\}$, $1 \leq i \leq B$, $1 \leq k \leq L$, and $h_i = \prod_k \sigma_{i,k}$. Introduce boxed activations $-1 \leq a_i \leq 1$ and chains

$$x_{i,0} = a_i, \quad x_{i,k} = \sigma_{i,k} x_{i,k-1}.$$

A public partial-sum chain defines $W = \sum_i x_{i,L}$. Order all BL signs as $\tau_1, \dots, \tau_{BL}$, and add the carrier $r_0 = a_B$, $r_m = \tau_m r_{m-1}$, $w = r_{BL}$. Maximize $2 \sum_i \Gamma^{-i} a_i$, with $\Gamma = 256$. All equalities have at

most three nonzeros per row and constant column incidence. There are $2B$ inequalities and $\Theta(BL)$ variables. The optimizer is unique, and direct elimination gives

$$W(\eta) = \sum_i h_i \rho(\eta \Gamma^{-i}), \quad w(\eta) = H \rho(\eta \Gamma^{-B}), \quad w^* = H := \prod_i h_i. \quad (70)$$

At $\eta_j = \Gamma^{j+1/2}$, the increment kernel (61) gives $h_j(W(\eta_j) - W(\eta_{j-1})) \geq 2d_0 - 1 \geq 13/16$. Thus the path has B separated constant-margin transitions. At its final checkpoint, $|w(\eta_B) - H| \leq 1/32$, so additive error $1/4$ in this one bounded coordinate computes parity of all BL signs. Both its quantum and classical query complexities are $\Theta(BL)$; reading all signs proves the upper bound and the parity lower proves the lower. All nonzero magnitudes, norms and sampling laws are public, and each queried matrix entry uses at most one source sign. Duplicate occurrences refer to the same fixed input bit. The barrier parameter $2B$ and public $O(B)$ -bit objective scales do not alter this query accounting.

Every feasible point of objective gap τ has $|w - H| = 1 - a_B \leq \tau/(2\Gamma^{-B})$. Equal objective weights remove this exponentially small gap requirement but also remove the scale separation. At $\eta_f = 16B\Gamma^B$, $|W(\eta_f) - \sum_i h_i| \leq 1/32$; additive error $1/4$ in this unnormalized sum also computes parity by rounding. Normalizing by B changes the required accuracy to $O(1/B)$. More generally, if a block bit has an exact coherent evaluator of cost q_0 , amplitude estimation gives its mean to error ϵ using $O(q_0 \min\{B, 1/\epsilon\})$ queries [12]. A mean at constant accuracy therefore need not retain a factor B , whereas the bounded parity carrier does.

The query lower bounds in this section concern specified final outputs or joint transcripts. In a static-oracle model an algorithm required only to return a final relation may compute it without generating an intermediate iterate. A geometric transition is not a new oracle input. A genuine chronological charge needs an online-output, fresh-input, or memory restriction, or a composition theorem for the required final function. Our parity and threshold constructions use the last option; they do not prove per-iteration lower bounds.

The Forrelation, direct-sum, robust recovery and XOR theorems used here are prior work. Optimization scalar encodings of Boolean composition also appear in Apers and Gribling [5]. Our contribution in this section is the explicit sparse realization and its quantitative output contracts: the increment leakage kernel, the ordinary predictor-decrement sensitivity kernel, and the threshold compiler with objective-gap stability. To the best of our knowledge, these particular access-preserving central-path realizations have not appeared in the cited literature; no priority claim is made for their individual query-complexity ingredients.

9 What repeated Newton queries can reuse

The preceding direct sums require independent information in the specified output. A large number of central-path movements alone is insufficient. We first give an explicit sparse KKT counterexample and then quantify one way in which low-dimensional solution structure permits reuse. Neither statement treats residual certification, dense output, or state preparation as free computation.

9.1 A fixed sparse KKT state at every central multiplier

Let $M \in \mathbb{R}^{n \times n}$ be invertible, with bidirectional sparse access, row and column sparsity at most s , and $\|M\| \leq 1$. For the one-cone program (41), write $v = M^{-1}e$, $a = \rho(\eta)$, and $q_a = 1 - a^2$. Retain both $u = Mx$ and the equality multiplier λ .

Proposition 9.1 (A common KKT ray). *At every exact central point, the augmented predictor equations are*

$$\begin{pmatrix} D_a & 0 & I \\ 0 & 0 & -M^T \\ I & -M & 0 \end{pmatrix} \begin{pmatrix} \dot{u} \\ \dot{x} \\ \dot{\lambda} \end{pmatrix} = \begin{pmatrix} 2e \\ 0 \\ 0 \end{pmatrix}, \quad D_a = \frac{2}{q_a}I + \frac{4a^2}{q_a^2}ee^T. \quad (71)$$

Their unique solution and normalized state are

$$(\dot{u}, \dot{x}, \dot{\lambda}) = \frac{q_a^2}{1+a^2}(e, v, 0), \quad \frac{(e, v, 0)}{\sqrt{1+\|v\|^2}}. \quad (72)$$

The KKT matrix is at most $(s+2)$ -sparse, with constant-overhead sparse-access simulation from M and M^T . Thus any one-shot lower bound for preparing $|v\rangle$ at sufficiently small constant state error transfers to each individual KKT checkpoint, up to constants.

Proof. Stationarity in x gives $M^T\lambda = 0$, so $\lambda = 0$. The other stationarity equation is $u/(1-\|u\|^2) = \eta e$; hence $u = ae$ and $x = av$. Differentiating gives $a' = q_a^2/(1+a^2)$ and (71). The second block row forces $\dot{\lambda} = 0$, the third forces $\dot{u} = M\dot{x}$, and $D_a e = 2(1+a^2)e/q_a^2$ proves (72). The remaining entries are public diagonal or identity entries because e is a basis vector. This proves sparsity and oracle simulation. Since $\|v\| \geq 1$, the x register has probability at least $1/2$; measuring that register extracts $|v\rangle$ with constant probability. Sufficiently small constant Euclidean state error perturbs this probability and the conditional state by only constant amounts. $\square$

This is a state-output comparison, not a claim that every scalar derived from that state is hard. It also imposes no uniform bound on the condition number of the augmented KKT matrix near the boundary.

The converse output distinction can be equally sharp. A signed equality chain $r_0 = \alpha$, $r_i = \sigma_i r_{i-1}$, with $|\alpha| \leq 1$ and objective $\alpha + 2r_L$, reduces to maximizing $(1+2H)\alpha$, where $H = \prod_i \sigma_i$. Its optimum is three for $H = 1$ and one for $H = -1$; two-sided relative value error below $1/2$ therefore costs $\Theta(L)$ quantum or classical queries. Yet its nonzero reduced-coordinate solution state lives in a one-dimensional Hilbert space and is input independent up to global phase. The discarded sign or scale is essential to scalar readout. The terminal optimizer coordinate r_L^* is actually one in both cases, so that coordinate cannot serve as the parity readout. The paired interval barrier has parameter one, but this does not supply the hidden reduced objective coefficient without acquiring the chain signs.

The same fixed instance can require arbitrarily many bounded local movements. For $\phi(u) = -\log(1-\|u\|^2)$, direct inversion of its radial Hessian gives

$$\|\nabla\phi(u)\|_{u,*}^2 = \frac{2\|u\|^2}{1+\|u\|^2} \leq 1.$$

Objective gap at most $0 < \epsilon \leq 1$ implies $e^T u \geq 1 - \epsilon/2$, so $1 - \|u\|^2 \leq \epsilon$ and $\phi(u) - \phi(0) \geq \log(1/\epsilon)$. The gradient bound makes the barrier-metric length of every interior path from the center to such a point at least $\log(1/\epsilon)$. A chord with starting Dikin norm at most $R < 1$ has length at most $\log(1/(1-R))$. One can see this directly for the ball: along a line $\phi(u+td)$ is the sum of two negative logarithms of affine factors. If $h^2 = d^T \nabla^2 \phi(u) d$, the absolute logarithmic slopes at zero have squared sum h^2 ; at $t \geq 0$ their denominators are at least $1-th$. Therefore the metric speed is at most $h/(1-th)$, whose integral from zero to one is $-\log(1-h)$. A round containing at most m such chords consequently obeys

$$\#\text{rounds} \geq \frac{\log(1/\epsilon)}{m \log(1/(1-R))}. \quad (73)$$

Pullback by M preserves this barrier metric on the equality slice. Equation (73) concerns this explicit primal barrier and bounded-Dikin chords; it is not an arbitrary algorithmic iteration lower bound.

Lemma 9.2 (An elementary hard ray). *There are four-sparse matrices M_σ , of dimension $\Theta(L)$ and condition number $\Theta(L)$, for which preparing $|M_\sigma^{-1}e\rangle$ to sufficiently small constant state error has quantum query complexity $\Theta(L)$. One classical read-and-solve step also uses $O(L)$ queries and arithmetic. These matrices may be used in Proposition 9.1.*

Proof. Take a path $p_0, \dots, p_L$ with hidden edge signs $\sigma_1, \dots, \sigma_L$. Attach a complete public binary tree with $h = 2^{\lceil \log_2 L \rceil}$ leaves at p_0 , and a disjoint copy at p_L . For every nonroot node z impose $y_z - s_z y_{\pi(z)} = 0$, where s_z is the appropriate hidden path sign or the public value one; impose $y_{p_0} = 1$. These equations define a triangular matrix A_σ with row sparsity two and column sparsity four. Its dimension is $N_L = L + 4h - 3$, and $\|A\| \leq \sqrt{\|A\|_1 \|A\|_\infty} \leq \sqrt{8}$. Set $M = A/\sqrt{8}$. The unscaled solution $A^{-1}e$ is one on the first tree and $H = \prod_i \sigma_i$ on the second, with prefix parities on the path. The solution $M^{-1}e = \sqrt{8}A^{-1}e$ has the same normalized state, since its common public factor cancels. This state has constant probability $2h/N_L \geq 2/5$ in the span of the two public uniform leaf states. Measuring their sum and difference identifies H whenever that subspace is observed. A constant number of sufficiently accurate preparations therefore computes parity, which needs $\Omega(L)$ queries [7]. Reading all signs and forward substitution gives the matching upper bound.

Every entry of A^{-1} is zero or a signed ancestor indicator. Its absolute row and column sums are $O(L)$, so $\|A^{-1}\| = O(L)$. The path submatrix of A^{-1} is, after diagonal sign changes, the cumulative-sum matrix; applying it to the uniform vector shows norm $\Omega(L)$. Since $1 \leq \|A\| = O(1)$, the condition is $\Theta(L)$. The augmented KKT graph also has a width-three tree decomposition: for each nonroot z use the bag $\{\lambda_z, u_z, x_z, x_{\pi(z)}\}$ and join bags along the dependency tree; omit the absent parent variable at the root. $\square$

After one $O(L)$ -query solve, the stored vector v determines every center and predictor in (72) by public scalar rescaling, regardless of the movement count in (73). Implicit classical output then needs only a scalar and the stored vector. Repeated dense writes or quantum state preparations still cost work, but no further raw input query is required. More generally, reading all $O(ns)$ addressable nonzero coefficient words of a fixed sparse matrix gives a finite cap on subsequent coefficient-query complexity. This cap applies to outputs mathematically determined by those words, with arbitrary storage and computation. It does not reconstruct unspecified behavior of an arbitrary unitary completion, nor bound arithmetic, gate, memory, or bit complexity.

The same reuse is directly relevant to scalar output. For any fixed public readout c , one has $c^T x(\eta) = \rho(\eta) c^T M^{-1}e$. One additive- ϵ estimate of $c^T M^{-1}e$ therefore supplies the entire sequence of these scalar centers to additive error ϵ , since $0 \leq \rho(\eta) < 1$. Choosing the plateau readout of Section 7 makes this one scalar classically hard; its reuse needs no transfer from a state-output lower bound.

The sharper one-shot state lower $\Omega(\kappa \log(1/\epsilon_{\text{state}}))$ of Mori et al. [33, Theorem 1] holds for a constant-sparsity family under coherent sparse position/value access, with $0 < \epsilon_{\text{state}} \leq 1/11$ measured in Euclidean distance between normalized solution states; right-hand-side preparation cost is treated as negligible. This concerns a different output from our scalar inverse forms. Its existence cannot justify multiplying one-shot hardness by a movement count. The elementary family above already disproves that general inference without relying on a precision-sensitive state-preparation theorem.

9.2 Residual-certified span reuse

Consider any sequence of nonsingular systems $H_t z_t = f_t \neq 0$ and unit rays $\psi_t = z_t / \|z_t\|$. Maintain a span of stored checkpoint vectors. At time t , minimize the *current* relative residual $\|H_t y - f_t\| / \|f_t\|$ over this span. If the residual is too large, solve the system once and append its ray. Every use of an old vector requires its image under the current operator; it is not assumed that old residuals remain valid.

Proposition 9.3 (Exact span). *If the full sequence of solution rays has span dimension r , exact residual minimization and exact checkpoint rays require at most r refresh solves, for any acceptance threshold in $(0, 1)$. This is sharp.*

Proof. If the current ray belongs to the stored span, that span contains the exact solution and its minimum residual is zero. Each rejection therefore adds a linearly independent ray. For sharpness take $H_t = I$, $f_t = e_t$, $t = 1, \dots, r$; every not-yet-stored vector has relative residual one. $\square$

For a robust version, set $A_t = (\|z_t\| / \|f_t\|) H_t$, so $A_t \psi_t = f_t / \|f_t\|$, and assume $\|A_t\| \leq B_0$. Necessarily $B_0 \geq 1$. These normalized operators are only for analysis; the algorithm can minimize the original relative residual and need not know $\|z_t\|$. Let $0 < \eta \leq 1$ be the desired residual scale. Suppose minimization returns a candidate with residual at most $\eta/8$ above the true minimum, and the test estimates that candidate's residual to additive error $\eta/8$. Accept if the reported value is at most $3\eta/4$. An accepted candidate has true relative residual at most $7\eta/8$, whereas rejection implies minimum residual greater than $\eta/2$.

Theorem 9.4 (Numerical span with normalized checkpoints). *Let D be the system dimension and let $1 \leq r \leq D$ be an integer. Suppose every refreshed stored vector $\hat{\psi}_t$ has norm one and $\|\hat{\psi}_t - \psi_t\| \leq \xi$. Define the Kolmogorov width*

$$d_r = \inf_{\dim S=r} \sup_t \text{dist}(\psi_t, S), \quad \Delta = \frac{\eta}{2B_0} - \xi > 0.$$

If

$$d_r + \xi < \frac{\Delta^{2-1/r}}{2r}, \tag{74}$$

then the residual procedure makes at most $2r - 1$ refreshes, on the event that all stated minimization, test and checkpoint errors hold.

Proof. For a stored span V , its best relative residual is at most $B_0 \text{dist}(\psi_t, V)$. Rejection therefore implies $\text{dist}(\psi_t, V) > \eta/(2B_0)$, and the next approximate stored vector has distance greater than Δ from V . Suppose $2r$ refreshes occur and form a matrix Y from their normalized columns. Its first QR diagonal has magnitude one and all later diagonals exceed Δ , giving

$$\sqrt{\det(Y^* Y)} > \Delta^{2r-1}.$$

Choose an r -space S with all column distances at most $d_r + \xi + \epsilon$, where $\epsilon > 0$ is arbitrary. The first r singular values of Y are at most $\sqrt{2r}$ since its columns have norm one. The remaining r are at most $\|(I - \Pi_S)Y\| \leq \sqrt{2r}(d_r + \xi + \epsilon)$, since $\Pi_S Y$ has rank at most r . Thus

$$\sqrt{\det(Y^* Y)} \leq (2r)^r (d_r + \xi + \epsilon)^r.$$

Letting $\epsilon \downarrow 0$ contradicts (74). Normalization of every stored column is used in this volume bound. $\square$

There is a stronger bound under a correspondingly stronger stability assumption. Suppose the exact rays have rank r , and whenever a new exact ray is dependent on *the exact versions of the previously selected checkpoint rays*, it has a representation in those rays whose coefficients have sum of absolute values at most Λ . If $B_0\Lambda\xi \leq \eta/2$, replacing that representation by the stored approximate rays has relative residual at most $\eta/2$. It cannot trigger a rejection. Hence at most r refreshes occur. Stability in an unspecified final basis is insufficient for this adaptive argument. This condition permits checkpoint error linear in η ; any resulting tomography complexity must still use the cost law of the particular tomography interface.

These statements specialize the established ideas of subspace recycling and greedy reduced bases [9, 37] to a residual-certified checkpoint contract. They give refresh counts, not unconditional quantum speedups. With explicit vectors of dimension D and R retained checkpoints, forming current images and solving their least-squares problem costs, for example, $O(R \text{mv}(H_t) + DR^2 + R^3)$ arithmetic per test. Quantum tests have their own input preparation, normalization and coefficient-conditioning costs. If the systems are expressed in a linear nullspace representation enforcing homogeneous feasibility equations, span combinations preserve those equations. For an affine feasibility constraint one must instead retain a particular feasible point and combine homogeneous directions. Neither observation supplies cone-scaling access for free.

9.3 When low numerical rank is justified, and what it does not buy

A useful sufficient regularity condition is holomorphic dependence. Suppose a unit-ray curve on $[-1, 1]$ extends holomorphically to a Bernstein ellipse of parameter $\chi > 1$, continuously to its boundary, with norm at most W . Its vector-valued Chebyshev coefficients satisfy $\|c_j\| \leq 2W\chi^{-j}$ for $j \geq 1$, by the usual contour integral on the ellipse. The degree- $(r-1)$ truncation lies in the span of r coefficient vectors, and its tail is bounded by

$$d_r \leq \sum_{j=r}^{\infty} 2W\chi^{-j} = \frac{2W\chi^{-(r-1)}}{\chi - 1}. \quad (75)$$

Thus a uniform ellipse and bound give a logarithmic numerical-rank scale. They are actual hypotheses, not consequences of strict complementarity. For example, the strictly complementary separable QP $\min_{x \geq 0} \frac{1}{2} \|x\|^2 + c^T x$, with distinct $c_i > 0$, has barrier central coordinates and predictors

$$x_i(\tau) = \frac{-c_i + \sqrt{c_i^2 + 4\tau}}{2}, \quad x'_i(\tau) = \frac{1}{\sqrt{c_i^2 + 4\tau}}.$$

The coordinate functions of the predictor are linearly independent: analytic continuation of a vanishing linear combination to a neighborhood of $-c_i^2/4$ forces its i th coefficient to vanish, since only that summand has that branch singularity. The normalized predictor rays therefore span the full dimension; common normalization does not change linear relations among coordinate functions on the real interval. Nor is there a uniform complex neighborhood as $\min_i c_i \downarrow 0$. For a normalized-ray example in dimension two, choose $c_2 = 2c_1$. The squared first coordinate of the normalized predictor is $(c_2^2 + 4\tau)/(c_1^2 + c_2^2 + 8\tau)$, with a nonremovable pole tending to zero from below. A uniform Bernstein ellipse for normalized rays on $[0, 1]$ is therefore impossible in this family.

Low rank alone also does not make certification cheap. With coefficient queries, take $H_0 = H_1 = I_D$, $f_0 = e_0$, and

$$f_1 = e_0 + \sum_{i=1}^{D-1} a_i e_i, \quad a_i \in \{0, 1\}, \quad \sum_i a_i \in \{0, 1\}.$$

The solution span has dimension at most two and $B_0 = 1$. The minimum relative residual from $\text{span}\{e_0\}$ is zero or $1/\sqrt{2}$. Any test separating these cases solves unique search, requiring $\Omega(D)$ randomized or $\Omega(\sqrt{D})$ quantum queries. For completeness, under a uniform marked location a Q -query randomized algorithm hits it with probability at most $Q/(D-1)$ until the transcript changes; the quantum hybrid bound changes the average final-state distance by at most $O(Q/\sqrt{D})$. An exact norm oracle for f_1 would reveal the case immediately, which is why the coefficient-query input is explicit here.

Even a single stored ray does not eliminate explicit-output cost. For $H = I$ and $f_b = D^{-1/2}((-1)^{b_1}, \dots, (-1)^{b_D})$, a phase query implemented by the canonical bit oracle prepares the amplitude state. The input for the following lower bound is this bit/value oracle; a phase-only unitary would not expose the overall sign. A classical vector within Euclidean error $1/4$ determines all but at most $D/16$ signs by rounding its coordinates. This requires $\Omega(D)$ classical queries and $\Omega(D)$ writes; its quantum query cost is also $\Omega(D)$. To see the latter without assuming exact recovery, a Q -query final state, as the signs vary, lies in a space of dimension at most $\sum_{j=0}^Q \binom{D}{j}$ by the multilinear query expansion [7]. Any one rounded output is correct for at most $V = \sum_{j \leq D/16} \binom{D}{j} \leq 2^{H_2(1/16)D}$ inputs. For any measurement, its average success on uniform inputs is at most $V \sum_{j \leq Q} \binom{D}{j} / 2^D$: bound each state's density matrix by the projector onto this common space and sum the measurement operators. Constant success consequently requires $Q = \Omega(D)$. Here H_2 is binary entropy. The argument separates inexpensive state preparation from a classical vector, even when the same rank-one target is requested at every checkpoint.

10 Structured cone systems and scalar estimation

A dense cone Hessian need not make its inverse forms difficult. We give two classical comparisons. The first uses a sparse positive definite base and a small number of corrections, with sampling access to the correction *source* vectors. It produces a scalar estimate or a solution sampler without materializing a dense Newton direction. The second uses an explicit factorization and returns a full direction. These are different output contracts and require different structural parameters.

The low-rank cone identities and their use in sparse factorizations are established techniques [3, 17, 24]. Dimension-independent sample-query matrix arithmetic is also established [18]. The result below combines a full-rank sparse base with possibly indefinite corrections. Its explicit access and error analysis is needed because sampling a nearly zero transformed correction column can be arbitrarily expensive. We never normalize such a column.

10.1 A sparse base with sampled low-rank corrections

We work over $\mathbb{R}$ in this section. Suppose $C \in \mathbb{R}^{m \times n}$ has row and column sparsity at most d , and public spectral bounds give

$$S = CC^T, \quad \kappa^{-1}I \preceq S \preceq I, \quad N = S + CXLX^TC^T, \quad aS \preceq N \preceq \beta S. \quad (76)$$

Here $0 < a \leq 1 \leq \beta$, $X \in \mathbb{R}^{n \times r}$ satisfies $\|X\| \leq 1$, and the explicitly supplied $r \times r$ symmetric matrix L obeys $\|L\| \leq \ell$, where $\ell \geq 1$. Neither invertibility nor a sign condition is imposed on L . Each column has a representation $X_j = T_j v_j$, with $\|v_j\| = 1$, supplied $\text{SQ}(v_j)$, and a known contraction T_j having at most one entry per row and column. Each T_j has constant-cost row and column sparse count, location, and value access, including reporting an absent entry, as in Definition 2.1. This includes diagonal maps and coordinate embeddings. The norm of X_j , or of CX_j , is not supplied. The nonzero right-hand side b has $\text{SQ}(b)$; scaling permits $\|b\| = 1$ throughout the proof.

Theorem 10.1 (Sparse base and low-rank corrections). *Under (76), row and column sparse access to C and the stated vector access suffice for a randomized setup with a good event of probability at least $1 - \zeta$. On that event its coordinate oracle represents one fixed vector $\tilde{y}$ with*

$$\|\tilde{y} - N^{-1}b\| \leq \epsilon \|N^{-1}b\|.$$

Each requested draw either returns a failure flag or an exact sample from $p_{\tilde{y}}$ conditional on not failing. On the good setup event its flag probability is at most any prescribed $\delta \in (0, 1/2)$. Setup, coordinate queries, and draws all have bounded cost, including on bad setup outcomes. With $R_ = 2 + (r + 1)\kappa\beta\ell/a$, a common bound is*

$$(d + 1)^{O(\sqrt{\kappa} \log(R_*/\epsilon))} \text{poly}(R_*, \epsilon^{-1}, \log(1/\zeta), \log(1/\delta)). \quad (77)$$

The draw law is asserted on the good setup event; bad setups count toward ζ . No exact output-norm oracle is asserted. The case $r = 0$ is Theorem 3.7, with capped rejection.

Proof. Put $D_* = 1 + \ell/a$ and $U_* = D_*\ell$. The matrices

$$K = X^T C^T S^{-1} C X, \quad h = X^T C^T S^{-1} b, \quad J = (I + LK)^{-1}$$

obey $0 \preceq K \preceq I$ and $\|h\| \leq \sqrt{\kappa}$. Indeed, $C^T S^{-1} C$ is an orthogonal projection. The identity

$$J = I - L X^T C^T N^{-1} C X, \quad \|J\| \leq D_* \quad (78)$$

follows by multiplying by $I + LK$; the norm estimate uses $N^{-1} \preceq a^{-1} S^{-1}$. Thus Woodbury's identity does not require L^{-1} and gives

$$z = J L h, \quad \|z\| \leq U_* \sqrt{\kappa}, \quad y = N^{-1} b = S^{-1} b - S^{-1} C X z.$$

Use Lemma 3.1 to obtain a polynomial $P = P(S)$ with $\|I - SP\| \leq \eta \leq 1$. Functional calculus gives

$$\|P\| \leq 2\kappa, \quad \|PC\| \leq 2\sqrt{\kappa}, \quad \|C^T PC\| \leq 2. \quad (79)$$

For example, writing $C = S^{1/2} V$ with $V V^T = I$ proves all three bounds. Let $K_P = X^T C^T P C X$ and $h_P = X^T C^T P b$. Their biases satisfy $\|K_P - K\| \leq \eta$ and $\|h_P - h\| \leq \eta\sqrt{\kappa}$.

Each entry of K_P is estimated by sampling from v_i and evaluating $T_i^T C^T P C T_j v_j$ locally. Specifically, if $I \sim p_{v_i}$, the variable $(T_i^T C^T P C T_j v_j)_I / (v_i)_I$ has the desired expectation and second moment at most 4; zero sampled denominators have probability zero. For $(h_P)_i$ use $T_i^T C^T P b$ instead; its second moment is at most 4κ . Median-of-means estimates with entry errors $\rho/(2r)$ and $\nu/(2\sqrt{r})$, respectively, give

$$\|\hat{K} - K\| \leq \rho, \quad \|\hat{h} - h\| \leq \nu$$

with joint probability $1 - \zeta$, provided $\eta \leq \rho/2$ and $\eta\sqrt{\kappa} \leq \nu/2$. One may symmetrize $\hat{K}$ without increasing its error. The total number of local evaluations is

$$O\left[\left(\frac{r^4}{\rho^2} + \frac{r^2\kappa}{\nu^2}\right) \log \frac{r+1}{\zeta}\right]. \quad (80)$$

All these estimates start at the supplied vectors, not at CX_j .

If $U_*\rho \leq 1/2$, perturbing the small inverse gives $\|(I + L\hat{K})^{-1}\| \leq 2D_*$ and, for $\hat{z} = (I + L\hat{K})^{-1} L\hat{h}$,

$$\|\hat{z} - z\| \leq 2U_*(\nu + \rho U_* \sqrt{\kappa}).$$

The fixed output $\tilde{y} = Pb - PCX\hat{z}$ consequently satisfies

$$\|\tilde{y} - y\| \leq \eta\kappa(1 + U_*) + 4U_*\sqrt{\kappa}\nu + 4U_*^2\kappa\rho. \quad (81)$$

For completeness, the polynomial bias in $S^{-1}b$ is at most $\eta\kappa$, that in $S^{-1}CXz$ is at most $\eta\sqrt{\kappa}\|z\|$, and the remaining term is bounded using (79). The explicit choices

$$\begin{aligned} \rho &= \frac{\epsilon}{48\beta U_*^2\kappa}, & \nu &= \frac{\epsilon}{48\beta U_*\sqrt{\kappa}}, \\ \eta &= \min \left\{ \frac{\epsilon}{12\beta\kappa(1 + U_*)}, \frac{\rho}{2}, \frac{\nu}{2\sqrt{\kappa}} \right\} \end{aligned}$$

make (81) at most $\epsilon/(4\beta)$. Since $N \preceq \beta I$, $\|y\| \geq 1/\beta$, which proves the stated relative error with room to spare. On the good event, $\|\hat{z}\| \leq 4U_*\sqrt{\kappa}$.

It remains to sample without division by small component norms. The raw trial in Lemma 3.6 applies also to a rectangular or singular transform T : if its rows have at most W entries and $B_T \geq \|T\|$, its output probability at coordinate i in one trial is

$$\frac{|(Tv)_i|^2}{WB_T^2\|v\|^2}. \quad (82)$$

The proof uses only the column norm upper bound and row Cauchy–Schwarz inequality, so it needs no lower singular-value bound. Choose a common $W = (d+1)^{O(\sqrt{\kappa}\log(R_*/\epsilon))}$ bounding both row and column support sizes and materialization costs of all needed transforms. Sparse walks through CC^T , and the one-entry maps T_j , give such a bound. Put $w_0 = Pb$, $w_j = PCT_jv_j$, $c_0 = 1$ and $c_j = -\hat{z}_j$. Take raw denominators $D_0 = 4W\kappa^2$ and $D_j = 4W\kappa$ for $j \geq 1$, and $Z = \sum_{j=0}^r c_j^2 D_j$. Select j with probability $c_j^2 D_j/Z$ and perform one raw trial for w_j . If it returns i , accept again with probability

$$\frac{|\sum_{j=0}^r c_j w_{j,i}|^2}{(r+1) \sum_{j=0}^r |c_j w_{j,i}|^2}.$$

The denominator is positive whenever this step is reached. Summing over j shows that the final raw probability is $|\tilde{y}_i|^2/((r+1)Z)$. On the good setup event,

$$Z \leq 4W\kappa^2(1 + 16U_*^2), \quad \|\tilde{y}\| \geq \frac{1}{2\beta}.$$

Thus the total acceptance probability is at least $\pi_* = [16(r+1)W\beta^2\kappa^2(1 + 16U_*^2)]^{-1}$.

To make runtime unconditional, setup returns a failure flag if its small core is singular, if the Frobenius norm of its inverse exceeds $2\sqrt{r}D_*$, or if $\|\hat{z}\| > 4U_*\sqrt{\kappa}$. These checks do not reject a good setup and use small dense arithmetic. Every draw is capped at $\lceil \pi_*^{-1} \log(1/\delta) \rceil$ trials. Failure of all trials returns a flag. Independent identical trials ensure that truncation leaves the conditional accepted law unchanged. On a bad setup the loop still terminates, even if $\tilde{y} = 0$. Each trial costs $O((r+1)W)$; (80), the small solve, and these caps prove (77). For nonunit b , scale the fixed vector back by its known norm; its sampling law is unchanged. $\square$

Corollary 10.2 (Structured positive inverse forms). *Under the same hypotheses, $q = b^T N^{-1}b$ admits a relative- ϵ estimate with failure probability at most ζ and cost of the form (77). This statement uses only setup and coordinate queries, so it has no per-draw failure parameter.*

Proof. Normalize b to unit norm. Then $q \geq 1/\beta$ and $\|N^{-1}b\| \leq \kappa/a$. Run the preceding setup with vector error $\epsilon a/(4\beta\kappa)$ and failure budget $\zeta/2$. Its scalar bias is at most $\epsilon/(4\beta)$. Independently sample $I \sim p_b$ and average $\tilde{y}_I/b_I$ by median of means. Conditionally on good setup its second moment is at most $\|\tilde{y}\|^2 \leq 4\kappa^2/a^2$. Thus $O(\beta^2\kappa^2a^{-2}\epsilon^{-2}\log(1/\zeta))$ coordinate evaluations achieve additional error $\epsilon/(2\beta)$. A union bound proves the claim. No positivity of the approximate linear map, or noncancellation promise for its individual correction terms, is used. $\square$

10.2 Lorentz and generalized-power barriers

At an interior Lorentz point (t, z) with $t > \|z\|$, let $H = \nabla^2 F_L(t, z)$ for $F_L(t, z) = -\log(t^2 - \|z\|^2)$. Put $r_z = \|z\|$, $s = t^2 - r_z^2 > 0$, $d_x = s/2$, and $\chi = (t + r_z)/(t - r_z)$. If $r_z > 0$, put $u_\pm = 2^{-1/2}(1, \pm z/r_z)$. Differentiation and inversion give

$$H^{-1} = d_x I + r_z(t + r_z)u_+u_+^T - r_z(t - r_z)u_-u_-^T, \quad \chi^{-1}d_x I \preceq H^{-1} \preceq \chi d_x I. \quad (83)$$

Indeed, the eigenvalues are $(t + r_z)^2/2$, $(t - r_z)^2/2$ and $s/2$ on the two radial eigendirections and their orthogonal complement. At $z = 0$ the correction vanishes. For L_c blocks, let $\mathcal{D} = \bigoplus d_x I$, $C = \mathcal{A}\mathcal{D}^{1/2}$ and $S = CC^T$. Use the orthonormal columns $(1, 0)$ and $(0, z/r_z)$ in each block for X . The normalized inverse correction L is explicit, block diagonal, and has norm at most $\chi_{\max} - 1$. Thus $r \leq 2L_c$, $a = \chi_{\max}^{-1}$ and $\beta = \chi_{\max}$ suffice in (76). Supplied block norms and $\text{SQ}(z)$ give all required source access. A known positive scaling of S normalizes its spectral upper bound to one and scales N and the desired output accordingly. This proves the Lorentz instance of both preceding results.

For generalized-power cones, let $x_i > 0$, $\alpha_i > 0$, $\sum_{i=1}^{m_x} \alpha_i = 1$, and

$$p = \prod_i x_i^{2\alpha_i}, \quad s = p - \|z\|^2 > 0, \quad \lambda = p/s, \quad 1 \leq \lambda \leq \Lambda.$$

The barrier

$$F(x, z) = -\log s - \sum_i (1 - \alpha_i) \log x_i \quad (84)$$

has parameter $m_x + 1$ [38]. Its use and low-rank Hessian structure precede this work [17]; closed-form inverse-Hessian operators are given by Kapelevich et al. [26]. Set $k_i = 1 + (2\lambda - 1)\alpha_i$ and $D_0 = \text{diag}(k_i/x_i^2, (2/s)I)$. With $c = \|\alpha\|^2$, write the two disjointly supported columns

$$X_1 = (\alpha_i/(\sqrt{c}\sqrt{k_i}))_i \oplus 0, \quad X_2 = 0 \oplus z/\|z\|.$$

At $z = 0$, omit the correction: $\lambda = 1$ and $H = D_0$. Otherwise direct differentiation of (84) gives

$$H = D_0^{1/2}(I + XBXT)D_0^{1/2}, \quad B = \begin{pmatrix} 4\lambda(\lambda - 1)c & -4\lambda\sqrt{c(\lambda - 1)/2} \\ -4\lambda\sqrt{c(\lambda - 1)/2} & 2(\lambda - 1) \end{pmatrix}. \quad (85)$$

For example, before normalization the correction is $4\lambda(\lambda - 1)aa^T - (4\lambda/s)(az^T + za^T) + (4/s^2)zz^T$, where $a_i = \alpha_i/x_i$ and the two groups are embedded separately. This verifies both the signs and the diagonal base.

Lemma 10.3 (Power-cone comparison and estimated core). *The inverse Hessian satisfies*

$$(4\lambda)^{-1}D_0^{-1} \preceq H^{-1} \preceq 4\lambda D_0^{-1}. \quad (86)$$

In $H^{-1} = D_0^{-1/2}(I + XL(t)X^T)D_0^{-1/2}$, the symmetric coefficient matrix is

$$t = \sum_i \frac{\alpha_i^2}{ck_i}, \quad L(t) = -(I + B \operatorname{diag}(t, 1))^{-1}B.$$

For every t in the interval

$$\mathcal{I}_\lambda = [1/(2\lambda), \min\{1, 1/(2\lambda c)\}], \quad (87)$$

this formula is defined, $\|L(t)\| \leq 10\Lambda^2$, and $\|L(t') - L(t)\| \leq 100\Lambda^4|t' - t|$. The true t belongs to this interval and can be estimated to error u with $O(u^{-2} \log(1/\zeta))$ samples from p_α .

Proof. Since $1 \leq k_i \leq 2\lambda$ and $k_i \geq 2\lambda\alpha_i$, $t \geq 1/(2\lambda)$ and $ct \leq 1/(2\lambda)$, proving the interval claim. In the orthonormal basis obtained by normalizing the two column directions of X , the normalized Hessian is

$$Q(t) = \begin{pmatrix} 1 + 4\lambda(\lambda - 1)ct & -4\lambda\sqrt{ct(\lambda - 1)/2} \\ -4\lambda\sqrt{ct(\lambda - 1)/2} & 2\lambda - 1 \end{pmatrix}.$$

Its trace is at most 4λ and its determinant is $2\lambda - 1 - 4\lambda(\lambda - 1)ct \geq 1$ on (87). Its leading diagonal is positive, so both eigenvalues lie in $[1/(4\lambda), 4\lambda]$. On the orthogonal complement the matrix is identity. This proves (86). For $G = \operatorname{diag}(t, 1)$,

$$L(t) = G^{-1/2}(Q(t)^{-1} - I)G^{-1/2}.$$

This identity proves symmetry and gives $\|L(t)\| \leq 2\lambda(4\lambda + 1) \leq 10\Lambda^2$. Differentiation of the inverse formula gives $L'(t) = L(t) \operatorname{diag}(1, 0)L(t)$, hence $\|L'(t)\| \leq 100\Lambda^4$. Finally, t is the expectation of $1/k_I \in [0, 1]$ for $I \sim p_\alpha$. Hoeffding's inequality proves the sample bound. Clipping an estimate to (87) never increases its error. Clipping merely to $[1/(2\Lambda), 1]$ would not preserve the determinant bound and is insufficient. $\square$

Corollary 10.4 (Conditional cone-system bound). *Let A have full row rank and row and column sparsity d , with constant-cost row and column sparse count, location, and value access, a public block partition and constant-cost conversion between block-local and global indices. For L_c Lorentz blocks assume supplied $\operatorname{SQ}(z)$ separately for each block and queryable axial coordinates; for L_c generalized-power blocks assume supplied $\operatorname{SQ}(\alpha)$ and $\operatorname{SQ}(z)$ separately for each block, coordinate queries to x , and each block's barrier scalars p, s . These vector interfaces include each block's norm, rather than merely the norm and sampling law of a concatenated vector. Radial-vector norm and coordinate access are available also when $z = 0$; sampling is requested only for nonzero radial blocks, and a zero block's radial correction is omitted. Supply $\operatorname{SQ}(b)$ and spectral bounds normalizing $S = ADA^T$ to $[\kappa_S^{-1}, 1]$, where $\mathcal{D}$ is respectively the scalar Lorentz base or $\bigoplus D_0^{-1}$. For $N = AH^{-1}A^T$, relative solution approximation, its capped sampler, and relative estimation of $b^T N^{-1}b$ have cost*

$$(d + 1)^{O(\sqrt{\kappa_S} \log(2L_c \Gamma \kappa_S / \epsilon))} \operatorname{poly}(L_c, \Gamma, \kappa_S, \epsilon^{-1}, \log(1/\zeta), \log(1/\delta)), \quad (88)$$

where $\Gamma = \max \chi$ for Lorentz cones and $\Gamma = \Lambda$ for power cones. The setup and draw contracts are those of Theorem 10.1; scalar estimation omits δ .

Proof. The Lorentz case has already been reduced to Theorem 10.1. For power cones take $C = A\mathcal{D}^{1/2}$. The columns in (85) are diagonal contractions of $\alpha/\|\alpha\|$ and embeddings of $z/\|z\|$; their disjoint supports across and within blocks give $\|X\| \leq 1$. They do not require the exact norm of X_1 . Lemma 10.3 gives $a = 1/(4\Lambda)$, $\beta = 4\Lambda$, $\ell = 10\Lambda^2$ and $r \leq 2L_c$. Only L requires approximation. Estimate each t and clip as in (87), obtaining a fixed block diagonal $\hat{L}$. If $\|\hat{L} - L\| \leq v$, then

$$\|S^{-1/2}(\hat{N} - N)S^{-1/2}\| \leq v, \quad \hat{N} = S + CX\hat{L}X^TC^T.$$

For $v \leq a/2$, therefore, $(a/2)S \preceq \hat{N} \preceq (\beta + a/2)S$. The resolvent identity also gives, for every b ,

$$\|\hat{N}^{-1}b - N^{-1}b\| \leq \|\hat{N}^{-1}\| \|\hat{N} - N\| \|N^{-1}b\| \leq \frac{2\kappa_S v}{a} \|N^{-1}b\|.$$

Take $v \leq \min\{a/2, \epsilon a/(8\kappa_S)\}$, and run the theorem on $\hat{N}$ with tolerance $\epsilon/4$. Since $\epsilon \leq 1/2$, the resulting total relative error is less than ϵ . The estimates of t use accuracy $v/(100\Lambda^4)$ and joint failure budget $\zeta/2$; the remaining setup uses $\zeta/2$. The lemma bounds $\|\hat{L}\| \leq 10\Lambda^2$ even on inaccurate estimates. These are polynomial costs, and the logarithm of every required tolerance has the order asserted in (88). For the scalar contract, run this vector construction at tolerance $\epsilon a/(4\beta\kappa_S)$ and apply the independent estimator from Corollary 10.2. $\square$

This is a one-system oracle theorem. For fixed d, L_c, Γ, κ_S and accuracy, growing the cone dimensions alone does not force a polynomial query cost for these outputs. Maintaining the supplied iterate access, forming b , or keeping the parameters bounded along a path is not included. The cyclic one-cone example in Section 7 has a growing sparse-base condition number and hence does not contradict the theorem. Standard three-dimensional cone blocks instead give bounded block sparsity directly, so Section 3 already applies when the resulting reduced system has the stated spectral bounds.

10.3 The power-cone scalar oracle cannot be omitted

The supplied value p is substantive. Fix uniform weights on $2M$ positive coordinates, grouped into pairs. Encode a hidden bit h_j by choosing pair (a, b) or (c, d) , where $a^2 + b^2 = c^2 + d^2$ and $ab \neq cd$. The norm is public. A squared-coordinate sample is simulated by choosing a uniform pair and querying its one bit; coordinate queries also need one bit. For the quantum comparison, fix the canonical coherent value and preparation interfaces: prepare a uniform pair index, query its hidden bit, apply the corresponding public two-coordinate rotation, and uncompute the bit. Equal pair norms make this a preparation of the normalized input vector. Coherent value queries, the preparation unitary and its inverse, and their controlled versions all use $O(1)$ hidden-bit queries. This specifies a particular unitary completion; no claim is made about arbitrary completions of classical SQ access. However,

$$p = [(ab)^{M-|h|}(cd)^{|h|}]^{1/M}$$

determines $|h|$ exactly and therefore its parity. The classical and quantum bounded-error parity lower bounds give $\Omega(M)$ queries for exact acquisition of p under the classical vector access or its specified canonical coherent version. There is also a constant-relative obstruction without a bound on the logarithmic range. Take

$$(a, b) = (\sqrt{1 - e^{-2\gamma M}}, e^{-\gamma M}), \quad (c, d) = (1/\sqrt{2}, 1/\sqrt{2}), \quad \gamma = 4.$$

Both squared pair norms equal one, and adjacent possible values of p have ratio

$$[(1/2)/(e^{-4M}\sqrt{1 - e^{-8M}})]^{1/M} > e^4/2 > 3.$$

A relative error less than $1/2$ therefore still determines the integer $|h|$ and its parity. The same canonical interface simulation applies. This proves the same linear lower bound, in ideal real arithmetic; it uses a logarithmic coordinate range growing with M . With a known bounded range of $\log x_i$, estimating its weighted mean can instead estimate p . Sampling weights proportional to α_i suffices, but ordinary SQ(α) samples proportional to α_i^2 and does not automatically supply that

oracle. Uniform weights permit uniform index sampling. These costs are part of acquiring the input to Corollary 10.4.

The aggregated logarithm cone illustrates the limit of algebra alone. For $s = v \sum_i \log(w_i/v) - u > 0$, $v, w_i > 0$, the Hessian of $-\log s - \log v - \sum_i \log w_i$ is diagonal plus rank at most three. To see this, let $h = (0, 0, (1/w_i)_i)$, and use the positive diagonal with entries 1, $m/(vs) + 1/v^2$, and $(1 + v/s)/w_i^2$, where m is the number of w coordinates. The remaining terms are $-e_u e_u^T + (\nabla s)(\nabla s)^T/s^2 - (e_v h^T + h e_v^T)/s$; ∇s belongs to $\text{span}\{e_u, e_v, h\}$. The inverse has the same rank bound for its departure from the inverse diagonal, by the inverse identity. This observation supplies no uniform spectral comparison or scalar-acquisition bound, so we make no corresponding parameter-free sampling claim.

10.4 Eccentricity profiles with full output

For explicit directions a useful comparison allows many cones, including a few with arbitrarily large eccentricity. Assume $A = [A_1 \cdots A_{L_c}]$ has full row rank, and each (t_j, z_j) is interior, with $t_j > \|z_j\|$ and $H_j = \nabla^2[-\log(t_j^2 - \|z_j\|^2)]$. Write $N = \sum_j A_j H_j^{-1} A_j^T$ and $S = \sum_j d_j A_j A_j^T$, with the notation of (83). For a threshold $\theta \geq 1$, retain the blocks $B_\theta = \{j : \chi_j > \theta\}$ exactly and put

$$P_\theta = \sum_{j \in B_\theta} A_j H_j^{-1} A_j^T + \sum_{j \notin B_\theta} d_j A_j A_j^T, \quad b_\theta = |B_\theta|.$$

Proposition 10.5 (Profile preconditioner and energy certificate). *The matrix P_θ is positive definite and*

$$\theta^{-1} P_\theta \preceq N \preceq \theta P_\theta, \quad \text{rank}(P_\theta - S) \leq 2b_\theta. \quad (89)$$

Preconditioned conjugate gradients reduce the N -energy error by a factor η in $O(\theta \log(2/\eta))$ iterations. If $q = b - N\hat{y}$, then

$$\theta^{-1} q^T P_\theta^{-1} q \leq \|\hat{y} - N^{-1}b\|_N^2 \leq \theta q^T P_\theta^{-1} q. \quad (90)$$

Proof. For each untreated block, $\theta^{-1} d_j I \preceq H_j^{-1} \preceq \theta d_j I$. The treated contributions are identical in the two matrices and positive semidefinite, so summing proves (89). Equation (83) gives the rank bound. The preconditioned spectrum lies in $[\theta^{-1}, \theta]$. The usual Chebyshev residual bound for conjugate gradients is $2((\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1))^k$ in the energy norm; $\kappa \leq \theta^2$ proves the iteration count. If $\theta = 1$, one iteration suffices. Inverting the Loewner bounds and using $\|\hat{y} - N^{-1}b\|_N^2 = q^T N^{-1} q$ proves the certificate. $\square$

The certificate has an exact Newton interpretation. For $H\Delta x + A^T \Delta y = r_d$ and $A\Delta x = r_p$, the reduced right-hand side is $b = AH^{-1}r_d - r_p$. An approximate multiplier induces $\widehat{\Delta x} = H^{-1}(r_d - A^T \widehat{\Delta y})$, and

$$\|\widehat{\Delta x} - \Delta x\|_H^2 = \|\widehat{\Delta y} - \Delta y\|_N^2. \quad (91)$$

Thus (90) controls the primal barrier metric without a raw Euclidean condition factor. It does not by itself certify a different full-KKT or Euclidean residual contract.

Here is an explicit implementation bound. Let A have m rows and n columns. Supply a width- τ elimination order for its structural row-intersection graph, and write $\bar{\tau} = \tau + 1$. Each column support is a clique of size at most $\bar{\tau}$, so assembling S costs $O(\bar{\tau} \text{nnz } A)$; its positive definite Cholesky factor costs $O(m\bar{\tau}^2)$ and occupies $O(m\bar{\tau})$ words. After deleting zero update columns, write $P_\theta = S + UJU^T$ with at most $2b_\theta$ columns and diagonal J with entries ± 1 . The small core $J + U^T S^{-1} U$ is nonsingular: the determinant identity and $P_\theta \succ 0$ prove this even when the update

is indefinite. Forming U , solving for $S^{-1}U$, forming its Gram matrix, and factoring the core give the following conservative exact-arithmetic setup and application costs:

$$C_{\text{setup}} = O(n + \bar{\tau} \text{nnz } A + m\bar{\tau}^2 + mb_{\theta}(\bar{\tau} + b_{\theta}) + b_{\theta}^3), \quad (92)$$

$$C_{\text{apply}} = O(m(\bar{\tau} + b_{\theta}) + b_{\theta}^2). \quad (93)$$

These counts use dense storage for the transformed correction columns. A matrix-free product by N costs $O(\text{nnz } A + n)$ by applying (83) blockwise. Hence the total solve cost is

$$C_{\text{prof}} = C_{\text{setup}} + O(\theta \log(2/\eta)[\text{nnz } A + n + C_{\text{apply}}]). \quad (94)$$

For an adaptive choice, sort the eccentricities in decreasing order. Keeping the largest b and choosing $\theta = \chi_{(b+1)}$, with $\chi_{(L_c+1)} = 1$, gives (94) with correction count at most b . Sorting costs $O(L_c \log L_c)$; evaluating the displayed bound for all b then costs $O(L_c)$. Ties do not affect the argument. This gives a tradeoff between correction count and the first untreated eccentricity, in the supplied coordinates.

The rank charge has a sharp worst-case interpretation. Take b genuine three-dimensional Lorentz blocks, $A = I$, and choose $t + r_z = \sqrt{2\chi}$, $t - r_z = \sqrt{2/\chi}$. Then $S = I$ and N has three orthogonal b -dimensional eigenspaces E_+, E_0, E_- with eigenvalues $\chi, 1, \chi^{-1}$. If $P = I + R \succ 0$, $R = R^T$, and $\text{rank } R < 2b$, then

$$\kappa(P^{-1/2}NP^{-1/2}) \geq \chi. \quad (95)$$

To prove this, let p_+, p_- be the positive and negative inertia of R . If $p_+ < b$, some $v \in E_+$ has $v^T R v \leq 0$, so its generalized Rayleigh quotient is at least χ . Since $\dim \ker R > b$, $\ker R \cap E_+^{\perp}$ contains a nonzero vector whose quotient is at most one. Otherwise $p_+ \geq b$ implies $p_- < b$; a vector in E_- has quotient at most χ^{-1} and one in $\ker R \cap E_-^{\perp}$ has quotient at least one. This proves (95). Thus guaranteeing condition at most θ^2 for every profile with b blocks of eccentricity $\chi > \theta^2$ requires rank at least $2b$ in this base-plus-update class. In the same $A = I$ family, allowing unequal block eccentricities, under the stronger Loewner contract (89), blocks with $\chi_j > \theta$ force at least b_{θ} positive and b_{θ} negative update directions by restriction to their high and low eigenspaces. The profile rank bound is sharp in that sense too. These are linear-algebra bounds within a specified preconditioner class, not oracle lower bounds for optimization.

Signed Woodbury arithmetic can be numerically unstable. An alternative uses $P_{\theta} = S + UU^T - VV^T$. Blockwise, $d_j I - r_j(t_j - r_j)u_{j,-}u_{j,-}^T \succ 0$, hence $S - VV^T \succ 0$. The augmentation

$$\begin{pmatrix} S & V & U \\ V^T & I & 0 \\ U^T & 0 & -I \end{pmatrix}$$

is symmetric quasidefinite: its positive principal block on the first two groups is positive definite, and its last block is negative definite. Adding its at most $2b_{\theta}$ hub vertices to every graph bag increases width by at most $2b_{\theta}$. Quasidefinite matrices admit pivot-free LDL^T factorization under every symmetric permutation [39]. A supplied width order therefore gives setup $O((m + b_{\theta})(\bar{\tau} + 2b_{\theta})^2)$ and subsequent application and storage $O((m + b_{\theta})(\bar{\tau} + 2b_{\theta}))$, in addition to coefficient formation. Either implementation may be used. Neither this exact factorization fact nor (94) is a uniform bit-complexity bound. Product-form factorizations address numerical cancellation in this setting [24]; working precision and residual verification still have to match the desired outer solve contract.

10.5 Latent sparsity and explicit-output comparisons

Dense cone blocks can also hide a small graph. For one Lorentz block, let $J_0 = \text{diag}(1, -I)$, $u = J_0(t, z)$ and $s = t^2 - \|z\|^2$. Then

$$H = -2J_0/s + 4uu^T/s^2 = D_x + ww^T, \quad w = 2u/s. \quad (96)$$

The diagonal D_x is indefinite. For all blocks, put these diagonals and embedded w columns into D_x, W . The equality Newton system has the same primal and multiplier solution as

$$\begin{pmatrix} D_x & W & A^T \\ W^T & -I & 0 \\ A & 0 & 0 \end{pmatrix} \begin{pmatrix} \Delta x \\ h \\ \Delta y \end{pmatrix} = \begin{pmatrix} r_d \\ 0 \\ r_p \end{pmatrix}. \quad (97)$$

Eliminating the hub block gives H exactly. Full row rank of A and $H \succ 0$ imply nonsingularity of both systems. Define a graph with the n scalar coordinate vertices, m equality-row vertices, and two hubs per cone; include every structural equality incidence and join both hubs to all coordinates of their cone. The matrix in (97) uses a subgraph of this fixed graph.

Proposition 10.6 (Supplied latent width). *If a width- τ_L tree decomposition of this graph, with $O((\tau_L + 1)(n + m + L_c))$ total bag entries, is supplied, then one Newton system can be solved exactly in*

$$O((n + m + L_c)(\tau_L + 1)^2)$$

field operations, after $O(n + \text{nnz } A)$ coefficient formation. This one-right-hand-side statement does not assert a reusable symmetric factor or numerical stability.

Proof. Use (96) to form (97) without square roots. Replace each vertex in each bag by its row and column copies. Every nonzero of the matrix is now covered in its row-column bipartite graph, both copies inherit a connected subtree, and the new width is at most $2\tau_L + 1$. Write $N_{\text{aug}} = n + m + L_c$ for the augmented matrix order and $k = O(\tau_L + 1)$ for the copied decomposition width plus one. The supplied total-bag-entry bound implies $|\mathcal{T}| = O(kN_{\text{aug}})$ decomposition nodes. The preprocessing in Section 3 of Fürer et al. [21] converts this compact decomposition to a nice decomposition with $O(N_{\text{aug}})$ nodes in $O(k(|\mathcal{T}| + N_{\text{aug}})) = O(k^2N_{\text{aug}})$ time. Their Corollary 3 then solves a consistent rectangular system in $O((\text{rows} + \text{columns})(\text{width} + 1)^2)$ field operations. Our system is square and nonsingular; applying that theorem proves the bound. Its arbitrary pivoting handles exact cancellations, so no zero-free prescribed pivot order is assumed. $\square$

For a reusable symmetric factor, there is a two-hub alternative. With $r_z > 0$, define

$$D = 2I/s, \quad u_+ = \frac{\sqrt{2r_z(t + r_z)}}{s}(1, -z/r_z), \quad u_- = \frac{\sqrt{2r_z(t - r_z)}}{s}(1, z/r_z).$$

Then $H = D + u_+u_+^T - u_-u_-^T$ and $u_-^T D^{-1}u_- = 2r_z/(t + r_z) < 1$, so $D - u_-u_-^T \succ 0$. Zero radial vectors give zero hub columns. For the product barrier let U, V collect the positive and negative columns. Replacing the dual zero block by $-\rho I$, $\rho > 0$, gives the expanded matrix

$$\begin{pmatrix} D & V & U & A^T \\ V^T & I & 0 & 0 \\ U^T & 0 & -I & 0 \\ A & 0 & 0 & -\rho I \end{pmatrix}.$$

It is quasidefinite with positive group (x, V) and negative group (U, y) . A width- τ_L order gives factorization $O((n + m + L_c)(\tau_L + 1)^2)$ and storage and subsequent solve cost $O((n + m + L_c)(\tau_L + 1))$. Forming these coefficients uses norms and square roots; these counts are in real arithmetic. If K_0 is the original KKT matrix, $w_0 = K_0^{-1}g \neq 0$, and $\rho \|K_0^{-1}\| < 1$, the Neumann bound gives

$$\frac{\|w_\rho - w_0\|}{\|w_0\|} \leq \frac{\rho \|K_0^{-1}\|}{1 - \rho \|K_0^{-1}\|}.$$

For $g = 0$ both solutions are zero. Thus a public inverse-norm bound can control regularization bias. Floating-point stability or successful refinement requires additional scaling and backward-error hypotheses.

The two-hub count is necessary for a positive diagonal base with signed rank-one stars on a generic block. If all coordinates of $u = J_0(t, z)$ are nonzero and the dimension is at least three, a representation $H = D' + \sigma vv^T$ with $D' \succ 0$ diagonal would require $\sigma(v_i/u_i)(v_j/u_j) = 4/s^2$ for every $i \neq j$. Three indices force all ratios to be equal and $\sigma = 1$, leaving $D' = -2J_0/s$, whose time diagonal is negative. This does not restrict more general sparse bases.

A concrete graph class has constant latent width. Let the coarse bipartite graph join cones to equality rows that use them, and assume each scalar column of A occurs in at most one row. If the coarse graph has width w , replace each cone vertex in each bag by its two hubs. Attach a leaf bag $\{u_j, v_j, x_i, y_k\}$ for a coordinate used by row k , to a bag containing the corresponding coarse edge; unused coordinates need only the first three vertices. Edge coverage and connectedness of vertex bags are preserved. Consequently

$$\tau_L \leq \max\{2w + 1, 3\}.$$

Coarse forests have $\tau_L \leq 3$, independently of block dimensions and number of cones. Materialized Hessians may nevertheless contain arbitrarily large cliques. The better of (94) and Proposition 10.6 is a valid exact-arithmetic full-output comparison: the profile method permits large latent width with few outliers; the latent-width method permits many eccentric blocks.

These comparisons concern accepted Newton solves, not a guarantee that arbitrary alternative directions follow the same trajectory. A trajectory replacement requires the structural and access assumptions uniformly over every iterate allowed by the robust outer algorithm, and requires the classical direction to meet that algorithm's stated residual contract. For example, even without cone structure, a consistent full-column-rank system $Kx = f$ of total row and column dimension D_K , with M nonzeros and condition κ_K , has a zero-start CGLS solution with

$$\|Kx_k - f\| \leq 2 \left(\frac{\kappa_K - 1}{\kappa_K + 1} \right)^k \|f\|.$$

This follows by applying the CG energy bound to $K^T K$, whose condition is κ_K^2 , and identifying that energy norm with the residual norm. It gives the familiar exact-arithmetic cost $O((M + D_K)\kappa_K \log(2/\eta))$ for relative residual η , and makes no new algorithmic claim. Direct sparse positive definite elimination with a supplied width- w order gives the separate $O(D(w + 1)^2)$ arm for order D . These elementary comparisons must be charged alongside a quantum solve when it returns the same explicit direction.

In particular, data with $n + m = D$, $\text{nnz } A \leq D^{1+o(1)}$, and supplied profile parameters $\bar{\tau}, \theta, b_\theta = D^{o(1)}$, or supplied latent width $\tau_L = D^{o(1)}$, admit $D^{1+o(1)}$ arithmetic per solve up to accuracy logarithms. In the latent-width case, the nonzero-count hypothesis follows already from the structural incidence graph: width τ_L bounds its edges, hence $\text{nnz } A$, by $O(D(\tau_L + 1))$, since $L_c \leq n$. Materializing a full direction already requires $\Omega(D)$ word writes. Under matched access and the preceding uniform outer hypotheses, these classes therefore exclude a polynomial dimension speedup obtained solely by replacing the linear solver in an architecture that materializes every iterate. This conclusion does not apply to scalar, sample, or quantum-state outputs. Nor does a dense assembled matrix or a large raw condition number by itself evade either structured comparison.

11 Conclusion

A scalar Newton quantity has its own complexity. Positive relative inverse forms admit local classical estimation with a linear conditioning factor in the variance, while sparse full-SQ families force both statistical and approximation-degree costs. Explicit composition shows when those costs occur on the same scalar and why a signal’s baseline cannot be ignored. Under plain block access, the known variable-time algorithm is optimal up to logarithms, but exact sparse entries obey a different comparison.

The optimization constructions make these distinctions concrete. Hardness can remain in an ordinary value, an optimizer coordinate, or a small Newton decrement with fully simulated formulation access. It can also accumulate in a sequence of requested scalars or a single endpoint. The same examples show the limits of such conclusions: an unperturbed value may be public, a normalized state may discard a hard sign or scale, and a hard Newton ray may be reused throughout a long trajectory. Consequently, any whole-optimization comparison must account for input acquisition, the particular outputs requested, and information retained between solves.

Structured systems provide a complementary conclusion. A sparse full-rank base with a few controlled corrections permits scalar estimation and sampling without constructing a dense inverse or normalizing small correction columns. Cone dimension alone then does not determine query complexity. For explicit directions, eccentricity profiles and latent sparsity give different classical solve bounds. Their assumptions must hold at the iterates actually used, and their residual guarantees must match the outer method. Together, the results provide concrete bounds and counterexamples with which to make such comparisons, while leaving the unmatched general parameter regimes stated in the introduction open.

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